Form-Wound Squirrel Cage Induction Motors— 250 Horsepower and Larger

API STANDARD 541
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Manufacturing, Distribution and Marketing Department

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Form-Wound Squirrel Cage Induction Motors—250 Horsepower and Larger

SECTION 1—GENERAL

1.1 Scope

1.1.1 This standard covers the minimum requirements for all form-wound squirrel cage induction motors above National Electrical Manufacturers Association (NEMA) 440 frame sizes (nominally 250 horsepower and larger) for use in petroleum industry services.

This standard may be applied to adjustable speed motors and induction generators with appropriate attention to the specific requirements of such applications. For adjustable speed applications, proper selection of the motor and drive are required to avoid the following conditions:

a. Motor RMS current exceeding the continuous sinusoidal nameplate rating due to excessive voltage harmonics or improper volts/hertz levels.
b. Excessive winding temperature due to insufficient cooling, excessive torque levels, improper volts/hertz levels, and increased losses due to harmonics.
c. Insufficient motor accelerating torque at reduced speeds due to insufficient volts/hertz levels or limitations in the drive's momentary current capacity.
d. Increased noise levels due to increased fan noise (above base speed), excitation of mechanical resonances, or magnetic noise caused by supply source harmonics.
e. Mechanical failure of the motor or coupling due to the torque pulsations, operation at or near mechanical resonances, or excess speed.
f. Winding failures due to repetitive high amplitude voltage spikes created by the drive system.
g. Damage to the motor and drive due to improper application of power factor correction capacitors.
h. Higher motor temperatures that may limit application in Division 2 hazardous areas.
i. Shaft voltages/currents due to harmonics that may cause bearing failure.

Note 1: A round bullet (●) at the beginning of a paragraph indicates that either a decision is required or further information is to be provided by the purchaser. This information should be indicated on the data sheets (see Appendix A); otherwise it should be stated in the quotation request or in the order.

Note 2: A square bullet (■) at the beginning of the paragraph indicates additional requirements for special purpose motors. Special purpose motors are motors that drive high-inertia loads, vertical motors that support high-thrust loads, motors that are part of a complete train requiring vibration sensitivity criteria, motors that operate in abnormally hostile environments, or motors that drive critical unspared equipment.

● **1.1.2** When the motor is specified as special purpose on the Purchaser's Induction Motor Data Sheets (Appendix A), the paragraphs that are marked with square bullets (■) shall also apply. Where a paragraph includes both a round and square bullet, that paragraph shall apply when the motor is specified as special purpose, or when indicated on the data sheets, if the motor has not been specified as special purpose.

1.1.3 This standard requires the purchaser to specify certain details and features. In order for this standard to be meaningful as an aid to procurement, the purchaser must complete the data sheets in Appendix A.

1.2 Alternative Designs

The vendor may offer alternative designs (see 6.1.10 for proposal requirements). Equivalent metric dimensions, fasteners, and flanges may be substituted as mutually agreed upon by the purchaser and the vendor.

1.3 Conflicting Requirements

In case of conflict between this standard and the inquiry or order, the information included in the inquiry or order shall govern.

1.4 Coordination Responsibility

The vendor shall be responsible for providing the purchaser with all details necessary for the installation and operation of the motor as an integral part of the complete driven-unit assembly (see 6.2 for the information required).

1.5 Definition of Terms

1.5.1 *Breakdown torque* of a motor is the maximum torque that it will develop with rated voltage applied at rated frequency without an abrupt drop in speed.

1.5.2 *Pull-up torque* of an AC motor is the minimum torque developed by the motor during the period of acceleration from rest to the speed at which breakdown torque occurs. For motors that do not have a definite breakdown torque, the pull-up torque is the minimum torque developed up to the rated speed.

1.5.3 *Locked-rotor torque* is the minimum average torque that the motor will produce with a locked rotor at any angular position of the rotor with its rated voltage and frequency applied.

1.5.4 *Accelerating torque* is the difference between the input torque to the rotor and the sum of the load and loss torques; the net torque available for accelerating the rotating parts.

1.5.5 A *cold start* is a motor start that occurs when the rotor and stator are initially at ambient temperature.

1.5.6 A *hot start* is any restart of the motor that occurs when the motor is at a temperature above ambient temperature.

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1.5.7 *Lateral critical speeds* correspond to resonant frequencies of the complete rotor, bearings, and bearing support system.

Note: The basic identification of critical speeds is made from the natural frequencies of the system and of the forcing phenomena. If the frequency of any harmonic component of a periodic forcing phenomenon is equal to or approximates the frequency of any mode of rotor vibration, a condition of resonance may exist. If resonance exists at a finite speed, that speed is called a critical speed. This standard is concerned with actual resonant speeds rather than various calculated values. Actual critical speeds are not calculated undamped values but are critical speeds confirmed by test stand data. Critical speeds above the maximum test speed shall be calculated damped values or shall be determined by externally applied rotor excitations.

1.5.8 *Torsional critical speeds* correspond to resonant frequencies of the complete mass-elastic system in the drive train, including couplings and driven equipment. The torsional analysis, which is normally provided by the driven equipment supplier, considers excitations of torsional natural frequencies, including but not limited to, the following:

a. Gear problems such as unbalance and pitch line runout.
b. Torsional transients such as motor start-ups.
c. Torsional excitation resulting from drivers such as motors and reciprocating engines.
d. Torsional excitations resulting from adjustable speed drives.

1.5.9 *Forcing phenomena for vibration excitation* are defined as follows. The exciting frequency may be less than, equal to, or greater than the synchronous frequency of the rotor. Potential excitations to be considered in the design of the system shall include, but are not limited to, the following sources:

a. Mechanical unbalance in the rotor system.
b. Oil-film instabilities (whirl).
c. Internal rubs.
d. Misalignment.
e. Loose rotor system components.
f. Hysteresis and friction.
g. Start-up condition frequencies.
h. Twice the line frequencies.
i. Electrical unbalance.
j. Mechanical pulsations produced by the motor load.
k. Short circuits (faults) on the electrical system.

1.5.10 *Service factor.* Induction motors that have a rated temperature rise in accordance with this standard and that are operated at rated voltage and frequency shall have a service factor of 1.0. In applications that require an overload capacity, a higher horsepower rating should be used to avoid exceeding the temperature rises for the class of insulation used and to provide adequate torque capacity.

1.5.11 *Trip speed* (in revolutions per minute) is the speed at which the independent emergency speed device operates to shut down an adjustable speed motor.

1.5.12 The *power factor* is the ratio of kilowatt input to kilovolt-ampere input.

1.5.13 The use of the word *design* in any term (such as design power, design temperature, or design speed) shall be avoided in the purchaser's specifications. This terminology should be used only by the equipment designer and the manufacturer.

1.6 Referenced Publications

The editions of the following standards, codes, and specifications that are in effect at the time of publication of this standard shall, to the extent specified herein, form a part of this standard. The applicability of changes in standards, codes, and specifications that occur after the inquiry shall be mutually agreed upon by the purchaser and the vendor.

AFBMA[1]

Std 7 *Shaft and Housing Fits for Metric Radial Ball and Roller Bearings*
Std 8.2 *Mounting Accessories, Inch Design — Locknuts, Washers, and Adapters*
Std 9 *Load Ratings and Fatigue Life for Ball Bearings*
Std 11 *Load Ratings and Fatigue Life for Roller Bearings*
Std 20 *Radial Bearings of Ball, Cylindrical Roller, and Spherical Roller Types, Metric Design: Basic Plan for Boundary Dimensions, Tolerances, and Identification Code*

AGMA[2]

9002 *Bores and Keyways for Flexible Couplings (Inch Series)*

ANSI[3]

C50.41 *Polyphase Induction Motors for Power-Generating Stations*

API

Std 614 *Lubrication, Shaft Sealing, and Control Oil Systems for Special Purpose Applications*
Std 670 *Vibration, Axial Position, and Bearing Temperature Monitoring Systems*
Std 671 *Special Purpose Couplings for Refinery Service*
Std 678 *Accelerometer-Based Vibration Monitoring System (out-of-print)*

ASME[4]

B1.1 *Unified Inch Screw Threads (UN and UNR Thread Form*
B1.20.1 *Pipe Threads, General Purpose (Inch)*
B16.1 *Cast Iron Pipe Flanges and Flanged Fittings, Class 25, 125, 250, and 800*

[1]Anti-Friction Bearing Manufacturers Association, 1101 Connecticut Avenue, N.W., Suite 700, Washington, D.C. 20036.
[2]American Gear Manufacturers Association, 1500 King Street, Suite 201, Alexandria, Virginia 22314.
[3]American National Standards Institute, 1430 Broadway, New York, New York 10017.
[4]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.

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B16.2 *Ring Joint Gaskets and Grooves for Steel Pipe Flanges*
B16.5 *Pipe Flanges and Flanged Fittings*
Boiler and Pressure Vessel Code, Section V, "Nondestructive Examination"; Section VIII, "Rules for Construction of Pressure Vessels"; and Section IX, "Welding and Brazing Qualifications"
B16.11 *Forged Steel Fittings, Socket Welding and Threaded*

ASTM[5]
A 278 *Gray Iron Castings for Pressure-Containing Parts for Temperatures up to 650°F*
A 345 *Flat-Rolled Electrical Steels for Magnetic Applications*
A 395 *Ferritic Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures*
A 515 *Carbon Steel Pressure Vessel Plates for Intermediate and Higher Temperature Service*
A 536 *Ductile Iron Castings*
E 94 *Guide for Radiographic Testing*
E 125 *Reference Photographs for Magnetic Particle Indications on Ferrous Castings*
E 142 *Method for Controlling Quality of Radiographic Testing*
E 709 *Practice for Magnetic Particle Examination*

AWS[6]
D1.1 *Structural Welding Code—Steel*

IEEE[7]
43 *Recommended Practice for Testing Insulation Resistance of Rotating Machinery*
85 *Test Procedure for Airborne Sound Measurements on Rotating Electric Machinery*
112 *Test Procedures for Polyphase Induction Motors and Generators*
286 *Recommended Practice for Measurement of Power-Factor Tip-up of Rotating Machinery Stator Coil Insulation*
303 *Recommended Practice for Auxiliary Devices for Motors in Class I — Groups A, B, C, and D, Division 2 Locations*
429 *Standard Test Procedure for the Evaluation of Sealed Insulation Systems for AC Electrical Machinery Employing Form-Wound Stator Coils*
522 *Guide for Testing Turn-to-Turn Insulation on Form-Wound Stator Coils for AC Rotating Electric Machines*

ISO[8]
1940/1 *Mechanical Vibration—Balance Quality Requirements of Rigid Rotors*
3448 *Standard Industrial Liquid Lubricants—ISO Viscosity Classification*

NEMA[9]
MG 1 *Motors and Generators*

NFPA[10]
70 *National Electrical Code*
497M *Classification of Gases, Vapors, and Dusts for Electrical Equipment in Hazardous (Classified) Locations*

SSPC[11]
SP 6 *Commercial Blast Cleaning*

TEMA[12]
Standards of the Tubular Exchanger Manufacturers Association

[5]American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.
[6]American Welding Society, 2551 N.W. 7th Street, Miami, Florida 33125.
[7]Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, New York 10017.
[8]International Organization for Standardization, ISO publications are available from ANSI.
[9]National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, D.C. 20037.
[10]National Fire Protection Association, Batterymarch Park, Quincy, Massachusetts 02169.
[11]Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, Pennsylvania 15213-2683.
[12]Tubular Exchanger Manufacturers Association, 25 North Broadway, Tarrytown, New York 10591.

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SECTION 2—BASIC DESIGN

2.1 General

2.1.1 The equipment (including auxiliaries) covered by this standard shall be suitable for the specified operating conditions and shall be designed and constructed for a minimum service life of 20 years and at least 3 years of uninterrupted continuous operation. It is recognized that this is a design criterion and that uninterrupted operation for this period of time involves factors beyond the vendor's control.

● **2.1.2** Motors shall be designed for continuous operation and long periods of inactivity in an atmosphere that is made corrosive by traces of chemicals normally present in a petroleum processing facility and in an environment that may include high humidity, storms, salt-laden air, insects, plant life, fungus, and rodents. Motors shall be suitable for operation, periods of idleness, storage, and handling at the ambient temperatures listed under "Site Data" on the data sheets (see Appendix A). If additional considerations are necessary in the proposal, the purchaser shall specify them.

● **2.1.3** Control of the sound pressure level of all equipment furnished shall be a joint effort of the purchaser and the vendor. The equipment furnished by the vendor shall conform to the maximum allowable sound pressure level specified by the purchaser. The measuring and reporting of sound-level data shall be in accordance with IEEE 85.

● **2.1.4** All equipment shall be designed to run safely at the overspeeds specified in NEMA MG1-20.44 or to the specified trip speed (including overshoot) of the connected equipment, whichever is greater.

2.1.5 The arrangement of the equipment, including number of bearings, terminal housings, conduit, piping, and auxiliaries, shall be developed jointly by the purchaser and the vendor. The arrangement shall provide adequate clearance areas and safe access for installation, operation, and maintenance.

2.1.6 The design of piping systems shall achieve the following:

a. Proper support and protection to prevent damage from vibration or from shipment, operation, and maintenance.
b. Proper stiffness and normal accessibility for operation, maintenance, and thorough cleaning.
c. Installation in a neat and orderly arrangement adapted to the contour of the machine without obstructing access openings.
d. Elimination of air pockets.
e. Complete drainage through low points without disassembly of piping.
f. Provision for easy removal of covers for maintenance and inspection.

● **2.1.7** Electrical components and installations shall be suitable for the area classification (Class, Group, and Division) specified by the purchaser on the data sheets and shall meet the requirements of NFPA 70, Articles 500, 501, and 502, as well as local codes. The purchaser will furnish with the inquiry any applicable local codes or regulations.

2.1.8 Oil reservoirs and housings that enclose parts such as bearings, shaft seals, instruments, control elements, and terminals shall be designed to minimize contamination by moisture, dust, and other foreign matter during periods of operation or idleness. To avoid condensation, the minimum inlet temperature of the bearing cooling water shall be above the ambient air temperature.

2.1.9 Equipment shall be designed to permit rapid and economical maintenance. Major parts, such as frame components and bearing housings, shall be designed (shouldered or doweled) and manufactured to ensure accurate alignment on reassembly. Easily removable covers shall be provided for maintenance and inspection of coil end turns.

2.1.10 Every effort shall be made to avoid requirements for special tools. However, when tools and fixtures not commercially available are required to disassemble, assemble, or maintain the unit, they shall be included in the quotation and furnished as part of the initial supply of the motor. For multiple-unit installations, the requirements for quantities of special tools and fixtures shall be mutually agreed upon by the purchaser and vendor.

2.1.11 When special tools are provided, they shall be packaged in separate, rugged boxes and marked as "special tools for (tag/item number)." Each tool shall be tagged to indicate its intended use.

2.1.12 The motor shall perform within the specified acceptance criteria on the test stand and on its permanent foundation. After installation, the performance of the combined units shall be the joint responsibility of the purchaser and the vendor who has responsibility for the train.

● **2.1.13** Many factors (such as oil-or-purge piping and ductwork loads, alignment at operating conditions, supporting structure, handling during shipment, and handling and assembly at the site) may adversely affect site performance. When specified, to minimize the influence of these factors, the vendor shall review and comment on the purchaser's piping and foundation drawing, and the vendor's representative shall observe a check of the piping and ductwork performed by parting the ductwork and flanges. The vendor's representative shall check alignment at the operating temperature and, when specified, shall be present during the initial alignment check.

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2.1.14 Spare parts for the motors and all furnished auxiliaries shall meet all the criteria of this standard.

2.2 Electrical Design

2.2.1 RATING AND VOLTAGE

● **2.2.1.1** NEMA standard ratings shall be selected. If the required rating falls between two listed ratings, the larger listed rating shall be selected.

2.2.1.2 In general, motors are rated in accordance with Table 1.

● **2.2.1.3** Motors shall be capable of continuous operation at a service factor and temperature rise in accordance with 2.3.1.1 when operated at their rated horsepower, voltage, and frequency, both mechanically and electrically. When a service factor of 1.15 is specified, the motor may have a temperature rise of an additional 10°C (18°F) over that specified in 2.3.1.1 when continuously operated at the service factor horsepower and the rated voltage and frequency.

2.2.2 STATOR WINDINGS

● **2.2.2.1** Where differential protection is specified, both ends of each stator-phase winding shall be brought out to the terminal housing that encloses the current transformers (see 3.1.8 for additional requirements).

2.2.2.2 Coil ends shall be braced to prevent insulation cracking and fatigue as a result of motion during operation and starting. Windings shall withstand the starting duties specified in 2.2.6 with a life of at least 5000 full-voltage starts. The motor windings shall be braced to withstand an external three-phase short circuit at full load and 110 percent of rated voltage.

2.2.3 LOAD REQUIREMENTS

2.2.3.1 Unless otherwise specified, the load torque characteristics and total load inertia referred to the motor shaft shall be in accordance with NEMA MG-1-20.42.

● **2.2.3.2** Should the loads have other characteristics than those listed in NEMA MG1-20, the purchaser shall fully specify the load characteristics of the driven equipment. These shall include the following:

a. The speed torque characteristics of the load under the most stringent starting conditions.
b. When required, the speed torque characteristics of the load during reaccelerating conditions, the length of maximum voltage interruptions or fault related collapse, and the expected voltage at motor terminals during reacceleration.
c. The total load inertia (WK^2) referred to the motor shaft speed, where W is the rotor weight and K is the radius of gyration. This total load inertia shall include all loads connected to the motor shaft, such as couplings, gearbox, and driven equipment.

2.2.4 VOLTAGE AND FREQUENCY VARIATIONS

Unless otherwise specified, the motor shall operate in accordance with the voltage and frequency variations dictated by NEMA MG 1-20.45.

2.2.5 STARTING AND RUNNING CONDITIONS

2.2.5.1 Unless otherwise specified, the motor shall start and accelerate the connected load to running speed in accordance with NEMA MG 1-20.45.

● **2.2.5.2** Should the power supply system have characteristics other than those listed in NEMA MG 1-20, the requirements for starting capability, speed torque, and acceleration time shall be determined with the following information, as applicable, furnished by the purchaser:

a. The expected voltage at motor terminals under specified inrush currents, or
b. The minimum available system short circuit in KVA, the X/R ratio, and the minimum motor terminal voltage during starting, with all values referenced at the motor terminals.

2.2.5.3 When the motor speed torque curve at the conditions specified in 2.2.5.2 a or b is plotted over the load speed torque curve, the motor-developed torque shall exceed the load torque by a minimum of 10 percent (motor-rated torque as base) at all locations throughout the speed range up to the motor breakdown torque point.

2.2.6 STARTING CAPABILITIES

2.2.6.1 Fixed-speed motors shall have the starting capabilities in Table 2.

Note: Typical process plant operations are such that a motor will have a period of initial use of about 2 months for pump and compressor run-in and initial plant operations, during which time the maximum starting capability may be used. A need for maximum capability may also occur during subsequent start-ups. Between these start-up periods, there are usually longer periods of continuous running.

● **2.2.6.2** Starting capabilities for motors in excess of those shown in Table 2 shall be jointly developed between the purchaser and the vendor.

2.2.6.3 The minimum safe stall (locked rotor) time shall be the greater of 150 percent of, or 5 seconds more than, the time required to accelerate the actual driven load with the

Table 1—Motor Ratings

Horsepower	Phases	Frequency (hertz)	Voltage
250-600	3	60	460
250-600	3	60	575
250-4,000	3	60	2,300
250-7,000	3	60	4,000
1,000-12,000	3	60	6,600
3,500-Above	3	60	13,200

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Table 2—Starting Capabilities

Capability	Number of Starts at a Service Factor of 1.0
Consecutive 2-second jogging applications 10 seconds apart.[a]	3
Consecutive starts with the motor coasting to rest between starts.[a]	2
Consecutive starts with the motor coasting to rest and remaining idle for 20 minutes.[a]	3
Evenly spaced starts in first hour before continuous running.[a]	3[b]

Note: Where the total load inertia referred to the motor shaft exceeds 200 percent of the value listed in NEMA MG 1-20.42, the number of starts shall be in accordance with the value listed in NEMA MG 1-20.43.
[a]The first start is a cold start.
[b]Where the total load inertia referred to the motor shaft does not exceed 66 percent of the value listed in NEMA MG 1-20.42, the number of starts permitted is four.

voltage values specified under Paragraph 2.2.1.2. If these conditions cannot be met, the vendor shall notify the purchaser so that a workable solution can be jointly developed.

2.2.7 PERFORMANCE

With their rated voltage and frequency applied, motors shall, as a minimum, operate with the characteristics listed below. Where these limits will have an adverse effect on other characteristics, particularly efficiency, the motor vendor shall state the effect and recommend preferred values.

a. The maximum full-load slip shall not exceed 3 percent.
b. The maximum lock-rotor current shall not exceed 650 percent of the full-load current.
● c. Unless otherwise specified, the minimum locked-rotor, pull-up, and breakdown torques shall be no less than the values listed in NEMA MG1-20.41.

2.3 Insulation Systems

2.3.1 GENERAL REQUIREMENTS

2.3.1.1 All motors shall have an epoxy base, vacuum-pressure impregnated, nonhygroscopic insulation system, including leads and connections to the windings. When bus bars are used, they shall be insulated. As a minimum, the insulation system shall meet the criteria for NEMA Class F insulation; however, the allowable temperature rise above ambient, 40°C (104°F), shall not exceed that listed in NEMA MG 1-20.40 for Class B insulation. Strand insulation shall adhere tightly to the strand. The integrity of strand and turn insulation shall be maintained during forming, winding, and vacuum pressure impregnation (VPI) treatment. For multiturn stator windings, additional turn insulation shall be used as required to maintain the integrity of turn insulation in the noses or other areas of coil deformation. The insulation system shall be capable of withstanding the surge test specified in 4.3.4.2. Special provisions shall be made to seal the leads where they exit the coil. Stator coils shall be tightly secured in their slots and securely braced outside the slots to withstand electromagnetic and mechanical stresses during start-up and under normal operating conditions.

2.3.1.2 The total insulation system shall be impervious to the operating conditions specified in 2.1.2. All insulation, including lead insulation, shall be impervious to attack by the lubricating oil specified. Leads shall be braced and protected from chafing against the motor frame and terminal box.

2.3.1.3 All stator insulation systems shall be service proven and shall have been subjected to thermal evaluation in accordance with IEEE 429.

2.3.2 SEALED INSULATION SYSTEMS

Stators shall have a sealed insulation system, as defined by NEMA MG 1-1.27.2, that is capable of withstanding a sealed winding conformance test specified in 4.3.4.4.

2.4 Mechanical Design

2.4.1 ENCLOSURES

2.4.1.1 General Requirements

The following general requirements apply to enclosures:

a. Enclosure parts shall be made of cast or nodular iron, cast steel, cast aluminum, steel plate, or aluminum plate. Purchaser-approved, fiber-reinforced materials may be used for parts such as covers or nonsupportive enclosure sections. All enclosure parts shall have a minimum rigidity equivalent to that of sheet steel with a nominal thickness of ⅛ inch (3.0 millimeters).
b. Air deflectors shall be made of corrosion-resistant material or shall have corrosion-resistant plating or treatment.
● c. All the enclosure's bolts, studs, and other fastening devices shall be made of corrosion-resistant materials. When specified, AISI 300 series stainless steel shall be used. Internal fastening devices shall use locknuts, lock washers, locking plates, or tie wires.
d. The enclosure shall be designed to facilitate cleaning and painting of the motor's interior.
e. If aluminum is used, it shall have a copper content of less than 0.2 percent.
f. The impact of potential risks due to possible circulating currents in the enclosure should be considered for motors using multisection enclosures installed in classified locations. Overheating or sparking due to possible circulating currents shall be avoided, where necessary, by securing together the conducting components in a sound electrical and mechanical

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manner, or by providing adequate bonding straps between the motor housing components.

2.4.1.2 Motor Enclosures and Corresponding NEMA Specifications

2.4.1.2.1 Table 3 lists types of motor enclosures and the NEMA specifications to which they shall conform. Motor enclosures shall also conform to the requirements of 2.4.1.2.2 and 2.4.1.2.3.

2.4.1.2.2 Dripproof, weather-protected Type I motors and weather-protected Type II motors shall meet the following criteria:

a. Ventilation and other openings shall be limited in size by design or by metal screens in accordance with 3.5.1 and NEMA MG 1-1.25.4.

Note: Dripproof motors are not recommended for the operating conditions specified in 2.1.2 (that is outdoor operation without a protective shelter). Purchasers applying this type of motor should expect reduced reliability (see 2.1.1).

b. Weather-protected Type I and Type II motors shall be constructed so that any accumulation of water will drain from the motor before reaching the level of the windings or other live parts.

● c. When abrasive dust conditions have been specified on the data sheets under "Site Data," electrical insulation shall be protected from the abrasive action of airborne particles. This protection shall be in addition to the VPI resin and the manufacturer's standard coating.

2.4.1.2.3 Totally enclosed motors shall meet the following criteria:

a. Enclosures shall completely enclose the motors. Designs in which the stator laminations form a part of the enclosure or in which the stator laminations are otherwise exposed to external cooling air are not acceptable.

b. Fan covers shall be made of cast or nodular iron, cast steel, cast aluminum, steel plate, or aluminum plate. Covers shall have a minimum rigidity equivalent to that of steel plate with a nominal thickness of ⅛ inch (3.0 millimeters). Purchaser-approved, fiber-reinforced materials may be used. The air intake opening shall be guarded by a grill or a metal screen fastened on the outside of the fan cover. Requirements for grills or metal screens are covered in 2.4.10.5.

c. Sheet metal covers or wrappers used to form air passages over the motor enclosure shall have a minimum rigidity equivalent to that of steel plate with a nominal thickness of ⅛ inch (3.0 millimeters).

d. Totally enclosed motors shall be equipped with a plugged, threaded drain connection located at the lowest point of the frame. This connection shall be shown on the motor outline drawing.

e. Requirements for air-to-air heat exchanger tubes are in 2.4.10.8.

Table 3—Motor Enclosures and Corresponding NEMA Specifications

Common Enclosure	NEMA MG 1 Designation	NEMA MG 1 Specifications
Dripproof	DP	1.25.1
Explosionproof	XP	1.26.3
Weather protected		
Type I	WP-I	1.25.8.1
Type II	WP-II	1.25.8.2
Totally enclosed		
Fan cooled	TEFC	1.26.2
Pipe ventilated	TEPV	1.26.6
Water/air cooled	TEWAC	1.26.8
Air/air cooled	TEAAC	1.26.9

f. Where an enclosure make-up air intake is required for supplying pressurized air to the bearing seals, the intake shall be provided with filters suitable for the site data given on the data sheets.

2.4.1.2.4 Totally enclosed water/air-cooled motors shall meet the following criteria:

● a. Cooling water systems shall be designed for the following conditions unless the vendor notifies the purchaser that conflict will arise affecting performance, size/cost, and integrity of the cooler. The purchaser shall approve the final selection.

Velocity over heat exchange surfaces	5-8 ft/s (1.5-2.5m/s)
Maximum allowable working pressure	≥ 75 psig (5 bar (ga))
Test pressure (minimum of 1.5 times the maximum allowable working pressure)	≥115 psig (7.9 bar (ga))

Maximum pressure drop	15 psig	1 bar
Maximum inlet temperature	90°F	32°C
Maximum outlet temperature	120°F	49°C
Maximum temperature rise	30°F	17°C
Minimum temperature rise	20°F	11°C
Fouling factor on water side	0.002hr-ft2-°F/Btu	0.35m2-K/kW

Note: The criterion for velocity over the exchange surfaces is intended to minimize water-side fouling; the criterion for minimum temperature rise is intended to minimize the use of cooling water.

● b. The location of the cooler, orientation of the water box inlet and outlet, materials and construction of the cooler, cleaning requirements, and means of leak detection shall be developed jointly by the purchaser and the vendor. Leak detectors shall be provided to sense tube leakage. For double tube coolers, these detectors shall sense inner tube leakage and, when specified, outer tube leakage.

c. Cooler designs shall be of the water-tube type (water in the tubes). U-tube construction is not permitted. The construction of the water box and header shall be such that leaking tubes can be readily plugged and all tubes are accessible for cleaning.

● d. The motor's interior shall be baffled or otherwise constructed to prevent cooler tube leakage from striking the motor winding directly. When specified, coolers mounted above the motor windings shall be of double tube construction, requiring no further baffling. Alternatively, single tube, top-

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mounted coolers with proper baffling approved by the purchaser may be used where experience has proved this construction to be adequate. Pedestal-mounted coolers or coolers mounted underneath the winding may be of single tube construction. Cooler materials shall be compatible with the water supply.

e. The interior of the motor base shall be constructed so that cooler leakage will collect and drain from the motor before reaching the level of the windings. In pressurized motors, a liquid seal shall be provided for drain holes. A level alarm to warn of cooler leakage shall be provided.

● f. A flow-sensing device shall be provided for mounting in the water supply piping to each cooler. When specified, this device shall have a local indicator calibrated in gallons per minute, as well as switches activated by low and high flow. The high-flow switch shall be used to activate an alarm. The low-flow switch shall be supplied with the motor manufacturer's recommended setting, below which the motor will overheat and winding damage will result within 15 minutes of the low-flow occurrence.

● g. When specified, air temperature sensors shall be provided to sense air temperature into and out of the coolers.

h. Requirements for heat exchanger tubes are in 2.4.10.8.

2.4.2 FRAME AND MOUNTING PLATES

2.4.2.1 The frame shall be of cast or nodular iron, cast steel, or welded steel plate construction with removable end bells or end plates to permit removal of the rotor and facilitate replacement of stator coils. The frame of the completely assembled machine on its permanent foundation, with the rotor installed and rotating on its oil film, shall be free from structural resonance within ±5 percent of the electric line frequency and the following frequency ranges:

$$N = nN_{op} \pm 0.20N_{op}$$

Where:

N = frequency range, in hertz.
N_{op} = operating speed frequency, in hertz.
n = 1, 2, and 3.

Note: Transfer of vibration from surrounding equipment is avoided by proper layout of the foundation, which is the responsibility of the purchaser. After the machine is erected, the natural frequency of the foundation must differ by at least ±20 percent from one and two times the running speed frequency and by at least ±20 percent from one and two times the electric line frequency.

2.4.2.2 The stress values used in the design of the frame shall not exceed the maximum allowable stress criteria specified in Section VIII, Division I, of the ASME Code for the material used. The conditions evaluated shall include short circuits, out-of-phase reclosing, thrust, handling, and specified seismic loading.

2.4.2.3 The motor frame, including transition base, if supplied with the motor, and the bearing supports, shall be designed to have sufficient strength and rigidity to limit changes of alignment caused by the worst combination of torque reaction, conduit and piping stress, magnetic imbalance, and thermal distortion to 0.002 inch (0.05 millimeter) at the coupling flange. (This is not to be confused with the normal repeatable thermal growth between ambient and operating temperatures). The purchaser shall provide the motor vendor with the piping and conduit loading.

2.4.2.4 Frames on horizontal motors shall be rigid enough to permit the machine to be moved by the use of the lateral, axial, and vertical jackscrews.

2.4.2.5 Horizontal motors shall be equipped with vertical jackscrews appropriately located to facilitate alignment. If size and weight prohibit the use of jackscrews, other provisions shall be made for vertical jacking.

● **2.4.2.6** When specified, the motor shall be furnished with soleplates or a baseplate.

2.4.2.7 The term *mounting plate* refers to both baseplates and soleplates.

2.4.2.7.1 The upper and lower surfaces of bearing pedestals and mounting plates shall be machined parallel.

2.4.2.7.2 Mounting plates shall be equipped with vertical jackscrews to permit leveling of the mounting plates.

2.4.2.7.3 When the motor weighs more than 1000 pounds (453 kilograms), the mounting plates shall be furnished with horizontal jackscrews the same size as or larger than the vertical jackscrews. The lugs holding these jackscrews shall be attached to the mounting plates so that they do not interfere with the installation or removal of the drive element and the installation or removal of shims used for alignment.

2.4.2.7.4 When centerline supports are provided, they shall be designed and manufactured to permit the motor to be moved by using the horizontal jackscrews.

● **2.4.2.7.5** When epoxy grout is specified on the data sheets, the vendor shall commercially sandblast, in accordance with SSPC SP 6, all the grouting surfaces of the mounting plates and shall precoat these surfaces with a catalyzed epoxy primer applied to degreased white metal. The epoxy primer shall be compatible with epoxy grout. The purchaser shall submit to the motor vendor the specifications for the epoxy primer.

2.4.2.7.6 Anchor bolts shall not be used to fasten the motor to the mounting plates.

2.4.2.7.7 Mounting plates shall be supplied with leveling screws. Mounting plates that are to be grouted shall have 2-inch radiused (50-millimeter radiused) outside corners (in the plan view). Mounting surfaces that are not to be grouted shall be coated with a rust preventative.

2.4.2.7.8 Mounting plates shall be designed to extend at least 1 inch (25 millimeters) beyond the outer sides of the motor feet.

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2.4.2.7.9 The vendor of the mounting plates shall furnish AISI 300 series stainless steel shim packs at least ⅛ inch (3.0 millimeters) thick between the motor feet and the mounting plates. All shim packs shall straddle the hold-down bolts.

● **2.4.2.7.10** Anchor bolts will be furnished by the purchaser.

2.4.2.7.11 Fasteners for attaching the components to the mounting plates and jackscrews for leveling the soleplates shall be supplied by the vendor.

2.4.2.7.12 The horizontal and vertical jacking bolts shall be ¾ inch minimum diameter with UNC threads.

2.4.2.8 Frame mounting surfaces on a horizontal motor shall be machined on a common plane parallel to a horizontal plane through the motor's theoretical centerline. The mounting surfaces shall be machined within 0.005 inch (0.13 millimeter) of a plane through the lowest foot, and each foot shall be parallel to that plane, in the transverse or longitudinal direction, within 0.002 inch per foot (0.17 millimeter per meter).

2.4.2.9 The mounting surface on a vertical motor shall be machined perpendicular to the motor's centerline, and this surface shall not deviate from that perpendicular plane by more than 0.002 inch per foot (0.17 millimeter per meter).

2.4.2.10 The machined finish of the mounting surface shall not exceed 250 microinches (6.4 micrometers) arithmetic average roughness (R_a). Hold-down or foundation bolt holes shall be drilled perpendicular to the mounting surface or surfaces and, when the machined surface is a cast or other unmachined, uneven surface, spot faced to a diameter three times that of the hole diameter.

2.4.2.11 The motor frame support or supports shall be provided with two pilot holes for dowels. The holes shall be as near the vertical as is possible and shall be located to provide adequate space for field drilling, reaming, and placement of dowels. Unless otherwise specified, only the supports or mounting feet on the drive end of horizontal motors shall be doweled. Vertical motors shall have a rabbeted fit to the base and two dowels.

2.4.2.12 Alignment dowels or rabbeted fits shall be provided to facilitate disassembly and reassembly of end bells or plates, bearing-housing mounting plates, and bearing housings. When jackscrews are used as a means of parting contacting faces, one of the faces shall be counterbored or recessed to prevent a leaking joint or an improper fit caused by marring of the face.

2.4.2.13 When tapered dowel pins are provided by the vendor, the top end of the dowel shall have an undercut shank threaded to the nominal diameter nearest the dowel's outside diameter. The first two threads shall be machined off, and the shank shall be beveled to prevent damage when the pin is driven. A hex nut shall be provided with each pin.

● **2.4.2.14** Lifting lugs, through holes, or eyebolts shall be provided for lifting major components and the assembled motor. Any special mechanisms for lifting major components and the assembled motor shall be supplied in the quantities shown on the data sheets.

2.4.2.15 All fabricated welded structural steel shall be post-weld stress relieved. This does not apply to sheet metal components.

2.4.3 FRAME CONNECTIONS

2.4.3.1 Unless otherwise specified, inlet and outlet connections for field piping, including those for air ducting, lubrication, cooling medium, instrumentation, conduit, bus ducting, and drains, shall have the vendor's standard orientation and size, with the exception of sizes 1¼, 2½, 3½, 5, 7, and 9 inches, which shall not be used.

2.4.3.2 Tapped openings not connected to piping shall be plugged with solid round head steel plugs furnished in accordance with ANSI B16.11. Plugs that may later require removal shall be of corrosion-resistant material. Threads shall be lubricated. Tape shall not be applied to threads of plugs inserted into oil passages. Plastic plugs are not permitted.

2.4.3.3 Bolting shall be furnished as specified in 2.4.3.3.1 through 2.4.3.3.3.

2.4.3.3.1 The details of threading shall conform to ASME B1.1.

2.4.3.3.2 Cap screws shall be supplied on all frame connections except oil piping, unless studs are specifically approved by the purchaser.

2.4.3.3.3 Adequate clearance shall be provided at bolting locations to permit the use of socket or box wrenches.

■ **2.4.3.4** Openings for piping connections, except bearing oil inlet lines, shall be at least ¾ inch (20 millimeters) nominal pipe size. Oil inlet lines shall be not less than ½ inch (12 millimeters). All pipe connections shall be flanged or machined and studded. Where flanged or machined and studded openings are impractical, threaded openings in sizes through 1½ inches (40 millimeters) nominal pipe size, fitted in accordance with 2.4.3.4.1 through 2.4.3.4.3 unless otherwise specified, will be used.

2.4.3.4.1 A pipe nipple, preferably not more than 6 inches (152 millimeters) long, shall be screwed into the threaded opening.

2.4.3.4.2 Pipe nipples shall be a minimum of Schedule 80.

2.4.3.4.3 Tapped openings and bosses for pipe threads shall conform to ASME B16.5.

■ **2.4.3.5** Piping flanges shall conform to ASME B16.2 or B16.5 as applicable, except as specified in 2.4.3.5.1 and 2.4.3.5.2.

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2.4.3.5.1 Cast iron flanges shall be flat faced and shall have a minimum thickness of Class 250 for sizes 8 inches and smaller.

2.4.3.5.2 Flat-faced flanges with full raised-face thickness are acceptable on frames other than cast iron.

■ **2.4.3.6** Machined and studded connections shall conform to the facing and drilling requirements of ASME B16.1 or B16.5. Studs and nuts shall be furnished installed.

■ **2.4.3.7** Tapped openings and bosses for pipe threads shall conform to ASME B16.5. Pipe threads shall be taper threads conforming to ASME B1.20.1.

■ **2.4.3.8** Openings for duct connections shall be flanged and bolted. Connection facings shall be adequate to prevent leakage with proper gaskets and bolts. Unless otherwise specified, gaskets and bolts shall be provided by the vendor.

■ **2.4.3.9** Studded connections shall be furnished with studs installed. Blind stud holes in casings shall be drilled deep enough to allow a preferred tap depth, in inches, of 1½ times the major diameter of the stud. The first 1½ threads at both ends of each stud shall be removed.

● ■ 2.4.4 EXTERNAL FORCES AND MOMENTS

Motor frames and housings are generally designed to accept small external forces and moments from duct, conduit, and piping connections. If the auxiliary equipment (that is, ducting, coolers, silencers, and filters) is not supplied by the motor vendor, it is the purchaser's responsibility to specify on the data sheets the external loads expected to be imposed on the motor housings from this equipment. The vendor shall design the motor frame to accept the specified loads, including those from the bearing housings. The vendor shall provide the following information, as required, on the purchaser's data sheets:

a. Maximum allowable external forces and moments, if applicable.
b. If expansion joints are required for thermal expansion or isolation from vibration, information and recommendations about the joints.

2.4.5 ROTATING ELEMENT

2.4.5.1 General

● **2.4.5.1.1** The rotating element shall be designed and constructed to withstand the starting duties specified in 2.2.6 with a fatigue life of at least 5000 full-voltage starts or as specified by the purchaser. Shaft repair by plating or metal spray is not allowed unless approved by the purchaser.

2.4.5.1.2 For all two-pole motors, and all motors operating above the first lateral critical speed, the shaft shall be one-piece, heat-treated forged steel, suitably ground. Suitable fillets shall be provided at all changes in diameter and in keyways. Welded shaft and bar shaft/spider construction are not allowed.

For motors operating below the first lateral critical speed and 1800 RPM or less, a hot-rolled shaft and a welded shaft and bar construction may be used provided that the vendor can demonstrate that this construction will meet the requirements of 2.4.10.3.5 and the vendor has a minimum of 2 years of successful operating experience with such a design. This design shall meet the weld quality requirements of 4.2.2.3.2 and magnetic particle inspection requirements of 4.2.2.4.

2.4.5.1.3 When noncontacting radial vibration and/or axial-position probes are furnished, or when provisions for probes are required as described in 3.8, the rotor shaft sensing areas to be observed by radial-vibration probes shall be concentric with the bearing journals. All shaft-sensing areas (both radial vibration and axial position) shall be free from stencil and scribe marks or any other surface discontinuity, such as an oil hole or a keyway, for a minimum of one probe-tip diameter on each side of the probe. These areas shall not be metalized, sleeved, or plated. The final surface finish shall not exceed 32 microinches (0.8 micrometer) R_a, preferably obtained by honing or burnishing. These areas shall be properly demagnetized or otherwise treated so that the combined total electrical and mechanical runout does not exceed 30 percent of the maximum allowed peak-to-peak unfiltered vibration amplitude (refer to Figure 1) or the following value, whichever is greater:

a. For areas to be observed by radial-vibration probes, 0.25 mil (5 micrometers).
b. For areas to be observed by axial-position probes, 0.5 mil (15 micrometers).

■ **2.4.5.1.4** The shaft forging shall be ultrasonically inspected before the rotating element is assembled (see 4.2.2.3).

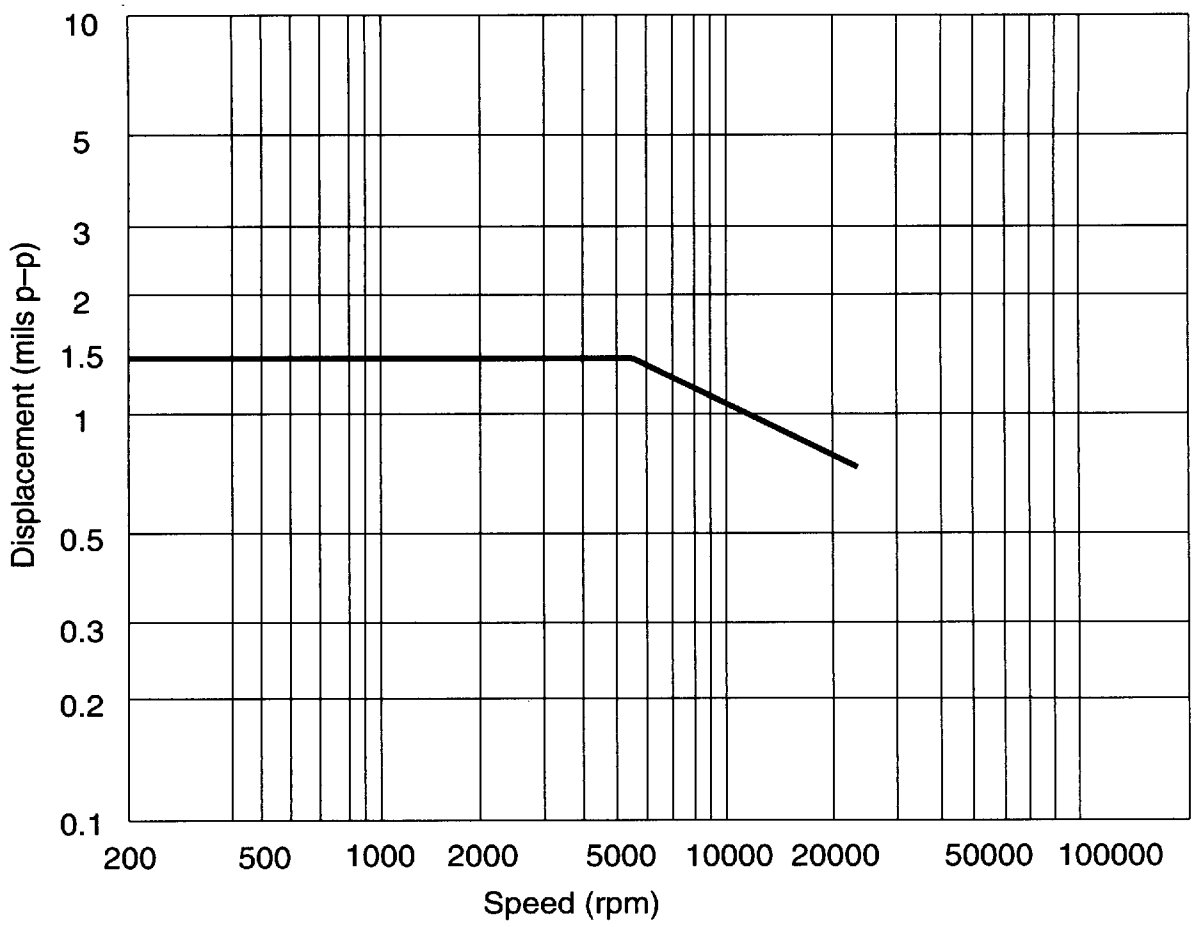
2.4.5.1.5 Shaft extensions at coupling fits shall conform to the requirements of API Standard 671 for tapered extensions and AGMA 9002 for cylindrical extensions.

2.4.5.2 Assembly

2.4.5.2.1 Rotor laminations shall have no burrs larger than 0.003 inch (0.076 millimeter). Laminations shall be distributed to minimize uneven buildup and evenly distribute magnetic properties in grain orientation. The method of assembly shall prevent scoring of the shaft surface, ensure positive positioning, and minimize bowing (all nontransient torque conditions shall be transmitted via rotor core/shaft interference fit). No shaft-straightening technique is permitted during or after fabrication of the rotor.

2.4.5.2.2 Fabricated-bar rotors shall be furnished with copper or copper alloy bars and end rings (see 6.1.9). If ap-

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Notes:
1. Vibration displacement at any frequency below running-speed frequency shall not exceed 0.1 mil or 20 percent of the measured unfiltered vibration displacement, whichever is greater.
2. Vibration displacement at any filtered frequency above running-speed frequency shall not exceed 0.5 mil peak-to-peak.
3. Vibration displacement filtered at running-speed frequency shall not exceed 80 percent of the unfiltered limit (runout compensated).

Figure 1—Shaft Vibration Limits (Relative to Bearing Housing Using Noncontact Vibration Probes): For All Hydrodynamic Sleeve Bearing Motors; With the Motor Securely Fastened to a Massive Foundation

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proved by the purchaser, cast or fabricated aluminum cage designs may be used if the vendor can demonstrate successful experience in meeting the starting duty requirements specified in 2.2.6, with a fatigue life of at least 5000 full voltage starts.

2.4.5.2.3 Shorting rings without circumferential joints are required for motors having synchronous speeds of 1500 revolutions per minute and higher and are preferred for slower speed motors. Jointed end rings are permitted for slower speed motors with the purchaser's approval.

2.4.5.2.4 To ensure good heat transfer to the rotor core and to limit vibration and fatigue of bars, all bars shall be maintained tightly in their slots. The rotor cage shall be maintained centered (e.g. swedged, center locked or pinned) to prevent axial movement.

2.4.5.2.5 The method by which the bars are attached to the current carrying end ring shall be selected to minimize localized heating and the nonuniform stresses that result. The bars shall be radially supported as necessary in the current carrying end ring (with the end ring slotted and/or grooved and bars inserted into that machined area) to prevent the braze or weld from being overstressed and maximize the joint contact area. The metal joining material shall not be subject to attack by hydrogen sulfide (that is, it shall be free from phosphorus). Inert gas welding, induction brazing, and multitorch full circle gas brazing are the acceptable methods. Outward bending of the ends of the rotor bars and articulation of the shorting ring shall be limited by design, material selection, or shrunk-on or fitted nonmagnetic steel retaining rings (see 6.1.9).

2.4.5.2.6 The material and processes used to fabricate copper and copper alloy bars and end rings shall be selected to minimize hydrogen embrittlement.

● **2.4.5.2.7** Rotors shall be designed to withstand overspeeds without permanent mechanical deformation (see 2.1.4). Overspeed requirements more stringent than those of NEMA MG 1-20.44 will be specified by the purchaser.

2.4.5.2.8 Fans may be integrally cast with the end rings or separately mounted. Separable fans shall be permanently indexed angularly and axially and mounted by one of the following methods:

a. Split hub on shaft.
b. Shrink fit on shaft or hub.
c. Spider or endplate mounted.

Slip-fitted fans secured to the shaft by means of set screws only are not acceptable. Removal and reassembly of the fans on the rotor shall not change the rotor balance enough to exceed the allowable residual unbalance limits.

2.4.5.2.9 Fans shall be capable of being balanced in accordance with 2.4.6.2.3. Welding is not an acceptable means of balancing a fan.

2.4.5.2.10 For speeds greater than or equal to 1800 revolutions per minute, the design of the stressed parts of fans shall include proper evaluation of stress concentration factors (SCF) for the geometry. The design of stressed rotating parts shall include fillets that will limit the SCF. Areas of concern include the fan, blade-to-disk intersections, keyways, and shaft section changes. All accessible areas of welds on fans shall be subjected to magnetic particle or liquid penetrant inspection.

2.4.6 DYNAMICS

2.4.6.1 Resonances

2.4.6.1.1 Lateral natural frequencies of the motor, which can lead to resonance amplification of vibration amplitudes, shall be removed from the operating speed frequency and other significant exciting frequencies (see 1.5.9) by at least 15 percent. Motors intended for continuous operation on adjustable speed drives shall meet this requirement over the specified speed range.

● **2.4.6.1.2** If the motor is to be supported in the field by a structure other than a massive foundation, the purchaser will specify this on the data sheets, and the motor vendor shall supply the following data, as a minimum, to the purchaser so that a system dynamic analysis can be made and an adequate foundation designed:

a. A detailed shaft section model with masses, mass elastic data including mass and rotational inertia (Wk^2), shaft section lengths, and inner and outer diameters.
b. For the minimum and maximum design bearing clearances and maximum oil operating temperature, an eight coefficient bearing model with damping and spring constants.
c. Horizontal and vertical bearing-housing stiffness.

Note: The rigidity of a foundation is a relative quantity. It must be compared with the rigidity of the machine bearing system. The ratio of bearing-housing vibration to foundation vibration is a characteristic quantity for the evaluation of foundation flexibility influences. One indication that a foundation is massive is if the vibration amplitudes of the foundation (in any direction) near the machine feet or base frame are less than 30 percent of the amplitudes that could be measured at the adjacent bearing housing in any direction.

When specified, a rotor dynamic analysis, with motor half-coupling weight, taking into account the effect of the foundation, shall be performed by the motor vendor in accordance with 2.4.6.1.3. The vendor shall identify the foundation data required from the purchaser to perform this analysis.

Note: A massive foundation is recommended. The natural frequencies of the foundation after the machine is erected must differ by at least ±20 percent from one and two times the running-speed frequency and by at least 20 percent from one and two times the electric-line frequency.

● ■ **2.4.6.1.3** When specified, the vendor shall provide a damped unbalanced response analysis for each machine to ensure acceptable amplitudes of vibration at any speed from zero to trip.

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2.4.6.1.3.1 The damped unbalanced response analysis shall include, but shall not be limited to, the following considerations:

a. Support (base, frame, and bearing housing) stiffness, mass, and damping characteristics, including effects of rotational speed variation. The vendor shall state the assumed support system values and the basis for these values (for example, tests of identical rotor support systems, assumed values).
b. Bearing lubricant film stiffness and damping changes due to speed, load, preload, oil temperatures, accumulated assembly tolerances, and maximum to minimum clearances.
c. Rotational speed, including the various starting speed detents, operating speed and load ranges (including agreed-upon test conditions if different from those specified), trip speed, and coast-down conditions.
d. Rotor masses, including the mass moment of coupling halves, stiffness, and damping effects (for example, accumulated fit tolerances, and frame effects).
e. Asymmetrical loading (for example, eccentric clearances).
f. For machines equipped with antifriction bearings, the vendor shall state the bearing stiffness and damping values used for the analysis and either the basis for these values or the assumptions made in calculating the values.

2.4.6.1.3.2 As a minimum, the damped, unbalanced response analysis shall include the following:

a. A plot and identification of the mode shape at each resonant speed from zero to trip, as well as the next mode occurring above the trip speed.
b. Frequency, phase, and response amplitude data at the vibration probe locations through the range of each critical speed, using the following arrangement of unbalance for the particular mode. This unbalance shall be sufficient to raise the displacement of the rotor at the probe locations to the vibration limit defined by the following equation:

$$L_v = \sqrt{\frac{12{,}000}{N}}$$

In *SI* units

$$L_v = 25.4\sqrt{\frac{12{,}000}{N}}$$

Where:

L_v = vibration limit (amplitude of unfiltered vibration), in micrometers (mils) peak to peak.
N = operating speed nearest the critical of concern, in revolutions per minute.

This unbalance shall be no less than two times the unbalance defined by the following equation:

$$U_b = \frac{4W}{N}$$

Where:

U_b = input unbalance from the rotor dynamic response analysis, in gram-millimeters (ounce-inches).
W = journal static weight load, in pounds, or for bending modes where the maximum deflection occurs at the shaft ends, the overhung weight load (that is, the weight outboard of the bearing), in kilograms (pounds).
N = operating speed nearest the critical of concern, in revolutions per minute.

The unbalance weight or weights shall be placed at the locations that have been analytically determined to affect the particular mode most adversely. For translatory modes, the unbalance shall be based on both journal static weights and shall be applied at the locations of maximum displacement. For conical modes, each unbalance shall be based on the journal weight and shall be applied at the location of maximum displacement of the mode nearest the journal used for the unbalance calculation, 180 degrees out of phase.
c. Modal diagrams for each response in item b above, indicating the phase and major axis amplitude at each coupling engagement plane, the centerlines of the bearings, the locations of the vibration probes, and each seal area throughout the machine. When an unbalance response test is specified (4.3.5.4), additional unbalance weights of $4U_B$ shall be placed at the balance planes in phase and also 180 degrees out of phase for correlation with the test data.
d. A stiffness map of the undamped rotor response from which the damped unbalanced response analysis specified in item c above was derived. This plot shall show frequency versus support system stiffness, with the calculated support system stiffness curves superimposed.
e. For machines whose bearing support system stiffness values are less than or equal to 3.5 times the bearing stiffness values, the calculated frequency-dependent support stiffness and damping values (impedances) or the values derived from modal testing. The results of the damped unbalanced response analysis shall include Bode plots that compare absolute shaft motion with shaft motion relative to the bearing housing.

● ■ **2.4.6.1.4** When specified, the driven-equipment vendor, the motor vendor, or both shall perform a steady-state and transient torsional and stress analysis of the motor and driven equipment, including gears, motor-driven pumps, and fan- or turbine-assisted units. The driven-equipment vendor shall be responsible for the complete and satisfactory performance of the units. The motor vendor shall be responsible for providing the physical data required for the torsional analysis to the purchaser or driven-equipment vendor, as specified, and in a timely manner to allow for any system modification that might be necessary (see 1.4). The torsional analysis shall include, but shall not be limited to, the following:

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a. A complete description of the method used to complete the analysis.
b. A graphic display of the mass-elastic system.
c. A tabulation identifying the mass moment and torsional stiffness for each component identified in the mass-elastic system.
d. A graphic display or expression of exciting pulsating torque or other torsional excitation versus speed or time.
e. A graphic display of torsional critical speeds and deflections (a mode shape diagram).

The torsional analysis shall confirm that the frequency of the torsional modes of the complete motor-driven train is at least 15 percent removed from any important excitation frequencies and both one and two times the line frequency. For trains with adjustable speed drivers, the operating speed includes operation up to the trip speed. If this requirement cannot be achieved, calculations shall be made to determine the maximum stresses, the frequency at which they occur, and the fatigue life of each element in the train. These calculations shall be given to the purchaser and to the driven-equipment vendor, the motor vendor, or both for approval. For adjustable speed drives, the torsional analysis shall also verify that the calculated shaft torque at the resonance points up to the maximum operating speed does not exceed the allowed maximum stress.

2.4.6.2 Balancing

● **2.4.6.2.1** All rotors shall be dynamically balanced in two or more planes. When specified, rotors operating at speeds in excess of the first actual lateral critical speed shall be balanced in at least three planes, including a rotor center plane. When a keyway is provided for a coupling hub or fan, the rotor shall be balanced with the keyway fitted with a crowned half-key, or its dynamic equivalent, to fill the usable length of the shaft keyway.

2.4.6.2.2 Balance weights added to the final assembly shall be readily removable and replaceable and made of AISI 300 series stainless steel or a purchaser-approved, corrosion-resistant material. If parent metal is to be removed to achieve dynamic or static balance, it shall be drilled out in a manner that maintains the structural integrity of the rotor and does not cause harmful or distortive hot spots during operation. Chiseling, sawing, or torch burning is not permitted. The use of solder or similar deposits for balancing purposes is not acceptable.

2.4.6.2.3 Rotor balancing shall not be accomplished by adding balance weights on the fan blades. After final balancing of two-pole and four-pole rotors, the fans, and other removable rotor components, shall be installed and component (step) balanced.

● **2.4.6.2.4** When specified, a balanced half-coupling furnished by the purchaser shall be mounted on the rotor, and the rotor balance shall be rechecked. Any increase in unbalance shall be reported to the purchaser. Any corrections to the rotor balance shall be mutually agreed upon by the purchaser and the motor vendor. Any corrections to the coupling balance shall be mutually agreed upon by the purchaser, coupling vendor, and motor vendor.

Note: Excessive radial shaft runout can cause high vibration after a balanced coupling has been mounted on the rotor. Shaft-extension radial runout should be checked against the vendor's drawings prior to making any corrections.

2.4.6.2.5 For the final balancing of the rotor in the balancing device, the maximum allowable residual unbalance in the correction plane shall be calculated from the following equation:

$$U_B = \frac{4W_r}{N_{mc}} \qquad (1)$$

In SI units,

$$U_B = \frac{6350\ W_r}{N_{mc}}$$

Where:

U_B = residual unbalance, in ounce-inches (gram-millimeters).
W_r = journal static loading, determined from the mass distribution in the rotor, in pounds (kilograms). (Typically one-half rotor weight.)
N_{mc} = maximum continuous speed, in revolutions per minute.

A balancing device is either a conventional balancing machine or the actual motor frame assembly with the rotor installed. When the motor frame is used as a balance device, the residual unbalance of the rotor shall be determined in accordance with 4.2.3.6 and Appendix C.

Note: Appendix C provides a method of determining the residual unbalance remaining in the completely assembled rotor and balancing machine sensitivity check.

2.4.6.2.6 Where a rotor is unsymmetrical, or the correction planes are unsymmetrically located, the allocation of residual unbalance between the correction planes by reference to journal static loading may not be appropriate. In this case, the proportionate allocation of residual unbalance to the correction planes should be determined by reference to ISO 1940/1. However, the total residual unbalance should be less than $4W/N_{mc}$, where W is the rotor mass and not the ISO 1940/1 balance grade.

2.4.6.3 Vibration

Motors shall be designed so that they meet the acceptance criteria stated in 4.3.3.

2.4.7 BEARINGS AND BEARING HOUSINGS

● **2.4.7.1** Unless otherwise specified, hydrodynamic radial bearings (sleeve or tilting pad, for example) shall be provided on all horizontal motors.

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Note: To limit bearing babbitt wear, bearings and lubrication should be evaluated for applications of hydraulic jacking means when applying hydrodynamic bearings in motors that require multiple starts per day.

● **2.4.7.2** Antifriction bearings shall be used for vertical motors and, when specified, for horizontal motors, provided that the following conditions are met:

a. The dN factor is less than 300,000. [The dN factor is the product of bearing size (bore) in millimeters and the rated speed in revolutions per minute.]
b. Standard antifriction bearings meet an L_{10} rating life of either 100,000 hours with continuous operation at rated conditions or 50,000 hours at maximum axial and radial loads and rated speed. (The L_{10} rating life is the number of hours at rated bearing load and speed that 90 percent of a group of identical bearings will complete or exceed before the first evidence of failure. See AFBMA Standard 9 or Standard 11, as applicable.)
c. Thrust bearings for vertical motors shall be rated for AFBMA L_{10} life of 5000 hours (minimum continuous service at 200 percent of the maximum up and down thrust the load may develop during starting or stopping or while operating at any capacity on the rated performance curve).

2.4.7.3 Hydrodynamic radial bearings shall be split for ease of assembly, precision bored, and of the sleeve or pad type, with steel- or bronze-backed babbitted replaceable liners, pads, or shells. These bearings shall be equipped with antirotation devices and shall be positively secured in the axial direction. The bearing design shall suppress hydrodynamic instabilities and provide sufficient damping to limit rotor vibration to the maximum specified amplitudes while the motor is operating loaded or unloaded at specified operating speeds, including operation at any critical frequency if that frequency is a normal operating speed. The liners, pads, or shells shall be in horizontally split housings and shall be replaceable. The bearings on each end of horizontal motors shall be identical. The design of the bearing housing shall not require removal of the lower half of end bells or plates, ductwork, or the coupling hub to permit replacement of the bearing liners, pads, or shells. Bearing temperatures measured with bearing metal temperature detectors shall not exceed 200°F (93°C) at rated operating conditions.

2.4.7.4 Ball-type thrust bearings shall be of the duplex-matched, single row, 40 degrees, angular-contact type (Series 7000) installed back to back. A spherical roller thrust bearing may be used for high thrust loads.

2.4.7.5 Thrust bearings for vertical motors shall be on top and preloaded. Multiple bearings to accommodate thrust in the same direction shall not be permitted.

Note: Spherical roller bearings often have springs designed to compress with the downthrust, and if the thrust is less than design, the rotor rides higher than normal and there will be increased vibration during no load testing. The vendor shall notify the purchaser on the proposal data sheet if this condition will exist.

2.4.7.6 Antifriction bearings shall be retained on the shaft and fitted into housings in accordance with the requirements of AFBMA Standard 7. However, the device used to lock the ball thrust bearings to the shaft shall be restricted by a nut with a tongue-type lockwasher; for example, Series W per AFBMA Standard 8.2.

2.4.7.7 Except for the angular-contact bearings and lower guide bearings in vertical motors, antifriction bearings shall have an internal clearance fit equivalent to AFBMA Symbol 3, as defined in AFBMA Standard 20. Single or double row bearings shall be of the Conrad type. Filling-slot (maximum load) antifriction bearings shall not be used.

2.4.7.8 Bearings shall be electrically insulated. For double-end drivers, the bearings and couplings on one end shall be electrically insulated. A shorting device shall be provided in the bearing housing on the drive end.

■ **2.4.7.9** Hydrodynamic thrust bearings for vertical motors shall be of the babbitted multiple segment type. Tilting-pad bearings shall incorporate a self-leveling feature that ensures each segment carries an equal share of the thrust load. With minor variation in pad thickness, each pad shall be designed and manufactured with dimensional precision (thickness variation) that will allow interchange of individual pads. The thrust collar shall be replaceable. Fretting and axial movement shall be prevented either by positively locking the collar to the shaft or by another method. The thrust faces of the collar shall have a surface finish of not more than 16 microinches (0.5 micrometer) R_a, and the total indicated axial runout of either thrust face shall not exceed 0.0005 inch (12 micrometers). Split thrust collars are not acceptable.

● **2.4.7.10** Hydrodynamic thrust bearings for vertical motors shall be sized for continuous operation at no more than 50 percent of the bearing manufacturer's continuous catalog rating. In addition to thrust from the rotor weight, the maximum axial force from the driven equipment transmitted through the coupling shall be considered a part of the duty of any thrust bearing. The momentary downthrust capability shall remain within the bearings catalog rating. Conservative thrust bearing ratings are required due to possible inaccurate pump thrust data.

Note: Operating thrust bearings at 50 percent of the rating will cause an oversize bearing to be used. This will result in an increase in the bearing losses.

2.4.7.10.1 For nonaxially locating gear- or spline-type couplings, where sliding may take place at the tooth mesh, the transmitted external axial force shall be calculated from the following formula:

$$F = 126{,}000\mu\frac{P_r}{N_r d}$$

In SI units,

$$F = 19{,}095\mu\frac{P_r}{N_r d}$$

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Where:

F = external force, in pounds (kilonewtons).
P_r = rated power, in horsepower (kilowatts).
N_r = rated speed, in revolutions per minute.
d = gear tooth Pitch Circle Diameter (PCD) in inches (millimeters). (Use d = 2 times the shaft diameter if coupling details are unknown.)
μ = coefficient of friction at the gear teeth. (Use μ = 0.25 unless a definite value is available.)

2.4.7.10.2 Thrust loads for diaphragm- and disk-type couplings shall be calculated on the basis of the maximum allowable deflection permitted by the coupling manufacturer.

2.4.7.11 Bearing housings for pressure-lubricated hydrodynamic bearings shall be arranged to minimize foaming. The drain system shall be adequate to maintain the oil and foam level below shaft end seals and to allow a sufficient oil level for oil flinger disk or oil ring operation. For all hydrodynamic bearings, when the inlet oil temperature is 120°F (49°C), the rise in oil temperature (inlet to drain) shall not exceed 50°F (28°C) under the most adverse specified operating conditions. When the inlet oil temperature exceeds 120°F (49°C), special consideration shall be given to bearing design, oil flow, and allowable temperature rise.

● **2.4.7.12** On horizontal motors, bearing housings for oil-lubricated, nonpressure-fed bearings shall have oil reservoirs of sufficient depth to serve as settling chambers. The oil sump temperature shall not exceed 180°F (82°C), based on the specified operating conditions and an ambient temperature of 104°F (40°C). The housings shall be provided with tapped and plugged fill openings at least ½ inch National Pipe Thread (NPT) in size. When specified, the housings shall be equipped with constant level sight-feed oilers at least 8 ounces (0.25 liter) in size, with a positive level positioner (not a set screw), transparent containers (not subject to sunlight- or heat-induced opacity or deterioration), protective wire cages, and supplemental support in addition to the piping. A permanent indication of the proper oil level shall be accurately located and clearly marked on the outside of the bearing housing with permanent metal tags, marks inscribed in the castings, or another durable means. If the oil level indicator breaks accidentally, the resulting drop in oil level shall not result in loss of bearing lubrication, that is, reduction of the oil level below the level required for oil ring operating.

2.4.7.13 Housings for ring-oil-lubricated bearings shall be provided with plugged ports positioned to allow visual inspection of the oil rings while the equipment is running. This is not required on TEFC motors.

● **2.4.7.14** The requirements of 2.4.7.14.1 through 2.4.7.14.4 apply when oil mist lubrication or purging is specified.

2.4.7.14.1 An oil mist inlet connection, ¼ inch nominal pipe size, shall be provided in the top half of the bearing housing. The pure or purge oil mist fitting connections shall be located so that oil mist will flow through the antifriction bearings.

2.4.7.14.2 A vent connection, ½ inch nominal pipe size, shall be provided on the housing or end cover for each of the spaces between antifriction bearings and the housing shaft closures.

2.4.7.14.3 When pure or purge oil mist lubrication is specified, shielded or sealed bearings shall not be used.

2.4.7.14.4 When pure oil mist lubrication is specified, oil rings or flingers (if any) and constant-level oilers shall not be provided, and a mark indicating the oil level is not required. However, when purge or condensing oil mist lubrication is specified, these items shall be provided and the oiler shall be piped so that it is maintained at the internal pressure of the bearing housing.

2.4.7.15 Bearing-housing mounting surfaces shall be flat and in the same plane, machined perpendicular to or parallel with the bearing bore, as required.

2.4.7.16 Bearing housings shall be positively located by cylindrical precision dowels or rabbeted fits. Bearing housings and support structures shall be designed so that, upon assembly, none of the air-gap measurements taken in at least three positions (spaced 90 degrees apart) at each end of the motor deviates from the limits given below for the air gap, as defined by the following equation:

$$D = [(H-L)/A]100$$

Where:

D = percentage deviation.
H = highest of the readings at one end of the motor.
L = lowest of the readings at the same end of the motor.
A = average of the readings at the same end of the motor.

The air gap between the exterior of the rotor and the interior of the stator must be measured at both ends of the motor. Measurements should be taken at the same positions on both ends. The percentage deviation shall not exceed 10 percent of the air gap. This data shall be recorded and made part of the final report. To allow for accurate measurement, stator surfaces at the measuring positions shall be free from resin buildup.

2.4.7.17 Bearing housings shall be machined for mounting vibration detectors as described in 3.8.

■ **2.4.7.18** Sufficient cooling, including an allowance for fouling, shall be provided to maintain the oil temperatures in accordance with 2.4.7.3, 2.4.7.11, and 2.4.7.12. Where water cooling is required, water jackets shall have only external

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connections between the upper and lower housing jackets and shall have neither gasketed nor threaded connection joints, which may allow water to leak into the oil reservoir. If cooling coils and external fittings are used, they shall be of AISI 300 series stainless steel or 90–10 copper nickel with welded or brazed joints without internal fittings. Coils shall have a nominal thickness of at least 0.042 inch (1.0 millimeters) and a diameter of at least ½ inch (12.7 millimeters).

● ■ **2.4.7.19** When specified, bearing-oil temperature indicators shall be provided on the bearing housing of nonpressure-fed bearings. If the sensor cannot be removed without loss of oil, a thermowell shall be provided.

2.4.7.20 At ambient temperature, the fit between the outside of the bearing shell and the bearing housing shall be line to line (zero clearance) and shall preferably be a light interference fit.

● **2.4.7.21 Shaft Seals**

Shaft seals shall conform to the following:

a. Frame shaft seals shall be of nonsparking materials, of the split type to allow replacement without removal of the rotor or coupling, and centerable about the shaft. Where aluminum is used, it shall have a copper content of less than 0.2 percent. Where end-shield-supported bearings are used, the inner seal shall be maintained at atmospheric pressure. Pressure balancing from the cooling fan shall be by use of copper or steel tubing, unless other materials are approved by the purchaser. Seals shall be designed to minimize the entry of fumes, dirt, and other foreign material into the stator housing. When specified, seals shall be constructed so that a purge gas can be introduced. If possible, self-aligning seals shall be used. When specified, the shaft seals shall be fabri- ● ■ cated from electrically nonconducting materials.

b. Bearing housings for horizontal motors shall be equipped with split labyrinth-type end seals and deflectors where the shaft passes through the housing. Lip-type seals shall not be used. The deflectors shall be made of nonsparking materials.

The design of the deflectors shall effectively retain oil in the housing and prevent entry of foreign material into the housing. No oil shall leak past the seals, during both stationary and operating conditions, while circulating lube oil.

2.4.8 LUBRICATION

2.4.8.1 Unless otherwise specified, bearings and the bearing housing shall use hydrocarbon oil and shall be arranged for a flinger- or ring-type lubrication system in accordance with the bearing manufacturer's recommendations for using hydrocarbon oil.

● **2.4.8.2** Oil flinger disks or oil rings shall have a minimum submergence of ¼ inch (6 millimeters) above the lower edge of a flinger or above the lower edge of the bore of an oil ring. Oil slingers shall have mounting hubs to maintain concentricity and shall be positively secured to the shaft. When specified, an oil ring/flinger shall be provided and sized to prevent bearing damage during coastdown without forced lubrication. If oil rings are not practical, as with tilting pad bearings, the vendor shall advise and obtain approval from the purchaser.

● **2.4.8.3** When specified for oil flinger disk lubricated bearings, thermostatically controlled heating devices shall be provided in the bearing housings. The heating devices shall have sufficient capacity to heat the oil in the bearing housing from the specified minimum site ambient temperature to the vendor's minimum required temperature in 4 hours. The thermostatic enclosure shall be compatible with the area classification requirements. (Refer to 2.4.8.8.8 for heat density limitations).

● **2.4.8.4** Where a pressurized or circulating lubrication system is required by a gear, the driven equipment, or both, the motor bearing oil may be supplied from that system, when specified. The purchaser will specify the supplier of the complete lubrication system.

● **2.4.8.5** Where oil is supplied from a common system to two or more machines (such as a compressor, a gear, and a motor), the oil's characteristics will be specified on the data sheets by the purchaser on the basis of mutual agreement with all vendors supplying equipment served by the common oil system.

Note: The usual lubricant employed in a common oil system is a hydrocarbon oil that corresponds to ISO Grade 32, as specified in ISO 3448.

2.4.8.6 The vendor shall state in the operating manual the amount of lubricating oil required and the specifications for the oil.

● ■ **2.4.8.7** Unless otherwise specified, pressurized oil systems shall conform to the requirements of API Standard 614.

■ **2.4.8.8** Pressure lubrication systems for the motor only that are other than those described in API Standard 614 shall consist of an oil pump with a suction strainer, a supply-and-return system, an oil cooler (when required), a full-flow filter, a low lube oil pressure shutdown switch, and other necessary instruments. The requirements of 2.4.8.8.1 through 2.4.8.8.9 shall apply.

2.4.8.8.1 Oil-containing pressure components shall be steel.

● **2.4.8.8.2** Oil return lines shall be sized to run no more than half full and shall be arranged to ensure good drainage (recognizing the possibility of foaming conditions). Horizontal runs shall slope continuously, at least ¼ inch per foot (42 millimeters per meter), toward the reservoir. If possible, laterals (not more than one in any transverse plane) should enter drain headers at 45 degree angles in the direction of the flow. Nonconsumable backup rings and sleeve-type joints shall not be used. Oil filters shall be capable of withstanding the specified oil supply header pressure. Pressure piping

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downstream of oil filters shall be free from internal obstructions that could accumulate dirt. Unless otherwise specified, oil supply piping and tubing, including fittings (excluding slip-on flanges), shall be AISI 300 series stainless steel.

● ■ **2.4.8.8.3** When specified, a main oil pump driven by the motor shaft and oil rings or flinger discs shall be provided for shutdown and emergency operation. See 2.4.8.8.5 for start-up considerations.

■ **2.4.8.8.4** Unless the oil pump is enclosed in a reservoir, it shall have a steel case.

● ■ **2.4.8.8.5** A separately driven, automatically controlled standby pump shall be provided by the purchaser for equipment that will operate at idling speeds or will require rapid starting.

● ■ **2.4.8.8.6** An oil cooler shall be provided to maintain the lube oil supply temperature at or below 120°F (49°C). The cooler shall be of a water-cooled, shell and tube type or of a suitable air-cooled type, as specified. Shell and tube coolers shall have water on the tube side. A removable bundle design is required for coolers with more than 5 square feet (0.46 square meter) of surface, unless otherwise specified. Removable bundle coolers shall be in accordance with Tubular Exchanger Manufacturers Association (TEMA) Class C and shall be constructed with a removable channel cover. Tubes shall not have an outside diameter of less than ⅝ inch (16 millimeters), and the tube wall shall not have a thickness of less than 0.049 inch (1.25 millimeters). U-bend tubes are not permitted. The vendor shall state the lube oil pressure at the cooler outlet so that the purchaser can provide water at lower pressure, if desired, to prevent contamination of the lube oil in the event of cooler failure. The cooler shall be equipped with vent and drain connections on the oil and water sides. The vendor shall include in the proposal complete details of any proposed air-cooled cooler. Internal oil coolers are not acceptable.

● **2.4.8.8.7** Full-flow filters with replaceable elements and filtration of 25 microns nominal or finer shall be supplied for babbitted bearings. Full-flow filters with replaceable elements and filtration of 10 microns nominal or finer shall be supplied for aluminum or microbabbitted bearings. The filters shall be located downstream of the cooler. For motor-driven centrifugal oil pumps, filter cases and heads shall be suitable for operation at the maximum discharge pressure at the driver's trip speed. For positive displacement pumps, filter cases and heads shall be suitable for operation at a pressure not less than the relief valve setting. Filters that have covers weighing more than 35 pounds (16 kilograms) shall have cover lifters. Filters shall not be equipped with a relief valve or an automatic bypass. Filter cartridge materials shall be corrosion resistant. Metal-mesh or sintered-metal filter elements are not acceptable. Flow shall be from the outside toward the center of the filter cartridge. When the filter design requires cartridges stacked two or more high, a center post and a cap for the top cartridge shall be used to secure the cartridges to the bottom of the filter housing. If the cartridge-to-cartridge joint is not self-aligning, a collar shall be used between the stacked cartridges to ensure alignment. The pressure drop for clean filter elements shall not exceed 15 percent of the total allowable dirty pressure drop, or 5 pounds per square inch (0.34 bar) at an operating temperature of 100°F (38°C) and normal flow. Cartridges shall have a minimum collapsing differential pressure of 70 pounds per square inch (4.8 bar). The filter shall be equipped with a valved vent and clean and dirty side valved drain connections. The dirty side connections shall be located lower in the housing than the filter element or cartridge support base. Where a specific filter element is desired, the purchaser will specify the manufacturer and model number of the element. When specified, duplex filters with a constant-flow switch valve shall be provided.

Note: Micron particle size implies the shape of a spherical bead; thus, a 10-micron particle is a sphere with a diameter of 10 microns. Within the element's recommended maximum pressure drop, 10 microns nominal implies that the efficiency of the filter on particles that are 10 microns or larger in diameter will be no less than 90 percent for the life of the element. Absolute micron particle ratings are different. A micron-absolute filter rating implies that no particles of the rating size or larger will pass; for example, a filter rating may be 10 microns nominal and 15 microns absolute.

● ■ **2.4.8.8.8** When specified, a removable steam-heating element external to the oil reservoir or a thermostatically controlled electric immersion heater shall be provided for heating the charge capacity of oil before start-up in cold weather. The heating device shall have sufficient capacity to heat the oil in the reservoir from the specified minimum site ambient temperature to the manufacturer's required start-up temperature within 4 hours. If an electric immersion heater is used, it shall have a maximum watt density of 15 watts per square inch (2.3 watts per square centimeter).

■ **2.4.8.8.9** An AISI 300 series stainless steel oil reservoir, separate from the base, shall be supplied with the following characteristics and appendages unless otherwise specified:

a. The capacity to settle moisture and foreign matter adequately, to avoid frequent refilling, and to provide adequate allowance for system rundown.
b. A retention time of 8 minutes, based on normal flow and total volume below the minimum operating level.
c. Rundown capacity that includes the oil contained in all components, bearing and seal housings, control elements, and vendor-furnished piping that drains back to the reservoir. The rundown capacity shall also include an allowance of at least 10 percent for the purchaser's interconnecting piping.
d. Provisions to eliminate air and to minimize flotation of foreign matter to the pump suction.
e. Fill connections, level indicators, and breathers suitable for outdoor use.
f. Sloped bottoms and connections for complete drainage.

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g. Cleanout openings as large as is practicable.
h. An interior that has been descaled and protected from rust by the manufacturer's approved process. (A permanent surface coating shall not be applied without the purchaser's specific approval.)

2.4.9 END PLAY AND COUPLINGS

2.4.9.1 Horizontal sleeve-bearing motors shall have a total end play of at least ½ inch (12.7 millimeters). The running center of the rotor shall not shift from either side of the stator's geometric center by more than 3/32 inch (2.4 millimeters).

2.4.9.2 Flexible couplings used with horizontal sleeve-bearing motors will be of the limited-end-float type. Unless otherwise specified, the end float shall be limited to 3/16 inch (4.8 millimeters).

2.4.9.3 When horizontal hydrodynamic bearings are provided, the motor shall have a permanent indicator to show the allowable limits of rotor movement after the coupling is installed and aligned. The indicator shall be durable, shall be adjacent to the drive end shaft shoulder, and shall show the magnetic center and the total end float of the rotor.

● **2.4.9.4** When specified, the motor half-coupling shall be mounted on the motor shaft with either a taper or a cylindrical fit. Taper fits shall be in accordance with API Standard 671 and shall be verified with a ring gauge supplied by the driven equipment manufacturer. Cylindrical fits shall be in accordance with AGMA 9002.

● **2.4.9.5** Testing with the contract coupling is preferred. If this is not practical, the mechanical running test shall be performed with a coupling and coupling hub idling adapter in place, which results in moments equal to (±10 percent) the moment of the contract coupling hub plus one-half that of the coupling spacer. The test coupling, including any necessary idling adapter or mass moment simulator, shall be furnished by the supplier specified on the data sheet. When all testing is completed, the idling adapters shall be furnished to the purchaser as part of the special tools.

2.4.10 MATERIALS

2.4.10.1 General

2.4.10.1.1 Minor parts that are not identified (such as nuts, springs, washers, gaskets, and keys) shall have corrosion resistance equal to that of specified parts that will operate in the same environment. All hardware used for the purchaser interface shall be in accordance with applicable American standards (for example, ANSI standard threads).

● **2.4.10.1.2** The purchaser will specify any corrosive agents present in the environment, including constituents that may cause stress corrosion cracking.

2.4.10.1.3 Where mating parts such as studs and nuts of 18-8 stainless steel or materials having similar galling tendencies are used, they shall be lubricated with a suitable antiseizure compound.

● ■ **2.4.10.1.4** When specified, the vendor shall propose ASTM optional tests and inspection procedures necessary to ensure that materials are satisfactory for the service. Such tests and inspections shall be listed in the proposal. The purchaser should consider specifying additional tests and inspections, especially for materials in critical service.

■ **2.4.10.1.5** If parts exposed to conditions that promote intergranular corrosion are to be fabricated, hard faced, overlaid, or repaired by welding, they shall be made of low carbon or stabilized grades of austenitic stainless steel.

2.4.10.1.6 Unless specifically approved by the purchaser, no component shall be repaired by plating, plasma spray, metal spray, impregnation, or similar methods.

2.4.10.2 Castings

2.4.10.2.1 The vendor shall specify the material grade of castings on the data sheets.

2.4.10.2.2 Castings shall be sound and free from porosity, hot tears, shrink holes, blow holes, cracks, scale, blisters, and similar injurious defects. Surfaces of castings shall be cleaned by sandblasting, chemical cleaning, or any other standard method. Mold-parting fins and remains of gates and risers shall be chipped, filed, or ground flush.

2.4.10.2.3 Ferrous castings shall not be repaired by welding, peening, plugging, burning in, or impregnating, except as specified in 2.4.10.2.3.1 and 2.4.10.2.3.2.

2.4.10.2.3.1 Weldable grades of steel castings may be repaired by welding, using a qualified welding procedure based on the requirements of Section IX of the ASME Code.

2.4.10.2.3.2 Cast gray iron or nodular iron may be repaired by plugging within the limits specified in ASTM A 278, A 395, or A 536. The holes drilled for plugs shall be carefully examined, using liquid penetrant, to ensure that all defective material has been removed. All necessary repairs not covered by ASTM specifications shall be subject to the purchaser's approval.

2.4.10.2.4 Fully enclosed cored voids, including voids closed by plugging, are prohibited.

■ **2.4.10.2.5** Nodular iron castings shall be produced in accordance with ASTM A 395. The grade shall be specified by the vendor.

2.4.10.3 Welding

2.4.10.3.1 Structural welding, including weld repairs, shall be performed by operators using procedures qualified in accordance with AWS D1.1. Other welding codes may be used if specifically approved by the purchaser.

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2.4.10.3.2 The vendor shall be responsible for the review of all repairs and repair welds to ensure that they are properly heat treated and nondestructively examined for soundness and compliance with applicable qualified procedures.

2.4.10.3.3 All butt welds shall be continuous full-penetration welds.

2.4.10.3.4 Intermittent welds, stitch welds, and tack welds are not permitted on any structural part of the motor, including the outer enclosure. If specifically approved by the purchaser, intermittent welds may be used where significant problem-free operating experience exists and well-established design procedures are available.

2.4.10.3.5 Welding of or to shafts is not acceptable on two-pole motors, for balancing purposes, or on finished shafts. Any shafts or spiders subjected to welding shall be post-weld stress relieved prior to finish machining.

● ■ 2.4.10.4 Low Temperature

For specified operating temperatures below -20°F (-29°C), steels shall have, at the lowest specified temperature, an impact strength sufficient to qualify under the minimum Charpy V-notch impact energy requirements of Section VIII, Division 1, UG-84, of the ASME Code. For materials and thicknesses not covered by the code, the purchaser will specify the requirements on the data sheets. The use of ASTM A 515 steel is prohibited.

2.4.10.5 Protective Grills or Metal Screens

Protective grills or metal screens shall be fabricated from not less than 0.049 inch (1.25 millimeter) AISI 300 series stainless steel with a maximum mesh of ¼ inch (6.4 millimeters).

● 2.4.10.6 Fans

Fan systems, blades, and housings shall be designed to prevent sparking as a result of mechanical contact or static discharge. They shall be made of corrosion-resistant, ductile materials to minimize failure from corrosion or fatigue. Aluminum (with a copper content of less than 0.2 percent), bronze, or reinforced thermosetting conductive plastic (to bleed off static charges) is typically used in fan systems. When specified, the vendor shall demonstrate to the purchaser's satisfaction that the fan system provides the nonsparking qualities and durability required.

2.4.10.7 Stator Lamination Core Plate

Stator lamination core plate shall be of at least C-5 quality in accordance with ASTM A 345. C3 quality plate shall not be used dispersed or as a coating over C5 quality plate. The stator core shall be capable of withstanding winding burnout for rewind at a temperature of 750°F (399°C) without damage or loosening.

2.4.10.8 Heat Exchangers

Heat exchangers shall conform to the following:

● a. Air-to-air exchanger tubes used in the tube-type motors shall be made of copper, copper-based alloy, aluminum, aluminum alloy containing no more than 0.2 percent copper, or AISI 300 series stainless steel.

● b. Unless otherwise specified, water-to-air heat exchanger tubes shall be not less than .625 inch (16 millimeters) diameter and 18 BWG (1.25 millimeters) wall thickness made of 90-10 CU-NI material. The purchaser has the responsibility to have cooling water chemistry checked for compatibility.

2.4.11 NAMEPLATES AND ROTATION ARROWS

● **2.4.11.1** Nameplates and rotation arrows shall be of AISI 300 series stainless steel or of nickel-copper alloy (Monel or its equivalent), as specified, securely fastened by pins of similar material, and located for easy visibility.

2.4.11.2 As a minimum, the data listed below shall be clearly stamped on the motor's nameplate(s):

a. Vendor's name.
b. Serial number.
c. Horsepower.
d. Voltages.
e. Phase.
f. Full load power factor/efficiency.
g. Frequency, in hertz.
h. For antifriction bearings, the manufacturer and model number.
i. For flood-lubricated bearings, the oil flow rate, in gallons per minute.
j. For pressure-lubricated bearings, the oil pressure required, in pounds per square inch gauge.
k. Full load amperes (FLA).
l. Locked rotor amperes (LRA).
m. Full load speed, in revolutions per minute.
n. Time rating.
o. Temperature rise, in degrees Celsius; the maximum ambient or cooling air temperature for which the motor was designed; and the insulation system's designation.
p. Service factor.
q. Starting limitations.
r. Location of the magnetic center, per 2.4.9.3.
s. For Class I or Class II, Division 2, locations, the NFPA 70 "T" identification number for space heaters.
t. Enclosure type.
u. Total motor weight and rotor weight.
v. Year of manufacture (i.e., 1994).

2.4.11.3 Separate connection diagrams or data nameplates shall be located near the appropriate connection box for the following:

a. Motors with more than three power leads.
b. Space heaters (operating voltage and wattage).

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c. Temperature detectors (resistance, in ohms, or junction type).
d. Vibration and position detectors (manufacturer and model number.)
e. Connections of proper rotation (including bidirectional).
f. Current transformer secondary leads (when provided), with polarity marks.

● **2.4.11.4** When specified, the purchaser's identification information shall be stamped on a separate nameplate.

SECTION 3—ACCESSORIES

3.1 Terminal Boxes

● **3.1.1** Terminal boxes and auxiliary equipment enclosures shall be constructed of cast or nodular iron, cast steel, cast aluminum, steel plate, or aluminum plate with a minimum rigidity equivalent to that of steel plate with a nominal thickness of ⅛ inch (3.0 millimeters). Minimum dimensions and usable volumes shall not be less than those specified in ANSI C50.41. When specified, larger boxes shall be provided for special cable terminations and other devices.

● **3.1.2** When specified, the terminal box for the main phase terminations shall be capable of withstanding the pressure build-up resulting from a three-phase fault of 50 percent of the specified maximum available KVA (one-half cycle after fault inception) for a duration of 0.1 second. For motors fed from fused motor starters, the box withstand capability shall be coordinated with the I^2t (ampere-squared seconds) let-through energy specified on the data sheet. If a rupture disc is used to relieve pressure build-up, it shall not compromise the environmental rating of the box and the discharge from the pressure release shall be directed away from the possible personnel traffic.

3.1.3 For motors rated at 601 volts and higher, accessory leads shall terminate in a terminal box or boxes separate from the motor terminal housing. However, secondary connections for current and potential transformers located in the terminal housing are permitted to terminate in the terminal housing if they are separated from power leads or buses by a suitable physical barrier to prevent accidental contact. For motors rated at 601 volts and higher, the termination of leads of accessory items that normally operate at 50 volts RMS or less shall be separated from other leads by a suitable physical barrier to prevent accidental contact or shall be terminated in a separate box.

● **3.1.4** As a minimum, terminal boxes and auxiliary equipment enclosures shall meet the requirements for NEMA Type 3R and for the area classification shown on the data sheets. (See IEEE 303 for areas other than nonclassified areas.) The terminal boxes shall be suitable for conductor entry as specified on the data sheets. Each terminal box shall have a bolted, gasketed cover that is fully accessible from the front and arranged for convenient access. (Explosionproof boxes shall not be gasketed unless specifically approved by the purchaser.) The terminal strips shall be moisture resistant. All vertical gasketed surfaces shall be provided with a drip shield at the top. The gasket material shall be impervious to oil attack.

3.1.5 Grounding for field wiring inside terminal box shall conform to the requirements of NEMA MG 1-11.06.3.

● **3.1.6** When specified, a terminal box shall be supplied with the following items:

a. Thermal insulation on the interior top side.
b. Space heaters.
c. Drains.
d. Breathers.
e. Provisions for purging.
f. Removable links.
g. Adequate space for termination of shielded cables.
h. Universal bushing studs or receptacles.
i. Arresters and surge capacitors.
j. Differential and phase current transformers.
k. Silver-plated bus connections.
l. Potential transformers.
m. Ground bus.

3.1.7 When surge protection is provided in accordance with 3.6.2, a low-impedance ground path shall be provided between the surge protection and the stator core. This low-impedance path may be provided by running a copper No. 4/0 American Wire Gauge (AWG) wire in parallel with the motor leads. This wire shall be as short as practical and shall bond the motor core to the terminal box by means of compression fittings at the ground point, as specified in 3.1.5. This additional wire shall not be used for motors with a separate mounted terminal box. In this case, the ground connection in the terminal box shall be grounded directly to the plant's ground grid.

Note: Direct connection of the motor stator core or frame and a separate mounted terminal box to ground can create a closed ground loop where voltages can be induced by magnetic stray fields of the motor. Small induced voltages can result in harmful high currents due to the low impedance.

● **3.1.8** When differential current transformers are provided in accordance with 3.6.3 and secondary leads are specified, the leads shall be routed, in a workmanlike manner, away from high-voltage motor leads and protected by a physical barrier to prevent accidental contact. These leads shall be terminated at an appropriate shorting and grounding terminal block housed in an auxiliary box. The auxiliary box shall be accessible with-

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out removal of the stator terminal box's cover. When self-balancing current transformers are furnished, six stator leads shall be brought out to the terminal box, and removable links shall be used to allow access to each end of the phase windings. Each link shall be installed so that it can be removed without disturbing other parts and connections.

3.1.9 Wiring and terminal blocks in all terminal boxes shall be clearly identified. The method for marking the wiring shall be a stamped terminal of the heat-shrinkable type. The terminal blocks shall be permanently and suitably stamped. Stator leads shall be identified in accordance with NEMA MG 1. Current transformer leads shall have polarity identification markings at the transformer and at the terminal block in the auxiliary terminal box. All wiring markings shall agree with the notations on the special nameplates required by 2.4.11.3.

3.1.10 All wiring shall have insulation that is impervious to oil attack.

● **3.1.11** A terminal head or boxes, as specified, shall be supplied for each bearing-vibration-sensing unit and each temperature-sensing unit's terminations with the exception of stator resistance temperature detectors (RTDs), which shall be in a separate terminal box.

3.2 Winding Temperature Detectors

● **3.2.1** Winding temperature detectors shall be supplied when specified or when required by 3.5.5. These devices shall be RTDs.

3.2.1.1 Unless otherwise specified, RTD elements shall be platinum, three-wire elements with a resistance of 100 ohms at 32°F (0°C). These elements shall have tetrafluoroethylene-insulated, stranded, tinned copper wire leads at least 22 AWG in size with a calibration ratio of 1.3926. The leads shall meet the requirements of NFPA 70.

● **3.2.1.2** Two sensing elements per phase shall be installed and suitably distributed around the circumference in the motor's stator winding slots. When specified, one lead of each of these elements shall be grounded in the terminal box.

3.2.2 To prevent damage, the leads for all detectors shall be protected during manufacture and shipment of the motor. The vendor's drawings shall show the location and number of each sensing element in the motor winding and its connection point on the terminal strip.

● 3.3 Bearing Temperature Detectors

3.3.1 When specified, bearing temperature detectors shall be provided in motors with hydrodynamic radial and thrust bearings. Detectors shall be installed so that they measure bearing metal temperature. Bearing temperature sensors shall be provided in accordance with API Standard 670 unless otherwise specified.

Note: Adoption of API Standard 670 requires two sensors per bearing for most bearings. If two temperature detectors are required, the use of dual element sensors should be considered where space is limited.

3.3.2 Where insulated bearings are specified, bearing temperature detectors shall be installed in such a way that they do not violate the integrity of bearing insulation.

3.4 Space Heaters

● **3.4.1** When specified, motors shall be equipped with completely wired space heaters whose leads are brought out to a separate terminal box. Heaters with exposed elements are prohibited. The heaters' sheath material shall be as specified. The heaters shall be installed inside the motor enclosure in a location suitable for easy removal and replacement. Heaters shall be located and insulated so that they do not damage motor components or finish.

● **3.4.2** Space heaters shall be low power density, one or three phase, with a frequency of 60 hertz and a voltage of 115, 230, or 460 volts, as specified, with all live parts covered.

● **3.4.3** Space heaters shall be arranged so that heat is radiated from both sides to provide uniform heating of the stator windings. The heaters shall maintain the temperature of the motor windings at approximately 9°F (5°C) above the ambient temperature. The surface temperature of the heater elements shall not be greater than 200°C or the value listed on the data sheet.

3.5 Screens and Filters

3.5.1 When airflow inlet and outlet screens are provided, see 2.4.10.5 for material. The screen holders shall be designed to permit easy removal and replacement of the screens while the motor is running, except when the screens are located downstream of the airflow through filters.

● **3.5.2** When specified, provisions for future airflow inlet filters in standard types and sizes shall be available as an optional item for Type I weather-protected motors.

3.5.3 Provisions for future airflow inlet filters in standard types and sizes shall be furnished in Type II weather-protected and forced-ventilation motors when the means of forced ventilation is mounted integrally with the motor. Replacement and servicing of the filters shall be easily accomplished while the motor is running, without allowing the intrusion of airborne dirt or debris.

● **3.5.4** When filters are specified, they shall be of the permanent type and shall meet the service requirements indicated on the data sheets. Filters shall be constructed of AISI 300 series stainless steel.

● **3.5.5** When filters or provisions for future filters are specified, winding temperature detectors shall be provided in accordance with 3.2. Connections shall be furnished for a specified device to measure the pressure drop across the filters.

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3.6 Alarms and Control Devices for Motor Protection

● 3.6.1 SWITCHES

Alarm and control devices shall be equipped with single-pole, double-throw switches with a minimum rated capacity of 10 amperes at 115 volts and 60 hertz. When specified, the vendor shall supply devices with DC-rated contacts. All devices shall be approved by the purchaser.

3.6.2 SURGE PROTECTION

● **3.6.2.1** When specified, surge capacitors shall be furnished on motors rated above 2 kilovolts. The surge capacitors shall be the last devices connected to the motor leads before the leads enter the stator. The connection leads to the capacitors shall be at least No. 4/0 AWG and shall be as short as is practicable. The capacitors shall be rated for the motor voltage and shall be installed in the motor terminal box. (See 3.1.7 for bonding requirements.)

● **3.6.2.2** When specified, metal oxide surge arresters shall be furnished. The surge arresters shall be rated for the system voltage and the method of system grounding specified on the data sheets. They shall be mounted in the terminal box, and all connection leads shall be as short as is practicable. (See 3.1.7 for bonding requirements.)

● 3.6.3 DIFFERENTIAL CURRENT TRANSFORMERS

When specified, differential protection current transformers shall be provided. The purchaser will advise the motor vendor of the size, type, and source of supply of the current transformers. (See 3.1.8 for installation requirements.)

3.7 Ground Connectors

Visible ground pads shall be provided at opposite corners of the machine frame. A ground connection point shall be provided by drilling and tapping the motor frame for a ½-inch national coarse (NC) thread bolt.

3.8 Vibration Detectors

● **3.8.1** Hydrodynamic bearing motors intended to operate at speeds greater than or equal to 1200 revolutions per minute, or when specified for other speeds, shall be equipped with noncontacting vibration probes (two per bearing) and a phase-reference probe, or shall have provisions for the installation of probes. Vibration and position equipment shall be furnished and installed in accordance with API Standard 670, except as described below. Shaft surface preparation in the probe area shall be in accordance with 2.4.5.1.3.

3.8.1.1 Noncontacting vibration probes shall be mounted through the bearing centerline, monitoring the shaft journal, where possible. If not possible due to oil ring interference or other bearing construction features, the probes shall be mounted inboard of the bearing (toward the rotor). Where either of the above is not possible due to motor construction characteristics, the vendor shall state this in the proposal and the probes may be mounted outboard of the bearing, but not near a nodal point.

3.8.1.2 The leads of the noncontacting vibration probes shall be physically protected by the use of conduit, or other purchaser-approved means, and shall be secured to prevent movement.

● ■ **3.8.2** When specified, motors with hydrodynamic bearings shall have provisions for the mounting of four radial-vibration probes in each bearing housing, and where hydrodynamic thrust bearings are provided, they shall have provisions for two axial-position probes at the thrust end.

Note: When the probes cannot be accessed during operation and the motor cannot be stopped conveniently to change defective probes, four probes at each bearing are recommended. Two of the probes are connected to the oscillator demodulators and the other two probes have their leads run to the oscillator demodulator terminal box and are not connected but held as spares.

● ■ **3.8.3** When specified, seismic vibration sensors, or provisions for such, shall be supplied in accordance with API Standard 670.

Note: Axial-position probes are normally applied to monitor thrust-loading and hydrodynamic thrust-bearing conditions in vertical motors. Axial probes are occasionally used to monitor a rotor's axial vibration. On horizontal motors, axial probes should not generally be applied because no thrust bearing is present and because axial probes used as vibration sensors will not generally accommodate the rotor's relatively large amount of axial motion. Noncontacting vibration systems are generally used on high-speed machines with hydrodynamic radial bearings, and accelerometer systems are generally used on units with antifriction bearings, which have high transmissibility of shaft-to-bearing force.

Note: Vibration detectors are not normally used on motors with 14 or more poles.

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SECTION 4—INSPECTION, TESTING, AND PREPARATION FOR SHIPMENT

4.1 General

● **4.1.1** Whenever the specification or purchase order calls for shop inspections and tests to be witnessed, observed, or performed by a purchaser's representative, the vendor shall provide sufficient advance notice to the purchaser before each inspection or test. At all other times the purchaser's representative, upon providing similar advance notice to the vendor, shall have access to all vendor and subvendor plants where work on or testing of the equipment is in progress. In each instance, the actual number of calendar days considered to be sufficient advance notice shall be established by mutual agreement between the purchaser and the vendor.

4.1.2 The vendor shall notify all subvendors of the purchaser's inspection and testing requirements.

● **4.1.3** The purchaser will specify the extent of his or her participation in the inspection and testing.

4.1.3.1 *Witnessed* means that a hold shall be applied to the production schedule and that the inspection or test shall be carried out with the purchaser or purchaser's representative in attendance. For mechanical running or performance tests, this requires written notification of a successful preliminary test.

4.1.3.2 *Observed* means that the purchaser shall be notified of the timing of the inspection or test; however, the inspection or test shall be performed as scheduled, and if the purchaser or purchaser's representative is not present, the vendor shall proceed to the next step. (The purchaser should expect to be in the factory longer than for a witnessed test.)

4.1.3.3 *Required* means that the paragraph in question applies or that certified documentation shall be recorded for the purchaser.

4.1.4 Unless otherwise specified, all required test and inspection equipment shall be provided by the vendor.

4.2 Inspection

4.2.1 GENERAL

4.2.1.1 The vendor shall keep the following data available for at least 5 years for examination by the purchaser or purchaser's representative upon request:

● a. When specified, certification of materials, such as mill test reports on shafts, forgings, and major castings.
b. Purchase specifications for all items on bills of materials.
c. Test data to verify that the requirements of the specification have been met.
d. Results of all quality control tests and inspections.
● e. When specified, final assembly maintenance and running clearances.

● **4.2.1.2** When specified, parts shall not be painted until the specified inspection of the parts is completed.

● **4.2.1.3** The purchaser will specify the following:

a. Parts that shall be subjected to surface and subsurface examination.
b. The type of inspection required, such as magnetic particle, liquid penetrant, radiographic, and ultrasonic examination.

4.2.2 MATERIAL INSPECTION

● ■ **4.2.2.1 General**

When radiographic, ultrasonic, magnetic particle, or liquid penetrant inspection of welds or materials is required or specified, the criteria in 4.2.2.2 through 4.2.2.5 shall apply unless other criteria are specified by the purchaser. Cast iron may be inspected in accordance with 4.2.2.4 and 4.2.2.5. Welds, cast steel, and wrought material may be inspected in accordance with 4.2.2.2. through 4.2.2.5.

■ **4.2.2.2 Radiography**

4.2.2.2.1 Radiography shall be in accordance with ASTM E 94 and ASTM E 142.

4.2.2.2.2 The acceptance standard used for welded fabrications shall be Section VIII, Division 1, UW-51 (continuous weld) and UW-52 (spot weld), of the ASME Code. The acceptance standard used for castings shall be Section VIII, Division 1, Appendix 7, of the ASME Code.

■ **4.2.2.3 Ultrasonic Inspection**

4.2.2.3.1 Ultrasonic inspection shall be in accordance with Section V, Articles 5 and 23, of the ASME Code.

4.2.2.3.2 The acceptance standard used for welded fabrications shall be Section VIII, Division 1, Appendix 12, of the ASME Code. The acceptance standard used for castings shall be Section VIII, Division 1, Appendix 7, of the ASME Code.

■ **4.2.2.4 Magnetic Particle Inspection**

4.2.2.4.1 Both wet and dry methods and magnetic particle inspection shall be in accordance with ASTM E 709.

4.2.2.4.2 The acceptance standard used for welded fabrications shall be Section VIII, Division 1, Appendix 6, and Section V, Article 25, of the ASME Code. The acceptability of defects in castings shall be based on a comparison with the photographs in ASTM E 125. For each type of defect, the degree of severity shall not exceed the limits specified in Table 4.

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Table 4—Maximum Severity of Defects in Castings

Type	Degree
I	1
II	2
III	2
IV	1
V	1
VI	1

■ 4.2.2.5 Liquid Penetrant Inspection

4.2.2.5.1 Liquid penetrant inspection shall be in accordance with Section V, Article 6, of the ASME Code.

4.2.2.5.2 The acceptance standard used for welded fabrications shall be Section VIII, Division 1, Appendix 8, and Section V, Article 24, of the ASME Code.

Note: Regardless of the generalized limits in 4.2.2, it shall be the vendor's responsibility to review the design limits of the equipment in the event that more stringent requirements are necessary. Defects that exceed the limits imposed in 4.2.2 shall be removed to meet the quality standards cited, as determined by the inspection method specified.

4.2.2.6 Pressure-containing parts of water cooling circuits (including auxiliaries) shall be tested hydrostatically with liquid at a minimum of 1½ times the maximum allowable working pressure but not less than 20 pounds per square inch gauge (138 kilopascals). The test liquid should be at a higher temperature than the nil-ductility transition temperature of the material being tested. The hydrostatic test shall be considered satisfactory when neither leaks nor seepage is observed for a minimum of 30 minutes.

4.2.3 INSPECTION

4.2.3.1 During assembly of the lubrication system and before testing, each component (including cast-in passages of the components) and all piping and appurtenances shall be cleaned chemically or by another appropriate method to remove foreign materials, corrosion products, and mill scale.

● **4.2.3.2** When specified for machines having circulating pressure oil systems with a rated pump capacity of 5 gallons per minute or more, the oil system furnished shall meet the cleanliness requirements of API Standard 614.

● **4.2.3.3** When specified, the purchaser may inspect for cleanliness the equipment and all piping and appurtenances furnished by or through the vendor before final assembly.

● ■ **4.2.3.4** When specified, the hardness of parts, welds, and heat-affected zones shall be verified as being within the allowable values by testing of the parts, welds, or zones. The method, extent, documentation, and witnessing of the testing shall be mutually agreed upon by the purchaser and the vendor.

● ■ **4.2.3.5** When specified, the purchaser's representative shall have access to the vendor's quality program for review.

● ■ **4.2.3.6** When specified, the following tests and inspections shall be performed:

a. Verification of residual unbalance of the completely assembled rotor in accordance with the method given in Appendix C.

Note: Alternative methods for residual unbalance verification may be required for rotors balanced in the motor frame assembly.

b. Inspection of stator prior to VPI.

4.3 Final Testing

4.3.1 GENERAL

● **4.3.1.1** The purchaser reserves the right to observe the testing, dismantling, inspection, and reassembly of equipment, as specified.

4.3.1.2 The vendor shall notify the purchaser not less than 10 working days before the date the equipment will be ready for testing.

4.3.1.3 The vendor shall provide calculated motor data from final witnessed testing immediately upon completion of the motor testing. The final results of critical parameters must be determined prior to the inspectors leaving the test facility.

■ **4.3.1.4** At least 6 weeks before the first scheduled test, the vendor shall submit to the purchaser, for review and comment, detailed procedures for all tests, including acceptance criteria for all monitored parameters. The following items shall be included in the test procedures booklet:

a. Types of tests (electrical or mechanical).
b. Testing sequence.
c. Detailed testing schedule.
d. Guarantee limits, such as overall vibration levels, limits of harmonic vibration components, frequency and amplification factors of critical speeds, motor efficiency and noise levels, and stator temperature rise.
e. Data measurements to confirm guarantee limits and proper operation of equipment components. This should include, but not be limited to, the following:
1. Power (KW), speed (RPM), and torque (ft-lbs).
2. Shaft and bearing vibration, filtered and unfiltered, for each vibration sensor.
3. Phase angle.
4. Journal bearing embedded temperatures.
5. Stator winding temperatures.
6. Cooling water flow and temperature.
7. Temperature on air inlets and discharge.
8. Lube oil flows, pressures, inlet, and drain temperatures for each bearing.
9. Voltage, current, power factor, and load.
10. All instrumentation and data points that are to be monitored in the field.

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f. Calculated lateral critical speed analysis.
g. A complete set of test data sheets that are to be used during the testing.
h. A listing of all alarm and shutdown levels.
i. Calibration sheets for all switches, vibration probes, and proximitors.
j. General arrangement drawings and bill of materials.
k. Residual rotor unbalance worksheet.
l. List of electrical test instrumentation and method of calibration.

● **4.3.1.5** Tests shall be made on the fully assembled motor, using contract components and accessories, and the prebalanced coupling half (including idling adapters), if specified. Terminal boxes may be excluded.

4.3.1.5.1 When the contract coupling hub is specified to be mounted by the vendor, it shall be mounted on the rotor along with any idler adapter or mass moment simulator at the completion of the final rotor-fan balance. If the rotor unbalance exceeds the allowable unbalance limit (4W/N per plane) after the coupling is mounted, the vendor and the purchaser shall mutually agree on the cause of the excess unbalance and take appropriate corrective action.

4.3.1.5.2 If it is not practical to mount the coupling with the rotor in the balance machine, the coupling hub and any idler adapter or mass moment simulator can be mounted after machine assembly but before the unbalance response test. Before mounting the coupling, the motor shall be properly mounted on a massive foundation and run until the bearing temperatures stabilize. The voltage shall be reduced to ¼ of rated to minimize electrically induced vibration and 1X filtered vibration data taken without the coupling mounted and again after mounting the coupling. The magnitude of the vectorial change in the 1X vibration on the shaft and bearing housings shall not exceed 10 percent of the vibration limits given in Figures 1 and 2.

During the test with ¼ rated voltage, the rotor shall be maintained on magnetic center. With the coupling mounted, the voltage shall be increased to rated and a complete set of filtered and unfiltered vibration data shall be taken. All data shall be within the limits given in Figures 1 and 2. If the vibration change or amplitude exceeds the allowable limits, the vendor and purchaser shall mutually agree on the appropriate corrective action. (See Note for 2.4.6.2.4)

4.3.1.6 If applicable, all oil pressures and viscosities shall be at the maximum operating temperature values recommended in the manufacturer's operating instructions for the specific unit being tested. Oil flow rates for each bearing housing shall be determined. (Accepted methods other than flowmeter may be used.) Test-stand oil filtration shall be 10 microns nominal or better.

4.3.1.7 All detection, protective, and control devices shall be checked, and adjustments shall be made as required prior to any running tests.

4.3.1.8 During the running tests, the mechanical operation of all equipment being tested and the operation of the test and purchased instrumentation shall be satisfactory.

4.3.1.9 If replacement or modification of bearings or seals or dismantling to replace or modify other parts is required to correct mechanical performance deficiencies, the mechanical vibration and unbalance response tests shall be repeated after these replacements or corrections are made.

4.3.1.10 Facilities to prevent entrance of oil into the motor shall be in operation throughout the test. Any violation of this condition requires termination of the test until the necessary correction is made.

4.3.1.11 The vendor shall maintain a complete, detailed log and plots of all final tests and shall submit the required number of copies to the purchaser, including data for bearing temperatures, rotor balancing, critical speeds, and vibration measurements taken over the operating speed range and the spectrum analysis. A description of the test instrumentation and certified copies of the instrument calibrations shall be kept available for the purchaser's review.

4.3.1.12 Unless otherwise specified, all test results shall be certified by the vendor and transmitted to the purchaser in reproducible form. Exceptions to the information specified in 4.3.1.11 shall be listed in the proposal.

● **4.3.1.13** When specified, before the start of testing, the manufacturer shall demonstrate the accuracy of the test equipment and/or automated data acquisition systems. The calibration and maximum deviation, from a recognized standard, at all phase angles and anticipated frequencies and harmonics, shall be demonstrated. A maximum deviation of no more than ½ percent, including all potential transformers, current transformers, test leads, shunts, voltage dividers, transducers, A to D converters, and computers that are part of the test set-up, shall be demonstrated. Every element of the test equipment set-up shall be included in the accuracy demonstration.

● **4.3.1.14** Prior to any mechanical running test, a check for "soft feet" shall be made. After the motor has been aligned, shimmed, and firmly secured to the test base, a dial indicator micrometer oriented in the vertical direction shall be attached at the mounting foot to be checked. The micrometer is then zeroed, the mounting bolt or bolts loosened at the foot, and the change in micrometer reading noted. If the micrometer reading exceeds .001 inch, the motor mounting requires cleaning or re-shimming. This soft foot check shall be performed at each motor mounting foot, with the other feet secured, until all micrometer change readings are less than .001 inch. If there are intermediate bases, this check shall be performed at each interface between the motor and test floor.

4.3.2 ROUTINE TEST

Each motor shall be given a routine (commercial) test to demonstrate that it is free from mechanical and electrical de-

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fects. This test shall be conducted in accordance with the applicable portions of IEEE 112 and NEMA MG 1. The test shall include the following items:

a. Measurement of no-load current (each phase).
b. Measurement of nominal no-load speed.
c. A determination of locked rotor current.
d. AC high-potential test on the stator.
● e. An insulation resistance test by megohmmeter and polarization index per IEEE 43. The insulation resistance measurement and polarization index shall be performed in accordance with Table 5. (The polarization index is the ratio of the 10-minute resistance value to the 1-minute resistance value.) When specified, the preferred values in Table 5 shall be used.
f. Measurement of winding resistance, using the Wheatstone bridge method.
g. Vibration test per 4.3.3.
h. A test of the bearing insulation.
i. A test of bearing temperature rise. The motor shall be operated at no load for at least 1 hour after the bearing temperatures have stabilized. Stable temperature is defined as a change of not more than 1°C in 30 minutes. The no-load run shall demonstrate that bearing operation is without excessive noise, heating, vibration, or lubrication leaks.
j. Inspection of the bearings and oil supply when furnished. Before the tests are run, each sleeve bearing's journal-to-bearing clearance and bearing-shell-to-bearing-cap crush and alignment shall be determined and recorded. After all running tests have been completed, the shaft journals and bearings shall be inspected by completely removing both the top and bottom halves of each bearing. The contact between the shaft journal and the bearing bore shall be a minimum of 80 percent of the axial length and symmetrical with no-edge loading. Where the lubricant is accessible, its condition shall be visually examined after the run.
k. Measurements of the air gap per 2.4.7.16.
l. Shaft voltage and current measurements.

4.3.3 VIBRATION TESTS

4.3.3.1 Vibration shall be measured to determine the mechanical performance and acceptability of all motors. This test is to be performed during the bearing temperature rise test (4.3.2.2, item i) and heat run or hot rotor test and tandem test when these have been specified.

4.3.3.2 During the shop running test of the assembled motor, vibration measurements of the motor and foundation shall be made with the motor properly shimmed (no soft feet, see 4.3.1.14) and securely fastened to a massive foundation (see Note 1 to 2.4.6.1.2) or test floor stand. Elastic mounts are not permitted.

4.3.3.3 Motors that are provided with noncontact probes or provisions for noncontact probes shall be tested to verify that the shaft-sensing areas meet the total electrical and mechanical runout requirements of 2.4.5.1.3. The probe track runout shall be measured with the rotor at slow roll (200-300 RPM) speed, where the mechanical unbalance forces on the rotor are negligible. A continuous unfiltered trace of the noncontact probe output shall be recorded for a 360 degree shaft rotation at each probe location with the shaft rotating in the assembled motor. The rotor shall be held at its axial magnetic center during recording. An alternate method of acceptance for measurement of electrical and mechanical runout is to roll the rotor in V-blocks at the journal centerline while measuring runout with a noncontacting vibration probe and a dial indicator at the centerline of the probe location and one probe tip diameter to either side. When this acceptance method is chosen, measurement and documentation of runout in the assembled motor are also required to check for changes in probe track runout.

Table 5—DC Test Voltages for Insulation Resistance and Determination of Polarization Index

Motor Voltage	Preferred	Acceptable
< 2300	1000	1000
2300 - 3999	2500	1000
≥ 4000	5000	2500

4.3.3.4 Vibration measurements shall be taken in the horizontal and vertical radial directions and the axial direction on the bearing housings of antifriction-bearing and hydrodynamic-bearing motors. All shaft radial vibration measurements shall be taken using noncontacting eddy-current probes when the motor is equipped with them or if provisions for noncontacting probes are specified. Where shaft noncontacting probes or provisions for probes are not specified and the shaft is inaccessible for vibration measurement, only bearing housing vibration measurements shall be taken. (See 2.4.5.1.3 for requirements at probe-sensing areas.) Shaft and bearing housing vibration data shall be recorded for the unfiltered, one-half running speed, one times running speed (and phase angle), and two times running speed and the one times and two times line frequency amplitudes.

4.3.3.5 Unfiltered and filtered radial and axial vibration, electrical input, and temperature data shall be recorded at 30-minute intervals during all mechanical running tests. If the vibration pulsates, the high and low values shall be recorded.

4.3.3.6 For two-pole motors, after the bearing temperatures have stabilized, filtered and unfiltered vibration readings at each position shall be recorded continuously for a period of 15 minutes. This data shall be continuously plotted or tabulated at 1-minute increments over the 15-minute period. If the vibration modulates, the high and low values of vibration and the frequency of the modulation shall be recorded.

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4.3.3.7 If mutually agreed upon by the purchaser and the vendor, the purchaser may use his or her monitoring or recording equipment in conjunction with the vibration transducers mounted on the machine to record the dynamic behavior of the machine during testing.

4.3.3.8 All purchased vibration probes, transducers, oscillator-demodulators, and accelerometers shall be in use during the test. If vibration probes are not furnished by the equipment vendor or if the purchased probes are not compatible with shop readout facilities, then shop probes and readouts that meet the accuracy requirements of API Standard 670 shall be used.

4.3.3.9 Shop test facilities shall include instrumentation with the capability of continuously monitoring and plotting revolutions per minute, peak-to-peak displacement, and phase angle. Presentation of vibration displacement and phase marker shall also be by oscilloscope or spectrum analyzer.

4.3.3.10 The vibration characteristics determined by the use of the instrumentation specified in 4.3.3.8 and 4.3.3.9 shall serve as a basis for acceptance or rejection of the machine.

4.3.3.11 During the shop test of the motor, operating at its rated voltage and rated speed or at any other voltage and speed within the specified operating speed range, the shaft displacement and bearing housing velocity of vibration shall not exceed the values shown in Figures 1 and 2. The shaft readings in Figure 1 include a maximum allowance for electrical and mechanical runout in accordance with 2.4.5.1.3. If a temperature test is specified (see 4.3.5.1.1, item e), the vibration shall be within the filtered and unfiltered limits shown in Figures 1 and 2 throughout the temperature range from the test ambient temperature to the total design temperature.

If a vibration at a particular station and frequency and at full-load steady state temperature exceeds the limits of Figures 1 and 2, and when specifically approved by the purchaser, the corresponding value taken at ambient temperature on the massive base and corrected for thermal effects will be used as the criterion for acceptance. The method allows for "responsive amplification" due to the setup at the dynamometer. Following the full-load run, the coupling will be quickly disconnected. The motor will be run with full voltage, and the vibration level at the particular station and frequency will be recorded. This hot value will be divided by the corresponding value in the same setup (at the dynamometer but uncoupled) but with the rotor at ambient temperature. This ratio will be used as a multiplier to be applied to the corresponding value recorded on the massive foundation.

The magnitude of vibration vector changes from no load to rated temperature shall not exceed 50 percent of the curve value of Figures 1 and 2. For motors that do not comply with this vector change allowance, while remaining within the limits shown in Figures 1 and 2, and when specifically approved by the purchaser, the vendor shall demonstrate the structural stability of the motor. The vibration test shall be repeated by letting the motor cool down to the no-load stabilized temperature and then reheating it to achieve a stable temperature while recording the vibration data. The magnitude of vector change in succeeding vibration amplitudes, for the cold motor under no load and for the hot motor at rated temperature, shall be within 10 percent of the allowable limits shown in Figure 1.

4.3.3.12 While the equipment is operating at maximum continuous speed and any operating temperature, sweeps shall be made for vibration amplitudes at frequencies other than running speed. These sweeps shall cover a frequency range from 25 percent of the running speed frequency to three times the line frequency. See Figures 1 and 2 for filtered vibration limits.

● **4.3.3.13** When specified, a data acquisition system shall be used to record the vibration data during testing. The recorded data shall be submitted to the purchaser together with the final test report.

4.3.3.14 Trim balancing may be performed, if approved by the purchaser. The residual unbalance test (4.2.3.6 and Appendix C) is required after trim balancing.

■ 4.3.4 STATOR TESTS

● 4.3.4.1 Stator Core Test

When specified, prior to insertion of the stator coils into the core, the stator core interlaminar insulation integrity shall be verified. Loss of lamination insulation produces circulating currents leading to excessive heating of the core (hot spots) eventually causing coil failure and melting of the core steel.

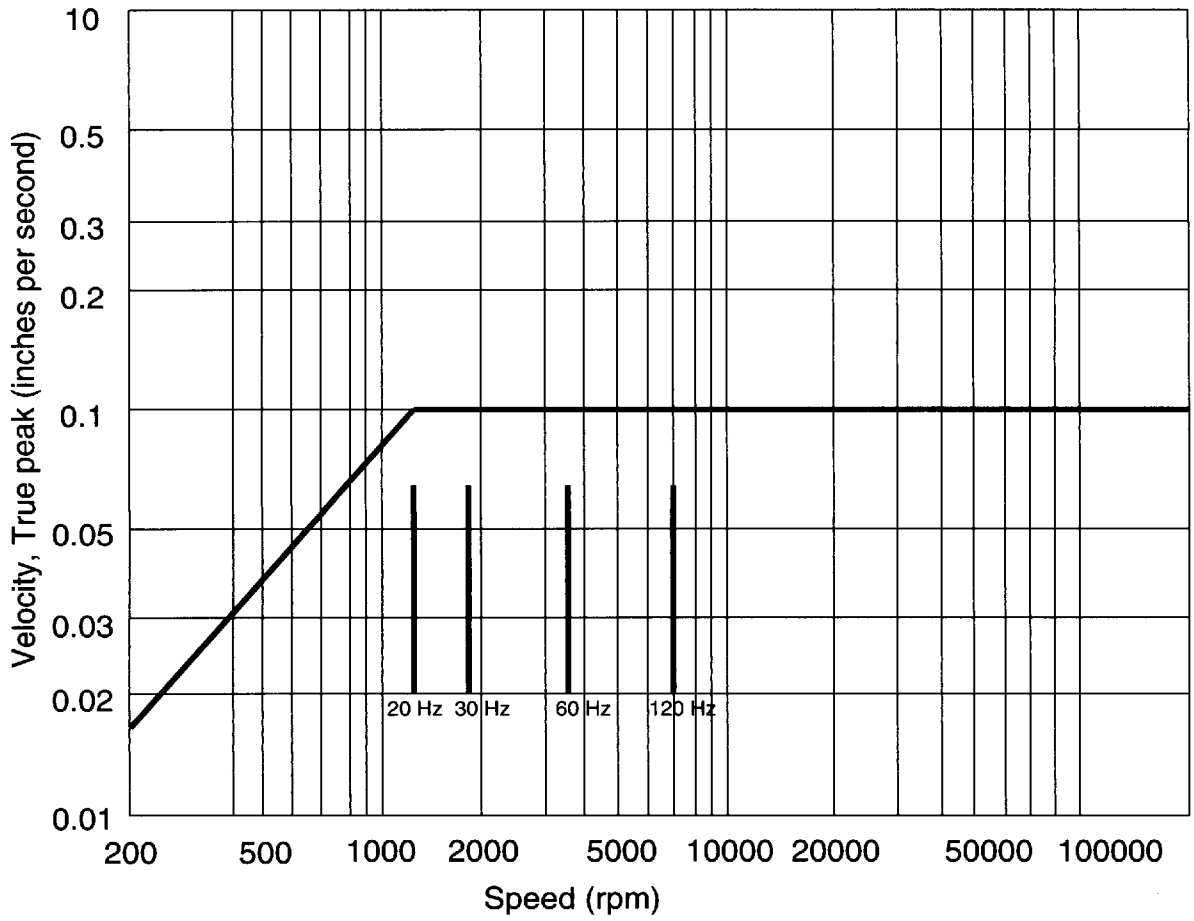
The test shall be performed by inducing rated flux density into the core by placing coils through it in a manner similar to a transformer winding. Rated flux shall be maintained for a minimum of ½ hour while continuously monitoring stator temperatures with an infrared camera or infrared thermometer. Interlaminar insulation deficiencies (hot spots) are defined as being any core location 5°C above the adjacent core temperature.

4.3.4.2 Surge Test

Surge comparison tests shall be made of the turn insulation in the fully wound stator, just before the coil-to-coil connections are made, at test levels and methods in accordance with IEEE 522.

● When specified, two additional stator coils for special surge tests of the main and turn insulation shall be manufac-

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Notes:
1. For unfiltered vibration limits, use the motor synchronous or maximum rotational speed in rpm.
2. For filtered limits, use vibration frequency in hertz.
3. Limits are for sleeve and anti-friction bearing motors.

Figure 2—Bearing Housing Vibration Limits: For Sleeve and Antifriction Bearing Motors; With the Motor Securely Fastened to a Massive Foundation

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tured at the same time as the complete stator winding. These coils shall be completely cured and tested as follows:

a. The test of the main insulation shall consist of three successive applications of a 1.2/50-microsecond impulse voltage with a crest value of 5 PU. The impulse voltage shall be applied to both terminals of the coil conductor while the conducting surfaces of the simulated slot portions of the coil are connected to earth.

b. The test of the turn insulation shall consist of not less than three voltage impulses within 1 minute, applied between the coil terminations.

● 4.3.4.3 Power Factor Tip-Up Test

A power factor tip-up test shall be performed, when specified. This test may be specified on a completely wound stator or on individual coils. The test method shall be per IEEE 286.

When this test is specified for the individual coils, the manufacturer shall make two sacrificial coils of the same design and dimensions as the coils for the main winding and shall process them with the main stator. A power factor tip-up test shall be performed on both sacrificial coils. The acceptance criteria shall be mutually agreed upon between the manufacturer and the user.

● 4.3.4.4 Sealed Winding Conformance Test

4.3.4.4.1 When specified, motor stators equipped with sealed insulation systems shall be tested in accordance with NEMA MG 1-20.49, by means of a water-immersion or spray test. These tests shall be in addition to all other tests.

Note: This test exposes certain parts of the insulation to stress levels that are in excess of what it sees during normal operation. The test can cause dielectric deterioration of weaknesses that might never occur in service. Any internal ionization or carbonization initiated during the test will further weaken the insulation. Before this test is specified, the risk of insulation degradation should be weighed against the level of quality assurance provided by a successful water test. Any test failures should be analyzed to determine the method of repair.

4.3.4.4.2 At the completion of the water-immersion or spray test, the stators shall be rinsed and dried, at which point any other required tests may be performed.

■ 4.3.5 SPECIAL TESTS

4.3.5.1 Complete Test

● **4.3.5.1.1** When specified, each motor shall be given the complete test described below in items a through g in addition to the tests specified in 4.3.2. This test shall be in accordance with the applicable portions of IEEE 112 and NEMA MG 1 and shall include the following items:

a. Determination of efficiency and power factor at 100 percent, 75 percent, and 50 percent of full load. If specified, the determination will include the service factor rating. The purchaser, in consultation with the manufacturer, shall specify which method (such as Method B, dynamometer, or Method F, equivalent circuit calculation) given in IEEE 112 must be used in determining the performance data.

Note: When comparing competitive bids, it is necessary that all manufacturers quote efficiencies determined by the same method. Not all manufacturers have facilities for performing Method B, dynamometer, testing for large motors. Alternate methods, such as Method F, can be used by all manufacturers. The manufacturers should be consulted when the capability of test facilities is in question.

b. Determination of the locked rotor power factor.

c. Determination of full load current and slip.

d. Determination of locked rotor and breakdown torque.

e. A heat run at maximum continuous rated service factor for a minimum of 4 hours.

Note: The heat run may actually be a temperature test. This would allow the use of dual-frequency heat runs and/or loading at the limit of the test facility. Vertical motors may be tested in a horizontal mounting provided that airflow is representative of the actual vertical mount. If the limits of the testing facility are below that of the horsepower rating of the motor during the heat run, an extrapolation of the expected temperature rise at full load shall be provided, utilizing a method approved by the purchaser.

f. A test for the determination of the speed torque curve.

g. Noise test in accordance with IEEE 85.

● ■ **4.3.5.1.2** For special purpose motors and when specified for general purpose motors, hydrodynamic bearings shall be removed, inspected, and reassembled after completion of the mechanical running tests. There shall be no more than ±0.0002 inch change of the bearing bore diameter and no metal transfer between the shaft and the bearing. Initial and final dimensional checks shall be made with the bearing at the same temperature (approximately ±2°C).

● **4.3.5.1.3** When specified, the motor's insulation will be tested by means of a DC high-potential test, to the maximum voltage listed in Table 6. The test procedure shall be to apply voltage in not less than four approximately equal steps, pausing 1 minute at each step and 5 minutes at the final voltage, taking 15 seconds to increase the voltage slowly at the beginning of each step. During the test, a microammeter shall be watched closely for the inception of any leakage-current advance.

Table 6 — DC High-Potential Test Voltage Levels

Motor Rated Voltage (kilovolts) E	DC High-Potential Test Voltage (kilovolts) $(2E + 1)(0.75)(1.75)$
2.3	7.4
2.4	7.6
4.0	11.8
4.16	12.2
6.6	18.6
6.9	19.4
13.2	36.0
13.8	37.5

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4.3.5.2 Rated Rotor Temperature Vibration Test

● **4.3.5.2.1** For motors that do not receive the complete test of 4.3.5.1.1, a heat run test in accordance with 4.3.5.1.1, item e, shall be performed when specified.

4.3.5.2.2 If the heat run test specified in 4.3.5.1.1, item e, cannot be performed, the vendor shall submit complete details of an alternative test that permits measurements of motor vibration throughout the test for at least 4 hours with the motor operating at rated speed and with the rotor at full load temperature.

4.3.5.3 TANDEM TEST

When motors are specified to operate in tandem with a turbine, the running test shall be conducted in the following sequence:

a. Operate at speeds from zero to maximum continuous speed in speed increments and run at maximum continuous speed until the bearing and lube oil temperature have stabilized.

b. Increase the speed to 110 percent of maximum continuous speed and run for a minimum of 15 minutes.

c. Reduce speed to maximum continuous speed and run for 4 hours.

Note: Caution should be exercised when operating at or near critical speeds.

● 4.3.5.4 UNBALANCE RESPONSE TEST

When specified, a 15-percent separation margin (see 2.4.6.1.1) shall be verified by attaching the motor to a massive foundation support and subjecting the motor to the following unbalanced response test. Special considerations may be required for supersynchronous machines. For critical speed considerations, a test shall be made with a coupling-half mass moment or moments equivalent to those of the contract coupling or couplings.

a. A deliberate unbalance of $4U_B$ per plane (see 2.4.6.2.5, Equation 1) shall be applied to the rotor. The weights shall be placed at the balance planes in phase and also 180° out of phase. In cases where an overhung mass is present, such as a fan or coupling, resulting in a bending mode with maximum deflections at the shaft end, the amount of unbalance to be added to the overhung mass shall be based on four times the allowable residual unbalance in the overhung mass. (For example, from API 671, the assembled coupling may be balanced to $40Wc/N$, where Wc is the weight of the coupling and N is the maximum continuous speed. In this case the amount of unbalance to be added to the coupling would be $160Wo/N$, where Wo is the weight of the overhung mass.)

The unbalance weights can be placed at any location on the balance planes or coupling. However, if the shaft displacement exceeds 90 percent of the limits defined in 4.3.5.4.b and c, the test shall be repeated with the weights moved to new positions 90 degrees from the original positions to determine the sensitivity of the rotor response to unbalance weight placement. The maximum response obtained shall be used as the basis for passing the test.

b. The machine shall be run to 120 percent of its rated speed with the unbalance weights attached and then allowed to coast to rest. The shaft vibration relative to the bearing housing shall be observed. Machines with defined separation margins shall meet the following criteria:

1. The shaft displacement relative to the bearing housing at any speed within the operating speed range or separation margin limits shall not exceed the smaller of the following value or 55 percent of the shaft-to-bearing and seal diametral running clearances:

$$D_s = 1.5\sqrt{\frac{12{,}000}{N}}$$

Where:

D_s = shaft displacement, in mils peak to peak.
N = operating speed nearest the resonant speed of concern, in revolutions per minute.

2. The shaft displacement relative to the bearing housing at any speed outside the operating speed range or separation margin limits shall not exceed 90 percent of the shaft-to-bearing diametral running clearance.

c. When machines do not comply with the separation margin of 2.4.6.1.1, and when specifically approved by the purchaser, a well-damped response is to be demonstrated. The motor shall be run to 120 percent of its rated speed with the unbalance weights described in item a applied to the rotor and then allowed to coast to rest. The shaft displacement over the entire speed range, from 0 to 120 percent, shall not exceed the following value:

$$D_s = 1.5\sqrt{\frac{12{,}000}{N_{\max}}}$$

Where:

D_s = shaft displacement, in mils peak to peak.
N_{max} = maximum rated speed, in revolutions per minute.

● 4.3.5.5 BEARING HOUSING NATURAL FREQUENCY TEST

When specified, bearing housings or end bell supports shall be checked for resonance on one machine of each group of identical machines. The resulting response shall be plotted for a frequency sweep of 0–300 percent of running speed. To eliminate the interaction between the bearing housing, the rotor shall be turned at a slow roll (approximately 300 RPM). The response plots shall be made on each bearing housing in the horizontal, vertical, and axial directions. The application of the excitation force must be made in these same directions.

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No significant resonance shall occur within ±15 percent of running speed multiples, up to its third harmonic, or between 40 percent and 60 percent of running speed. A significant resonance is defined as a peak that lies within 6dB in amplitude (displacement) of the fundamental bearing housing resonance.

On variable speed machines, the vendor shall propose an alternate method in the proposal to verify that the natural frequency of the bearing housing will not be within the operating speed range of the motor. This method must be approved by the purchaser.

4.4 Preparation for Shipment

● **4.4.1** Each unit shall be suitably prepared for the type and mode of shipment specified. The preparation shall make the equipment suitable for at least 6 months of outdoor storage from the time of shipment, with no disassembly required before operation, except for inspection of bearings and seals. Machines that are disassembled for shipment or storage shall be provided with marine-type plywood over all openings and sloped for proper watershed when protected with exterior covering.

4.4.2 The vendor shall provide the purchaser with the instructions necessary to preserve the integrity of the storage preparation after the equipment arrives at the job site and before start-up.

4.4.3 The equipment shall be prepared for shipment after all testing and inspection have been completed and the equipment has been released for shipment by the purchaser. As a minimum, the preparation shall include that specified in 4.4.3.1 through 4.4.3.9.

● **4.4.3.1** Exterior surfaces, except for machined surfaces, shall be coated with manufacturer's standard paint unless otherwise specified.

4.4.3.2 Exterior machined surfaces, except for corrosion-resistant material, shall be coated with a suitable rust preventive.

4.4.3.3 After thorough cleaning, internal areas of bearings and carbon steel oil systems' auxiliary equipment shall be coated with a suitable oil soluble rust preventive.

4.4.3.4 Flanged openings shall be provided with metal closures at least 3/16 inch (4.8 millimeters) thick, with synthetic rubber gaskets and at least four full-diameter bolts.

4.4.3.5 Threaded openings shall be provided with steel caps or solid shank steel plugs. In no case shall nonmetallic caps or plugs be used.

Note: These plugs are for shipping purposes; permanent plugs are covered in 2.4.3.2.

● **4.4.3.6** When specified, the equipment shall be mounted on a rigid skid or base suitable for handling by forklift, truck, or crane. This skid shall extend beyond all surfaces of the motor.

4.4.3.7 The rotor shall be blocked to prevent axial and radial movement.

4.4.3.8 Space heater leads shall be accessible without disturbing the shipping package and shall be suitably tagged for easy identification.

● **4.4.3.9** When specified, the normal running bearings shall be removed and shipped in protective crates, and the motor shall be equipped with special shipping bearings for shipment by rail or ship.

4.4.4 Lifting points and lifting lugs shall be clearly marked. Each motor shall be properly identified with item and serial numbers. Material shipped in separate crates shall be suitably identified with securely affixed, corrosion-resistant metal tags indicating the item and serial number of the equipment for which it is intended. The recommended lifting arrangement shall be identified on boxed equipment.

4.4.5 The fit-up and assembly of machine-mounted piping, coolers, and other equipment shall be completed in the vendor's shop before shipping, unless specifically approved by the purchaser.

4.4.6 Exposed shafts and shaft couplings shall be wrapped with waterproof, moldable waxed cloth or vapor phase inhibitor paper. The seams shall be sealed with oilproof adhesive tape.

4.4.7 Components (both individual pieces and packaged sets) shipped with mounted preassembled piping, tubing, or wiring shall comply with the requirements of the Occupational Safety and Health Administration.

4.4.8 Auxiliary piping connections furnished on the purchased equipment shall be impression stamped or permanently tagged to agree with the vendor's connection table or general arrangement drawing. Service and connection designations shall be indicated.

4.4.9 Bearing assemblies shall be fully protected from the entry of moisture and dirt. If vapor phase inhibitor crystals in bags are installed in large cavities to absorb moisture, the bags must be attached in an accessible area for ease of removal. Where applicable, bags shall be installed in wire cages attached to flanged covers, and bag locations shall be indicated by corrosion-resistant tags attached with stainless steel wire.

4.4.10 One copy of the manufacturer's standard installation instructions shall be packed and shipped with the equipment.

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SECTION 5—GUARANTEE AND WARRANTY

The details of the guarantee and warranty will be developed jointly by the purchaser and the vendor during the proposal period.

SECTION 6—VENDOR'S DATA

6.1 Proposals

● **6.1.1** The evaluation factor. Motors will be evaluated on the basis of life-cycle cost (LCC) (purchase price plus present worth of losses) when the evaluation factor (dollars per kilowatt) is shown on the data sheets.

6.1.2 The vendor's proposal shall include the information specified in 6.1.3 through 6.1.19.

6.1.3 The vendor shall provide copies of the Vendor's Induction Motor Proposal Data Sheet (see Appendix B) with complete vendor information entered.

6.1.4 The vendor shall provide complete performance curves and data to fully define the envelope of operations and the point at which the manufacturer has rated the equipment, including the following items:

● a. The motor's average torque versus speed during starting at rated voltage and, when specified, at 80 percent of rated voltage, or any other specified value.

● b. The motor's current versus speed during starting at rated voltage and, when specified, at 80 percent of rated voltage, or any other specified value.

c. The inertia of the motor's rotor.

● d. Estimated times for acceleration at rated voltage, 80 percent of rated voltage, or any other specified voltage values.

e. The locked rotor (stalled) withstand time, with the motor at ambient temperature and at its maximum rated operating temperature, at 70, 80, 90, and 100 percent of rated voltage.

● f. Expected efficiency as determined in accordance with IEEE 112, Method B, C, E, or F, or by certified data from previously tested designs. The purchaser, in consultation with the vendor, shall specify the method to be used. For Method F (equivalent circuit calculations), the vendor shall state the method used for determining rotor resistance and stray-load loss. For purposes of quotation, when Method F is proposed, the stray-load losses shall be taken as 0.9 percent of the rated motor output for motors 2500 horsepower and above and 1.2 percent for all sizes below 2500 horsepower. For proposed stray load loss values less than these, the supplied values shall be based upon actual reverse rotation test data on a similar machine.

● g. When specified, the power factor and the guaranteed efficiencies at ½, ¼, ¾, and full load calculated in accordance with IEEE 112.

● h. When specified for motors that drive reciprocating compressors or pumps requiring a variable torque during each motor revolution, the maximum current variation under actual operating conditions, the calculated motor efficiency under these conditions based on an expected minimum efficiency, and the method of calculating the efficiency under operating conditions. The effect of voltage drop in the power supply system due to motor current shall be considered. The purchaser will furnish the value for the minimum short circuit kilovolt-amperes of the power supply system. Where lower limits are not specified, current pulsations under actual operating conditions shall be within the limits stated in NEMA MG 1-20.82. Compressor crank-effort diagrams and other relevant data will be supplied by the purchaser for this determination. For motors with pulsating current, efficiency under actual operating conditions will be used to evaluate the LLC of the motor.

Note: To verify performance, it may be necessary to check motor current pulsations under load in the field using an oscilloscope in accordance with NEMA MG 1-20.82.

● i. When witnessed tests are specified, the extra cost for such tests.

6.1.5 The vendor shall provide utility requirements, such as water, air, and lube oil, including the quantity of lube oil required at the supply pressure and the heat load to be removed by the oil. (Approximate data shall be defined and clearly identified as such.) This information shall be entered on the data sheets.

6.1.6 The vendor shall provide net weights and maximum erection weights with identification of the item. These data shall be stated individually where separate shipments, packages, or assemblies are involved. These data shall be entered on the data sheets.

6.1.7 The vendor shall provide a preliminary dimensional outline drawing showing the location of inlet and discharge connections and the direction of rotation when the motor is viewed from the end opposite the drive end.

6.1.8 The vendor shall provide a schematic diagram of the lube oil system.

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6.1.9 The vendor shall provide typical drawings and literature to fully describe the details of the offering or offerings. The vendor shall show shaft sealing and bearing details, internal construction, rotor construction, and the method of attaching the rotor bar to the shorting ring.

6.1.10 The vendor shall provide a specific statement that the complete motor and all auxiliary equipment are in strict accordance with this standard. If the motor and auxiliary equipment are not in strict accordance, the vendor shall include a specific list that details and explains each deviation. Deviations may include alternative designs or systems equivalent to and rated for the specified duties.

6.1.11 The vendor shall provide an explicit statement of the proposed guarantee and warranty (see Section 5).

6.1.12 The vendor shall provide a statement of the fixed number of weeks required to effect shipment after receipt of the order and all engineering data. Separate times shall be stated for multiple shipments, as in the case of separate packages or assemblies or multiple units.

6.1.13 The vendor shall provide a statement of the promised time or times after placement of the order for transmittal of the contract data (see 6.2). This information shall be presented in the form of an explicit schedule.

6.1.14 The vendor shall provide an itemized list of the special tools included in the offering. The vendor shall list any metric items included in the offering.

● **6.1.15** The vendor shall provide a separate price for each optional test that is specified and a packaged price for all the tests specified on the data sheets.

● **6.1.16** When specified, the vendor shall provide an outline of all necessary special weather and winterizing protection required for the motor and its auxiliaries for start-up, operation, and idleness. The vendor shall quote separately the protective items he or she proposes to furnish.

6.1.17 The vendor shall provide catalog cut sheets or similar data that describe all the auxiliary equipment.

● **6.1.18** When specified, the vendor shall provide a statement of the rate for furnishing a competent erection supervisor, as well as an estimate of the length of time the supervisor's services will be required under normal conditions.

● **6.1.19** When specified, materials shall be identified in the proposal with their applicable, AISI, ANSI, ASTM, or ASME numbers, including the material grade. When no such designation is available, the vendor's material specification, giving physical properties, chemical composition, and test requirements, shall be included in the proposal.

6.2 Contract Data

6.2.1 GENERAL

6.2.1.1 The following paragraphs specify the information to be furnished by the vendor. These data shall be tabulated on the Vendor's Induction Motor Contract Data Sheets (see Appendix B) and forwarded to the address or addresses noted on the order.

6.2.1.2 The data shall be identified on transmittal letters and in title blocks or pages with the following information:

a. The purchaser or user's corporate name.
b. The job or project number.
c. The equipment name and item number.
d. The purchase order number.
● e. Any other identification specified in the purchase order.
f. The vendor's identifying shop order number, serial number, or other reference required to identify return correspondence completely.

6.2.1.3 The vendor shall complete the Vendor Drawing and Data Requirements form (see Appendix D), detailing the schedule for transmission of drawings, curves, and data as agreed to at the time of the order, as well as the number and type of copies required by the purchaser.

6.2.2 DRAWINGS

● **6.2.2.1** The purchaser will state in the inquiry and in the order the number of prints and/or reproducibles required and the times within which they are to be submitted by the vendor (see 6.1.12).

● **6.2.2.2** The purchaser will promptly review the vendor's drawings when they are received; however, this review shall not constitute permission to deviate from any requirements in the order unless specifically agreed upon in writing. After the drawings have been reviewed, the vendor shall furnish certified copies in the quantity specified. Drawings shall be clearly legible.

6.2.2.3 The drawings furnished shall contain enough information so that when they are combined with the manuals specified in 6.2.4, the purchaser may properly install, operate, and maintain the ordered equipment. As a minimum, the following details shall be provided:

a. Overall dimensions and weights for each separately installed piece. Maintenance clearances and weight-handling capability required for erection and maintenance shall be included.
b. The direction of rotation.
c. As applicable, the size, type, location, and identification of all the purchaser's connections, including power, control, and instrument wiring; supply and drain details for lubrication oil and cooling water; and inlet and discharge details for cooling or purge air, as well as frame vents and drains. Connections plugged by the vendor shall be identified.
d. When applicable, the make, size, and type of couplings.
e. A list of any special weather-protection and climatization features supplied by the vendor and required by the purchaser.

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f. A list of auxiliary or other equipment furnished by the vendor for mounting by the purchaser.

g. Rigging provisions for removal of parts that weigh more than 300 pounds (136 kilograms).

■ h. Complete information to permit adequate foundation design by the purchaser. This shall include the following items:

1. The size and location of foundation bolts.
2. The weight distribution for each bolt/subsoleplate location.
3. Any unbalanced forces or moments generated by the unit or units in the specified operating range.
4. The location of the center of gravity.
5. Foundation forces as a result of worst case transient conditions.

6.2.2.4 The vendor shall supply schematic diagrams and bills of materials for each auxiliary system in the vendor's scope of supply, including control systems, as well as dimensional outline drawings for accessories and instruments. The bills of materials shall include and identify all components by make, type, size, capacity rating, materials, and other data as applicable.

6.2.2.5 Each drawing and diagram shall have a title block in the lower right corner, showing certification, reference to all identification data specified in 6.2.1.2, the revision number and date, and the drawing title, all visible when the drawing is folded to 8½ × 11 inches (a metric equivalent size is acceptable). Bills of materials shall be similarly identified.

6.2.2.6 A complete list of vendor drawings shall be included with the first-issue major drawings. This list shall contain the titles and a schedule for transmission of all the drawings to be furnished by the vendor.

6.2.3 DATA

6.2.3.1 The vendor shall provide full information to enable completion of the "as designed" contract data sheets. The vendor should correct and fill out the contract data sheets and submit copies to the purchaser.

6.2.3.2 The vendor shall make the following information available to the purchaser:

a. Completed "as built" contract data sheets.

● b. A record of shop test data (which the vendor shall maintain for at least 5 years after the date of shipment). When specified, the vendor shall submit certified copies of the test data to the purchaser before shipment.

c. When specified, the calculated rotor-response curves.

d. The rotor-balance report.

e. Complete winding data with the instruction manuals. The data shall be sufficient to permit the owner to have a set of stator coils built if required. The data shall include the following:

1. Number of coils, winding connection, and throw.
2. Total copper weight, copper strand sizes, and details of both turn and ground wall insulation.
3. Turns per coil and number of parallels.
4. Length of iron including vents.
5. Stator bore diameter, slot depth and width, plus depth below wedge.
6. Finished coil dimensions in slot, plus details of semiconducting finish and stress or gradient paint treatment at the coil end turns, if any.

6.2.4 INSTRUCTION MANUALS

● **6.2.4.1** The number of manuals, the specific information, and the detail required for each purchase will be defined in the purchasing document included with the inquiry.

■ **6.2.4.2** The vendor shall provide written instructions and a cross-referenced list of all drawings to enable the purchaser to install, operate, and maintain the complete equipment ordered. This information shall be compiled in manuals with title pages containing section titles and complete lists of the included reference drawings by title and drawing number. If the instruction manuals apply to more than one model or series of equipment, the instructions shall clearly indicate the specific sections that apply to the equipment involved.

■ **6.2.4.3** The installation manual shall include any special information required for proper installation design and actual installation that is not on the drawings (which shall be compiled in a separate manual). This manual shall be forwarded at a time mutually agreed upon in the order but not later than the final issue prints. It shall contain information such as special alignment procedures, utility specifications (including quantity), and all installation design data.

■ **6.2.4.4** Operation and maintenance manuals shall be forwarded no more than 2 weeks after successful completion of all specified tests. If required, these manuals shall include a section of special instruction for operation at specified extreme environmental (such as temperature) conditions. The following items shall be included in the manual:

a. Instructions covering start-up, normal shutdown, emergency shutdown, operating limits, and routine operational procedures.

b. Outline and sectional drawing, schematics, and illustrative sketches in sufficient detail to identify all parts and to clearly show the operation of all equipment and components and the method of inspection and repair. Standardized sectional drawings are acceptable only if they represent the motor's actual construction.

● ■ **6.2.4.5** When specified, one complete set of photographs showing the assembly of the machine shall be provided. Each step of the bearing assembly shall be individually photographed.

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APPENDIX A—
PURCHASER'S INDUCTION MOTOR DATA SHEETS

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PURCHASER'S INDUCTION MOTOR DATA SHEET

JOB NO. _____ ITEM NO. _____
PAGE 1 OF 4 BY _____
DATE _____ REVISION _____

APPLICABLE TO: O PROPOSAL O PURCHASE O AS BUILT
FOR _____
SITE _____
SERVICE _____
VENDOR _____

MODEL _____
UNIT _____
DRIVEN EQUIPMENT _____
NO. REQUIRED _____
SERIAL NO. _____

NOTE: IN CASE OF CONFLICTS, THE ORDER OF PRECEDENCE SHALL BE THE PO, THESE DATA SHEETS AND THEN API541.

MOTOR DESIGN DATA

BASIC DATA:

TYPE OF MOTOR (1.1.2) O GENERAL PURPOSE O SPECIAL PURPOSE
O VOLTS _____ PHASE _____ HERTZ _____
O NAMEPLATE HP (2.2.1.1) _____ SERVICE FACTOR (2.2.1.3) _____
O SYNCHRONOUS RPM _____
O INSULATION CLASS AND TYPE _____
O TEMPERATURE RISE (2.2.1.3): _____ °C ABOVE _____ °C AMBIENT
AT _____ SF BY O RESISTANCE O RTD
O MIN % OVERSPEED (2.1.4, 2.4.5.2.7) _____
O BEARING TYPE: O HYDRODYNAMIC (2.4.7.1) O ANTIFRICTION (2.4.7.2)
O MAX DRIVEN-EQUIPMENT VERTICAL THRUST (2.4.7.2) (2.4.7.10) _____
_____
O DIFFERENTIAL PROTECTION TO BE APPLIED (2.2.2.1)

SITE DATA (2.1.2)

AREA CLASSIFICATION (2.1.7):
O CLASS _____ GROUP _____ DIVISION _____ O NONCLASSIFIED
O IGNITION TEMPERATURE _____ TEMP. ID NO. _____
O ELEVATION, FT _____
O AMBIENT TEMPERATURE MAX, °F _ MIN, °F _____
O RELATIVE HUMIDITY: MAX, % _____ MIN, % _____
O INDOOR O OUTDOOR O HEATED O UNHEATED
O ROOF OVER MOTOR O NO ROOF OVER MOTOR
O NONMASSIVE FOUNDATION (2.4.6.1.2), DESCRIPTION _____
_____
O MAX SOUND PRESSURE LEVEL (2.1.3) _____ DBA _____

UNUSUAL CONDITIONS:

O ABRASIVE DUST (2.4.1.2.2, ITEM c) _____
O EXTERNAL FORCES AND MOMENTS (2.4.4) _____
O SEISMIC LOADING (2.4.2.2) _____
O CORROSIVE AGENTS (2.4.10.1.2) _____
O OTHER _____

ELECTRIC SYSTEM CONDITIONS:

O MAX S.C. KVA AT MOTOR BUS (3.1.2) _____
O I^2t LET-THROUGH ENERGY (3.1.2) _____
O MIN S.C. KVA AT MOTOR BUS (2.2.5.2, ITEM b) _____
O X/R RATIO (2.2.5.2, ITEM b) _____
O TYPE OF SYSTEM GROUNDING O RESISTANCE O REACTANCE
O UNGROUNDED OSOLID
GROUND FAULT AMPERES _____

STARTING (2.2.3, 2.2.5.2):

O TORQUES IN EXCESS OF NEMA MG 1-20.41 (2.2.7 ITEM C) _____
O FULL VOLTAGE O % REDUCED VOLTAGE AND TYPE _____
O LOADED (100%) O PARTIALLY LOADED (%) _____
O OTHER _____

STARTING (2.2.3, 2.2.5.2) (CONTINUED):

O LOAD CURVE 2.2.3.2, ITEM a _____
O VOLTAGE AT LOCKED ROTOR (2.2.5.2, ITEM a)(MIN % OF RATED): _____
O NUMBER OF FULL-VOLTAGE STARTS, IF NOT 5000 (2.4.5.1.1): _____
O CAPABILITY IN EXCESS OF TABLE 2 (2.2.6.2)
SPECIFY _____

MOUNTING (2.4.2):

O HORIZONTAL O VERTICAL O SHAFT UP O SHAFT DOWN
O FOOT MOUNTED O FLANGE MOUNTED, NEMA TYPE _____
O BASEPLATE FURNISHED BY (2.4.2.6) _____
O SOLEPLATE FURNISHED BY (2.4.2.6) _____

ENCLOSURE (2.4.1):

O OPEN--DRIPPROOF
O WEATHER PROTECTED (2.4.1.2.2) O TYPE I O TYPE II
O TEFC (2.4.1.2.3) O TEWAC (2.4.1.2.4) O TEPV
O EXPLOSION PROOF O OTHER TYPE _____
O TEAAC-TUBES (2.4.10.8, ITEM a): O COPPER O COPPER ALLOY
O ALUMINUM O STAINLESS STEEL O ALUMINUM ALLOY
O AISI 300 SERIES HARDWARE (2.4.1.1, ITEM c) _____

TEWAC HEAT EXCHANGER:

O TUBE MATERIAL (2.4.10.8, ITEM b) _____
O TUBE CONSTRUCTION (2.4.1.2.4, ITEM d) O DOUBLE TUBE O SINGLE TUBE
O AIR TEMPERATURE SENSOR (2.4.1.2.4, ITEM g) O YES O NO
O RTD TYPE _____ O TC TYPE _____
O COOLING WATER CONDITIONS PER 2.4.1.2.4, ITEM a? O YES O NO
IF NO, SPECIFY DIFFERENCES _____
_____
O FLOW SENSOR LOCAL INDICATOR (2.4.1.2.4, ITEM f)
O OUTER TUBE ON DOUBLE TUBE COOLERS LEAK DETECTION
(2.4.1.2.4, ITEM b): TYPE _____

DRIVE SYSTEM:

O DIRECT CONNECTED O GEAR (2.2.3.2, ITEM c)
O TYPE OF COUPLING (2.4.9.4) _____
FURNISHED BY _____
O ROTATION REQ'D BY DRIVEN EQUIPMENT WHEN FACING MOTOR
OPPOSITE DRIVE END: O CLOCKWISE O COUNTERCLOCKWISE
O DRIVEN-EQUIPMENT Wk^2(2.2.3.2, ITEM c) _____ LB-FT^2@ _____ RPM
O LOAD REACCELERATION REQUIRED (2.2.3.2, ITEM b)? O YES O NO
IF YES, PROVIDE:
MAX VOLTAGE INTERRUPTION, (CYCLES) _____
VOLTAGE AT MOTOR TERMINALS ON RECLOSURE _____
LOAD SPEED-TORQUE REFERENCE CURVE NO. _____
O OTHER _____
_____

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MOTOR DESIGN DATA (CONTINUED)

RECIPROCATING COMPRESSOR LOAD:

O CRANK-EFFORT DIAGRAM CURVE NO. _____

MISCELLANEOUS:

O SPECIAL LOW-TEMPERATURE MATERIAL REQUIREMENTS (2.4.10.4): _____

O EVALUATION FACTOR (EF), $/kW (6.1.1) _____

O FEDERAL, STATE, OR LOCAL CODES OR SPECIAL RULES (2.1.7): _____

MISCELLANEOUS (CONTINUED):

O LIST OF MATERIALS TO BE IDENTIFIED WITH APPLICABLE ANSI, ASTM OR ASME NUMBERS (6.1.19): _____

O SPECIAL IDENTIFICATION (6.2.1.2, ITEM e): _____

ACCESSORY EQUIPMENT

WINDING TEMPERATURE DETECTORS (3.2.1, 3.5.5):

O RTD TYPE: NO./PHASE (3.2.1.2) _____
O RESISTANCE MATERIAL (3.2.1.1) _____ OHMS _____
O 3-WIRE O 2-WIRE GROUND 1 LEAD O YES O NO
O OTHER _____

BEARING TEMPERATURE DETECTORS (3.3):

O PROVISION ONLY O PER API 670 O MANUFACTURER'S STD
O RTD TYPE: MATERIAL _____ O OHMS _____
O OTHER _____
O TERMINAL HEAD OR BOX (3.1.11) _____
O BEARING HOUSING TEMPERATURE INDICATORS REQUIRED (2.4.7.19) _____

LUBRICATION SYSTEM:

O SELF LUBE O PRESSURE LUBE
O BEARING OIL RINGS FOR COASTDOWN REQUIRED (2.4.8.2)
O OIL MIST (2.4.7.14)
O BEARING CONSTANT-LEVEL SIGHT FEED OILERS REQUIRED (2.4.7.12):
TYPE _____
O SPECIAL SEALS FOR GAS PURGE (2.4.7.21) _____
O LUBE SYSTEM COMMON WITH DRIVEN EQUIPMENT (2.4.8.4, 2.4.8.5):
SUPPLIED BY _____
TYPE OF OIL (2.4.8.5) _____ PRESSURE (2.4.8.8.2) _____
O PRESSURE LUBE SYSTEM (2.4.8.7):
O PER API 614 O MANUFACTURER'S STD
O MAIN OIL PUMP REQ'D (2.4.8.8.3)
O STANDBY OIL PUMP REQ'D (2.4.8.8.5)
O FILTER ELEMENT (2.4.8.8.7):
MANUFACTURER _____ MODEL NO. _____
O DUPLEX OIL FILTERS REQ'D (2.4.8.8.7) O OIL COOLER REQ'D (2.4.8.8.6)
O OIL HEATER REQUIRED (2.4.8.8.8)

VIBRATION DETECTORS (3.8):

O NON CONTACTING PROBES (3.8.1)
O PROVISIONS ONLY (3.8.1) O INSTALLED
PROBE TYPE _____ MODEL _____
OSCILLATOR/DEMODULATOR MODEL _____
NO. PER BEARING O TWO (3.8.1) O FOUR (3.8.2)

VIBRATION DETECTORS (3.8) (CONTINUED):

O BEARING HOUSING SEISMIC SENSORS (3.8.3)
O PROVISIONS ONLY O INSTALLED
SENSOR TYPE _____ MODEL _____
OUTPUT TRANSDUCER MODEL _____
NO. PER BEARING _____ LOCATION (H,V,A) _____
O VIBRATION SWITCH (3.8.3) O MANUAL RESET O ELECTRIC RESET
SWITCH TYPE _____ MODEL _____

MAIN CONDUIT BOX

O BOX LOCATION _____ O OVERSIZED BOX (3.1.1) _____
O MAIN SUPPLY LEADS: CONDUCTOR SIZE _____ TYPE _____
INSULATION _____ NUMBER PER PHASE _____
ENTER FROM (3.1.4): O ABOVE O BELOW O SIDE (R/L) _____
CONDUIT TYPE _____ SIZE _ NUMBER _____
O CURRENT TRANSFORMERS FOR DIFFERENTIAL PROTECTION (3.1.8 & 3.6.3)
ACCURACY CLASS _____
MOUNTED BY _____
QUANTITY _____ RATIO _____
O SURGE CAPACITORS (3.6.2.1): O .25μF/PHASE O .50μF/PHASE
MOUNTED BY _____
O SURGE ARRESTERS (3.6.2.2):
MOUNTED BY _____ RATED KV _____
O PHASE CURRENT TRANSFORMERS (3.1.6, ITEM j) FOR LOCAL AMMETER:
ACCURACY CLASS _____
QUANTITY _____ RATIO _____
O VOLTAGE TRANSFORMERS (3.1.6, ITEM l) FOR LOCAL VOLTMETER:
ACCURACY CLASS _____
MOUNTED BY _____
QUANTITY _____ RATIO _____
FUSES REQUIRED _____
O BUSHING STUDS OR RECEPTACLES (3.1.6, ITEM h) _____
O SPACE FOR STRESS CONES (3.1.6, ITEM g) _____
O THERMAL INSULATION (3.1.6, ITEM a) _____
O SPACE HEATERS (3.1.6, ITEM b) _____
O VOLTS _____ PHASE _____ HERTZ _____
O MAX SHEATH TEMPERATURE, °C _____

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ACCESSORY EQUIPMENT (CONTINUED)

MAIN CONDUIT BOX: (CONTINUED)

- O BREATHERS (3.1.6, ITEM d) _____
- O DRAINS (3.1.6, ITEM c) _____
- O PROVISION FOR PURGING (3.1.6, ITEM e) _____
- O REMOVABLE LINKS (3.1.6, ITEM f) _____
- O SILVER-PLATED BUS JOINTS (3.1.6, ITEM k) _____
- O GROUND BUS (3.1.6 ITEM m)
- O FAULT WITHSTAND (3.1.2) _____
- O OTHER TERMINAL BOX REQUIREMENTS (3.1.6) _____

AIR FILTERS (3.5.4):

- O PROVISION ONLY - WPI (3.5.2) O REQUIRED - WPI OR WPII (3.5.4)
- FILTER TYPE _____
- O DIFFERENTIAL PRESSURE DEVICE REQUIRED (3.5.5):
- TYPE _____

SPACE HEATERS (3.4.1, 3.4.2):

- O VOLTS _____ PHASE _____ HERTZ _____
- O MAX SHEATH TEMPERATURE °C (3.4.3) _____

BEARING HEATERS (2.4.8.3, 2.4.8.8.8):

- O TYPE _____
- VOLTS _____ PHASE _____ HERTZ _____

MISCELLANEOUS

- O EPOXY GROUT TO BE USED (2.4.2.7.5) TYPE _____
- O NAMEPLATE MATERIAL (2.4.11.1): O STAINLESS STEEL O MONEL
- O SPECIAL NAMEPLATE FOR IDENTIFICATION (2.4.11.4): _____
- O QUANTITY OF SPECIAL TOOLS REQUIRED (2.1.11, 2.4.2.14) _____
- O PROOF OF NONSPARKING, CORROSION-RESISTANT FAN (2.4.10.6)
- O SHAFT SEALS OF ELECTRICALLY NON-CONDUCTING MATERIALS (2.4.7.21)
- O LOCATION OF AUX. JUNCTION BOXES _____
- O SWITCHES WITH CONTACTS RATED FOR DC (3.6.1) O YES O NO

OTHER _____

COUPLING (2.4.9.4)

- O TAPER FIT O CYLINDRICAL FIT
- O COUPLING SUPPLIED BY _____
- O VENDOR _____ MODEL _____
- O COUPLING DATA SHEET NUMBERS _____
- O MOUNTED BY: O MOTOR MFR O DRIVEN-EQUIPMENT MFR OPURCHASER
- O IDLING ADAPTER/MASS MOMENT SIMULATOR SUPPLIED BY (2.4.9.5) _____
- COUPLING TO BE USED FOR TEST (4.3.1.5) O CONTRACT O SHOP O NONE

PAINTING:

- O VENDOR'S STANDARD O EXTRA COAT AT SHIPMENT
- O USER'S SPECIAL PAINT (4.4.3.1) _____

SHIPMENT:

- O DOMESTIC O EXPORT
- O EXPORT BOXING REQUIRED
- O OUTDOOR STORAGE FOR MORE THAN 6 MONTHS (4.4.1) _____
- O SPECIAL SHIPPING BEARINGS (4.4.3.9)
- O MOUNTED ON SKID (4.4.3.6)

OTHER REQUIREMENTS:

- O ANY EXTERNAL FORCES ON MOTOR HOUSING THAT MAY AFFECT SITE PERFORMANCE (2.1.13, 2.4.4) _____
- O DATA REQUIRED ON CURRENT VARIATION AND EFFICIENCY FOR VARIABLE TORQUE LOAD (6.1.4, ITEM h)
- O VENDOR'S REPRESENTATIVE TO BE PRESENT DURING INITIAL EQUIPMENT ALIGNMENT CHECK (2.1.13)
- O SPECIAL WEATHER PROTECTION REQUIREMENTS (6.1.16)
- O VENDOR'S REVIEW AND COMMENTS ON THE PURCHASER'S FOUNDATION DRAWINGS
- O GUARANTEED EFF. AND PF REQUIRED (6.1.4, ITEM G) _____
- O ERECTION SUPERVISOR COSTS (6.1.8)
- O ASSEMBLY PHOTOGRAPHS REQUIRED (6.2.4.5)
- O OTHER _____

COMMENTS: _____

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ANALYSIS, SHOP INSPECTION, AND TESTS (4.1.3, 4.3.1.1, 6.1.15)

	REQUIRED	WITNESSED	OBSERVED	
LATERAL CRITICAL SPEED ANALYSIS (2.4.6.1.3) BY _____	O			1
DESIGN REVIEW MEETING	O			2
TORSIONAL ANALYSIS (2.4.6.1.4) BY _____	O			3
SHOP INSPECTION (4.1.1)	O			4
DATA PRESERVATION (4.2.1.1)	●			5
MATERIAL CERTIFICATIONS (4.2.1.1, ITEM a)	O			6
ASSEMBLY, MAINTENANCE AND RUNNING CLEARANCES (4.2.1.1, ITEM e)	O			7
PAINTING DEFERRED (4.2.1.2)	O			8
SURFACE & SUBSURFACE INSPECTION OF PARTS (4.2.2.1, 4.2.1.3, ITEM a)	O	O	O	9
RADIOGRAPHIC TEST (4.2.2.2) PARTS: _____	O	O	O	10
ULTRASONIC TEST (4.2.2.3) PARTS: _____	O	O	O	11
MAGNETIC PARTICLE TEST (4.2.2.4) PARTS: _____	O	O	O	12
LIQUID PENETRANT TEST (4.2.2.5) PARTS: _____	O	O	O	13
HARDNESS TEST (4.2.3.4) PARTS: _____	O			14
OPTIONAL ASTM TESTS TO BE PROPOSED BY VENDOR (2.4.10.1.4)	O	O	O	15
BALANCE IN MINIMUM OF 3 PLANES (2.4.6.2.1)	O	O	O	16
CHECK BALANCE WITH HALF COUPLING (2.4.6.2.4)	O	O	O	17
INSPECTION FOR CLEANLINESS PER API 614 (4.2.3.2, 4.2.3.3)	O	O	O	18
REVIEW OF QUALITY CONTROL PROGRAM (4.2.3.5)	O			
STATOR INSPECTION PRIOR TO VACUUM-PRESSURE IMPREGNATION (4.2.3.6 ITEM b)	O	O	O	20
RESIDUAL UNBALANCE TEST (4.2.3.6 ITEM a AND APPENDIX C)	O	O	O	21
ROUTINE TEST (4.3.2)	●	O	O	22
POLARIZATION INDEX (4.3.2.1, ITEM e) VOLTAGE O 1000V O 2500V O 5000V ●	O	O		23
VIBRATION RECORDING (4.3.3.13)	O	O	O	24
STATOR CORE TEST (4.3.4.1)	O	O	O	25
SPECIAL SURGE TEST ON SAMPLE COILS (4.3.4.2.1)	O	O	O	26
POWER FACTOR TIP-UP TEST (4.3.4.3) O WOUND STATOR O SACRIFICIAL COILS	O	O	O	27
SEALED WINDING CONFORMANCE TEST (4.3.4.4)	O	O	O	28
DC HIGH-POTENTIAL TEST (4.3.5.1.3)	O	O	O	29
COMPLETE TEST (4.3.5.1.1)	O	O	O	30
RATED ROTOR TEMPERATURE VIBRATION TEST WHEN COMPLETE TEST NOT SPECIFIED (4.3.5.2.1)	O	O	O	31
BEARING INSPECTION (4.3.5.1.2) AT COMPLETION OF TEST	O	O	O	32
UNBALANCE RESPONSE TEST (4.3.5.4)	O	O	O	33
BEARING HOUSING NATURAL FREQUENCY TEST (4.3.5.5)	O	O	O	34
CERTIFIED DATA PRIOR TO SHIPMENT (6.2.3.2, ITEM b)	O			35
TEST EQUIPMENT SUPPLIED BY PURCHASER (4.1.4) _____				36
TEST EQUIPMENT/DATA ACQUISITION ACCURACY TO BE DEMONSTRATED PRIOR TO TEST (4.3.1.13)				37
OTHER _____	O	O	O	38
_____	O	O	O	39
_____	O	O	O	40
_____	O	O	O	41
_____	O	O	O	42
_____	O	O	O	

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APPENDIX B—VENDOR'S DATA SHEETS

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VENDOR'S INDUCTION MOTOR PROPOSAL DATA
- REQUIRED WITH PROPOSAL -

JOB NO. _____ ITEM NO. _____
PAGE 1 OF 1 BY _____
DATE _____ REVISION _____
PROPOSAL NO. _____
CONTRACT NO. _____
PURCHASED BY _____
ULTIMATE USER _____

O VENDOR _____
O HORSEPOWER _____ FULL LOAD SPEED (RPM) _____
O FRAME NO. _____ ENCLOSURE _____
O LOCKED ROTOR:
O AMPS AT RATED VOLTAGE _____ O AMPS AT 80% VOLTAGE _____
O POWER FACTOR _____
O MOTOR ROTOR Wk^2 (6.1.4, ITEM c) _____
O FULL-LOAD TORQUE (FLT). FT-LB _____
O LOCKED-ROTOR TORQUE, % FLT _____
O PULL-UP TORQUE, % FLT _____
O BREAKDOWN TORQUE, % FLT _____
O SPEED-TORQUE (6.1.4, ITEM a) CURVE NO. _____
O SPEED-CURRENT (6.1.4, ITEM b) CURVE NO. _____
O ESTIMATED STARTING TIME AT _____ % VOLTAGE (6.1.4, ITEM d), CURVE NO. _____
O LOCKED-ROTOR WITHSTAND TIME, COLD (6.1.4, ITEM e) AT:
70% VOLTAGE, SEC _____ 80% VOLTAGE, SEC _____
90% VOLTAGE, SEC _____ 100% VOLTAGE, SEC _____
O LOCKED-ROTOR WITHSTAND TIME AT RATED TEMPERATURE (6.1.4, ITEM e) AT:
70% VOLTAGE, SEC _____ 80% VOLTAGE, SEC _____
90% VOLTAGE, SEC _____ 100% VOLTAGE, SEC _____
O DESIGN DATA (6.1.4, ITEM f) NO LOAD CURRENT _____ AMPS

LOAD	1/4	1/2	3/4	FULL	SF
AMPERES					
EFFICIENCY					
POWER FACTOR					
O GUAR. EFFICIENCY (6.1.4, ITEM g)					

IEEE METHOD _____
O BEARING TYPE _____

O SPECIAL WEATHER PROTECTION (6.1.16) _____
O UTILITY REQUIREMENTS (6.1.5):
O BEARING OIL REQUIREMENTS: TYPE OIL _____
GPM _____ PSI _____
O HEAT LOAD TO BE REMOVED, kW _____
O COOLING WATER REQUIREMENTS:
GPM _____ PSI _____
O VENTILATION AIR REQUIREMENTS:
SCFM _____ PSI _____
O RECIPROCATING COMPRESSOR LOAD:
O CRANK-EFFORT DIAGRAM CURVE NO. _____
O MAXIMUM CURRENT PULSATIONS, % FL (6.1.4, ITEM h) _____
O CALCULATED EFFICIENCY BASED ON CURRENT-PULSATION OPERATING CONDITION (6.1.4, ITEM h) _____
O NET WEIGHT (6.1.6) _____ ROTOR WEIGHT (6.1.6) _____
O SHIPPING WEIGHT (6.1.6) _____ MAX ERECTION WEIGHT (6.1.6) _____
O DIMENSIONS, FT/IN (6.1.7): AIR GAP _____
LENGTH _____ WIDTH _____ HEIGHT _____
O MAX SOUND PRESSURE LEVEL AT 1 METER, dB(A) (2.1.3) _____
O VIBRATION TEST USING RESPONSIVE AMPLIFICATION METHOD (4.3.3.11)
O WHEN ROTOR DYNAMIC ANALYSIS IS SPECIFIED, LIST OF FOUNDATION DATA REQUIRED FROM PURCHASER (2.4.6.1.2) _____
O LIST OF OPTIONAL ASTM TESTS PROPOSED BY VENDOR (2.4.10.1.4) _____
O ROTOR/SHAFT DETAIL (6.1.9)

COMMENTS: _____

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VENDOR'S INDUCTION MOTOR CONTRACT DATA SHEET

JOB NO. _____ ITEM NO. _____
PAGE 1 OF 2 BY _____
DATE _____ REVISION _____
PROPOSAL NO. _____
CONTRACT NO. _____

O AS DESIGNED O AS BUILT

GENERAL INFORMATION

FOR _____
SITE _____
SERVICE _____
VENDOR _____

MODEL _____
UNIT _____
DRIVEN EQUIPMENT _____
NO. REQUIRED _____
SERIAL NO. _____

MOTOR DATA

NAMEPLATE HP (2.2.1.1)_____ SERVICE FACTOR (2.2.1.3)_____
VOLTS _____ PHASE _____ HERTZ _____
FULL LOAD SPEED (RPM) _____
FRAME SIZE _____ MANUFACTURER'S TYPE _____
ENCLOSURE TYPE _____
ROTATION WHEN FACING MOTOR OPPOSITE DRIVE END:
O CLOCKWISE O COUNTERCLOCKWISE O BI DIRECTIONAL

LOAD	1/4	1/2	3/4	FULL	SF
AMPERES					
EFFICIENCY					
POWER FACTOR					

NO LOAD CURRENT _____
LOCKED ROTOR (CURRENT) AT RATED VOLTAGE _____
LOCKED ROTOR (CURRENT) AT ___% RATED VOLTAGE _____
ALLOWABLE TIME AT LOCKED ROTOR:
LOCKED-ROTOR WITHSTAND TIME, COLD (6.1.4, ITEM e) AT:
70% VOLTAGE, SEC _____ 80% VOLTAGE, SEC _____
90% VOLTAGE, SEC _____ 100% VOLTAGE, SEC _____
LOCKED-ROTOR WITHSTAND TIME AT RATED TEMPERATURE (6.1.4, ITEM e) AT:
70% VOLTAGE, SEC _____ 80% VOLTAGE, SEC _____
90% VOLTAGE, SEC _____ 100% VOLTAGE, SEC _____
THERMAL CAPABILITY CURVES;
CURVE NUMBERS: _____
INSULATION SYSTEM:
CLASS _____
TEMPERATURE RISE (2.2.1.3): _____ °C ABOVE _____ °C AMBIENT
AT _____ SF BY O RESISTANCE O RTD
INDUCTION MOTOR EQUIVALENT CIRCUIT DATA:
LOCKED ROTOR AT RATED VOLTAGE
STATOR R _____ STATOR X _____ MAGN X _____
ROTOR R _____ ROTOR X _____ MAGN R _____
RATED LOAD AT RATED VOLTAGE
STATOR R _____ STATOR X _____ MAGN X _____
ROTOR R _____ ROTOR X _____ MAGN R _____

PHASE-TO-PHASE RESISTANCE _____ OHM, @ _____ °C 1
SUBTRANSIENT REACTANCE X''_d _____ 2
TOTAL WINDING CAPACITANCE TO GROUND _____ μF 3
RESIDUAL VOLTAGE OPEN-CIRCUIT TIME CONSTANT: 4
MOTOR ONLY: _____ SECONDS 5
MOTOR WITH SURGE CAPACITORS: _____ SECONDS 6
BEARING, COUPLING END: 7
MANUFACTURERS' TYPE _____ PART NO. _____ 8
BEARING, OUTBOARD END: 9
MANUFACTURERS' TYPE _____ PART NO. _____ 10
BORE LENGTH _____ BORE DIAM _____ MIN, _____ MAX 11
DESIGN CLEARANCE WITH SHAFT: _____ MIN, _____ MAX 12
BEARING LOADING: _____ PSI 13
THRUST BEARING, TOP (BOTTOM): 14
MANUFACTURERS' TYPE _____ PART NO. _____ 15
BEARING LUBRICATION: O SELF O PRESSURE O FLOOD 1
PRESSURE/FLOOD LUBRICATION: 1
OIL _____ GPM @ _____ PSIG _____ KW LOSSES 18
COOLING WATER _____ GPM @ _____ PSIG 19
TORQUE % OF RATED: 20
LOCKED ROTOR _____ PULL UP _____ 21
BREAKDOWN _____ 22
ROTOR Wk^2: _____ LB-FT^2 @ _____ RPM 23
TEST REPORT REFERENCE NO. _____ 24
MOTOR OUTLINE DRAWING NO. _____ 25
STARTING PERFORMANCE THERMAL LIMIT CURVE NO. _____ 26
SPEED TORQUE, SPEED CURRENT, SPEED TIME, SPEED PF 27
NET WEIGHT _____ ROTOR WEIGHT _____ 28
SHIPPING WEIGHT _____ MAX ERECTION WEIGHT _____ 29
MAX MAINTENANCE WEIGHT _____ 30
AREA CLASSIFICATION: 31
CLASS _____ GROUP _____ DIVISION _____ 32
MAX SOUND PRESSURE LEVEL AT 1 METER, dBA _____ 33
34
35

ACCESSORIES

36

WINDING TEMPERATURE DETECTOR: 37
RESISTANCE RATING, OHM/MATERIAL _____ 38
O 2-WIRE SYSTEM O 3-WIRE SYSTEM
RECOMMENDED ALARM SETTING, °C _____ 39
RECOMMENDED TRIP SETTING, °C _____ 40
THERMAL TEMPERATURE SWITCH CONTACTS: 41
O NORMALLY OPEN O NORMALLY CLOSED 42
OPENING SETTING, °C _____ CLOSED SETTING, °C _____

BEARING TEMPERATURE DETECTOR:
O RTD _____ OHMS _____ MATERIAL _____ OTHER _____ 43
RECOMMENDED ALARM SETTING, °C _____ 44
RECOMMENDED TRIP SETTING, °C _____ 45
VIBRATION DETECTORS (3.8): 46
NON CONTACTING PROBES (3.8.1)
O PROVISIONS ONLY (3.8.1) O INSTALLED
PROBE TYPE _____ MODEL _____
OSCILLATOR/DEMODULATOR MODEL _____
NO. PER BEARING O TWO (3.8.1) O FOUR (3.8.2)

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VENDOR'S INDUCTION MOTOR CONTRACT DATA SHEET

JOB NO. _____ ITEM NO. _____
PAGE 2 OF 2 BY _____
DATE _____ REVISION _____
PROPOSAL NO. _____
CONTRACT NO. _____

O AS DESIGNED O AS BUILT

ACCESSORIES (CONTINUED)

BEARING HOUSING SEISMIC SENSORS (3.8.3)
O PROVISIONS ONLY O INSTALLED
SENSOR TYPE _____ MODEL _____
OUTPUT TRANSDUCER MODEL _____
NO. PER BEARING _____ LOCATION (H,V,A) _____
VIBRATION SWITCH (3.8.3) O MANUAL RESET O ELECTRIC RESET
SWITCH TYPE _____ MODEL _____
SPACE HEATERS FOR MAIN CONDUIT BOX:
VOLTAGE, V _____ O 1-PHASE O 3-PHASE
WATTAGE, kW _____ MAX SURFACE TEMPERATURE, °C ____
AIR FILTERS: MANUFACTURER _____
TYPE _____

DIFFERENTIAL PRESSURE SWITCH
TYPE _____ SETTING _____
CURRENT TRANSFORMERS:
DIFFERENTIAL CLASS _____ RATIO _____ NO. MOUNTED _____
PHASE CLASS _____ RATIO _____ NO. MOUNTED _____
POTENTIAL TRANSFORMERS:
CONNECTION _____ RATIO _____ NO. MOUNTED _____
FUSES _____ SIZE _____ NO MOUNTED _____
SURGE PROTECTION:
LIGHTNING ARRESTER TYPE _____ VOLTAGE _____
CAPACITOR TYPE _____ CAPACITANCE _____

PRESSURE LUBE SYSTEM

MAIN PUMP: O INTEGRAL O SEPARATE
IF SEPARATE: O VERTICAL O HORIZONTAL
TYPE OF DRIVE O STEAM TURBINE O MOTOR
STANDBY PUMP: O VERTICAL O HORIZONTAL
TYPE OF DRIVE: O STEAM TURBINE O MOTOR
TURBINE DRIVER PRESSURE, PSIG _____

TURBINE DRIVER TEMPERATURE RANGE, °F _____
MOTOR VOLTS _____ PHASE _____ HERTZ _____
FILTERS: O SINGLE O DUPLEX
PRESSURE CONTROLLER NO. _____
RELIEF VALVE NO. _____ SETTING, PSIG _____
COOLERS: O TWIN O SINGLE
CONSTRUCTION MATERIALS _____

AIR COOLERS

NUMBER OF SECTIONS _____ LOCATION _____
SECTION RATING, % OF FULL LOAD _____
CONSTRUCTION MATERIAL DATA SHEET REFERENCED:
O AIR/AIR _____ O AIR/WATER _____
LEAK DETECTOR TYPE _____
AIR FLOW SCFM _____ PSIG _____
WATER PRESSURE, PSIG: MAX _____ MIN _____

WATER FLOW, GPM: MAX _____ MIN _____
FLOW SENSING DEVICE TYPE _____ VOLTAGE _____
AUXILIARY RELAY CONTACTS:
O NORMALLY OPEN O NORMALLY CLOSED
HIGH-FLOW ALARM SETTING, GPM _____
LOW-FLOW SHUTDOWN SETTING, GPM _____

OTHER INFORMATION

DRAWING REFERENCES _____
ACCESSORIES OUTLINE _____
WIRING DIAGRAM _____
SPECIAL WEATHER PROTECTION _____
WINDING / ROTOR REPAIR DATA:
ALL DIMENSION UNITS ARE: _____
COIL INFORMATION: TOTAL COPPER WEIGHT ___LBS
COPPER STRAND SIZE _____INSUL _____
NUMBER PER COIL TURN_____
TURN INSULATION DESCRIPTION _____
NUMBER OF TURNS PER COIL _____
COIL GROUND-WALL INSULATION DESCRIPTION _____
_____
FINISHED COIL DIMENSION IN SLOT REGION, W x H x L -
(ALSO GIVE TOLERANCES): _____
_____
TOTAL NUMBER OF STATOR SLOTS _____

WINDING THROW _____
WINDING CONNECTION _____
SLOT FILLER DESCRIPTION / THICKNESS: TOP _____
SIDE _____, BETWEEN TOP / BOTTOM COILS _____
SPECIAL END-TURN BRACING _____
OTHER INFO: _____
_____
ROTOR INFORMATION: NUMBER OF ROTOR BARS _____
BAR ALLOY _____LENGTH_____
CROSS-SECT DIMENSION, (ALSO GIVE TOLERANCES):
_____
SHORT CIRCUIT END-RING ALLOY _____
ROUGH FORGING OR CAST DIMENSIONS
OD _____ID _____THICKNESS _____
NON-MAGNETIC RETAINING RING ALLOY _____
_____
SHAFT MATERIAL _____
GRADE OF CASTINGS_____

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APPENDIX C—PROCEDURE FOR DETERMINATION OF RESIDUAL UNBALANCE

C.1 Scope

This appendix describes the procedure to be used to determine residual unbalance in machine rotors. Although some balancing machines may be set up to read out the exact amount of unbalance, the calibration can be in error. The only sure method of determining residual unbalance is to test the rotor with a known amount of unbalance.

C.2 Definition

Residual unbalance is the amount of unbalance remaining in a rotor after balancing. Unless otherwise specified, it shall be expressed in ounce-inches or gram-millimeters.

C.3 Maximum Allowable Residual Unbalance

C.3.1 The maximum allowable residual unbalance per plane shall be calculated using Equation 1 in 2.4.6.2.5 of this standard.

C.3.2 If the actual static weight load on each journal is not known, assume that the total rotor weight is equally supported by the bearings. For example, a two-bearing rotor weighing 2720 kilograms (6000 pounds) would be assumed to impose a static weight load of 1360 kilograms (3000 pounds) on each journal.

C.4 Residual Unbalance Check

C.4.1 GENERAL

C.4.1.1 When the balancing-machine readings indicate that the rotor has been balanced to within the specified tolerance, a residual unbalance check shall be performed before the rotor is removed from the balancing machine.

● **C.4.1.2** To check residual unbalance, a known trial weight is attached to the rotor sequentially in 6 (or 12, if specified by the purchaser) equally spaced radial positions, each at the same radius. The check is run in each correction plane, and the readings in each plane are plotted on a graph using the procedure specified in C.4.2.

C.4.2 PROCEDURE

C.4.2.1 Select a trial weight and radius that will be equivalent to between one and two times the maximum allowable residual unbalance [that is, if U_{max} is 1440 gram-millimeters (2 ounce-inches), the trial weight should cause 1440–2880 gram-millimeters (2–4 ounce-inches) of unbalance.]

C.4.2.2 Starting at the last known heavy spot in each correction plane, mark off the specified number of radial positions (6 or 12) in equal (60 degree or 30 degree) increments around the rotor. Add the trial weight to the last known heavy spot in one plane. If the rotor has been balanced very precisely and the final heavy spot cannot be determined, add the trial weight to any one of the marked radial positions.

C.4.2.3 To verify that an appropriate trial weight has been selected, operate the balancing machine and note the units of unbalance indicated on the meter. If the meter pegs, a smaller trial weight should be used. If little or no meter reading results, a larger trial weight should be used. Little or no meter reading also generally indicates that the rotor was not balanced precisely enough or that the balancing machine is not sensitive enough. If this occurs, the balancing machine can be checked for sensitivity by using the procedure outlined in C.5 and Figure C-1. A completed example is shown in Figure C-2.

C.4.2.4 Locate the weight at each of the equally spaced positions in turn, and record the amount of unbalance indicated on the meter for each position. Repeat the initial position as a check. All verification shall be performed using only one sensitivity range on the balance machine.

C.4.2.5 Plot the readings on the residual unbalance work sheet and calculate the amount of residual unbalance (see Figure C-3). The maximum meter reading occurs when the trial weight is added at the rotor's heavy spot; the minimum reading occurs when the trial weight is opposite the heavy spot. Thus, the plotted readings should form an approximate circle (see Figure C-4). An average of the maximum and minimum meter readings represents the effect of the trial weight. The distance of the circle's center from the origin of the polar plot represents the residual unbalance in that plane.

C.4.2.6 Repeat the steps described in C.4.2.1 through C.4.2.5 for each balance plane. If the specified maximum allowable residual unbalance has been exceeded in any balance plane, the rotor shall be balanced more precisely and checked again. If a correction is made in any balance plane, the residual unbalance check shall be repeated in all planes.

C.5 Balancing Machine Sensitivity Check

C.5.1 Using the maximum allowable residual unbalance (U_{max}) determined for the specific rotor being tested, prepare trial weights for unbalance factors of $\frac{1}{2}U_{max}$, U_{max}, $2U_{max}$, and $4U_{max}$.

C.5.2 Sequentially install each trial weight at the phase angle of unbalance for the rotor being tested. Record the balancing machine readings on the form shown in Figure C-1.

C.5.3 Plot the readings on the graph shown in Figure C-1. Then draw a best-fit straight line through the four points. If the line intersects the vertical axis below $\frac{1}{2}U_{max}$, the balancing machine is sensitive enough and the rotor should be rebal-

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API STD*541 95 ■ 0732290 0545817 273 ■

anced using the residual unbalance verification procedure. (See Figure C-2 for an example plot.) If the line intersects the vertical axis above ½U_{max}, the balancing machine is not sensitive enough and the results are unacceptable.

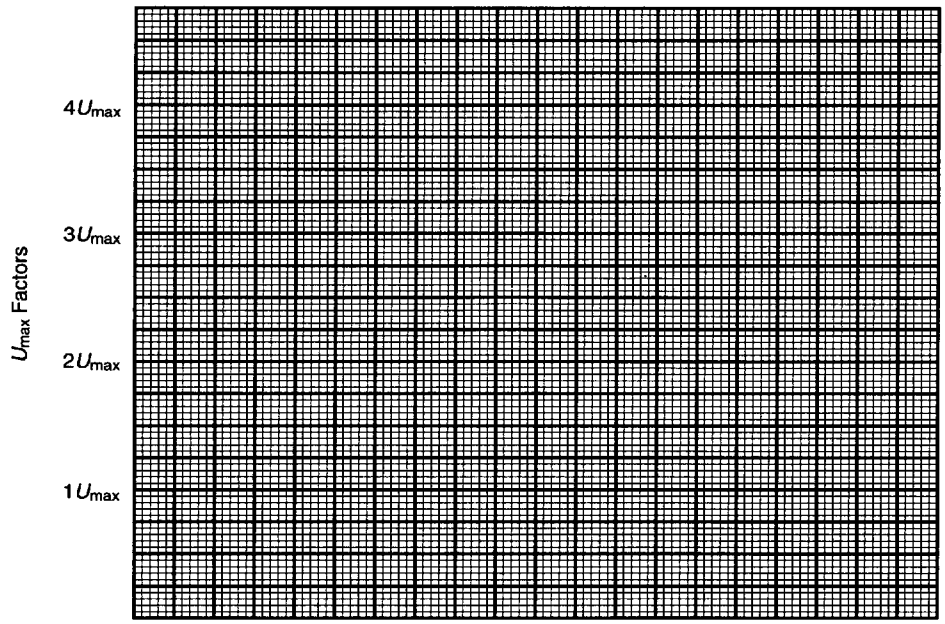
C.5.4 The following three options are available when the results of the balancing machine sensitivity check are unacceptable:

a. Have the balancing machine serviced in accordance with the manufacturer's procedure and repeat the sensitivity check.

b. Use a more sensitive balancing machine.

c. Perform the sensitivity check at the balancing machine's highest allowable speed. This speed must then be used when the rotor is balanced.

U_{max} Factors	Balancing Machine Readout[a]
$^1/_2U_{max}$	
$1U_{max}$	
$2U_{max}$	
$4U_{max}$	

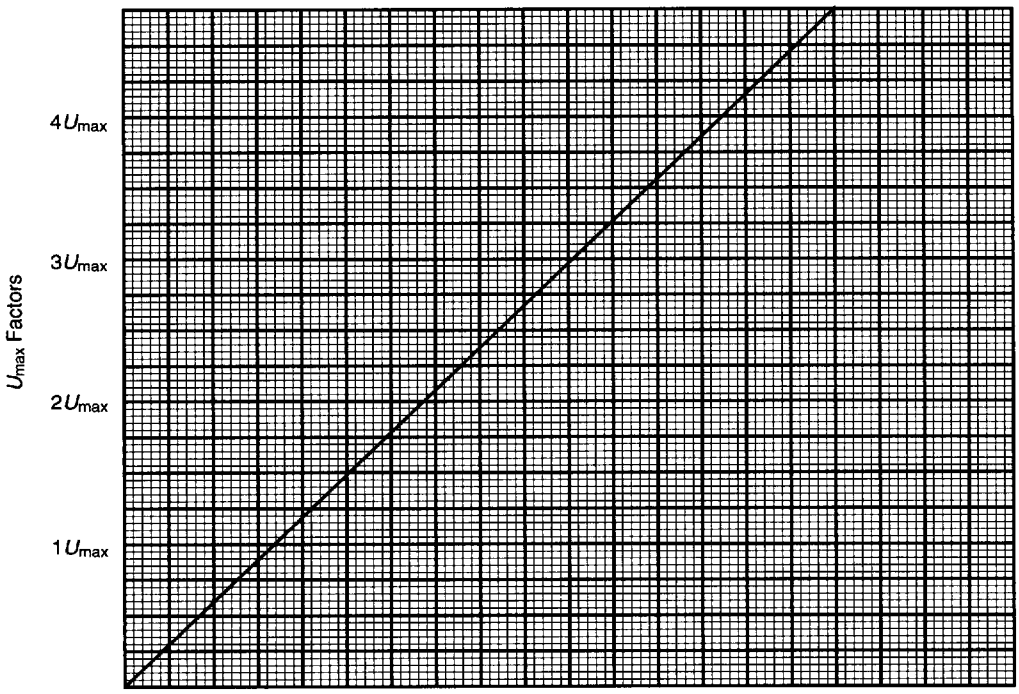


[a]Use balancing machine readouts to scale graph.

Figure C-1—Sensitivity Check Worksheet

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U_{max} Factors	Balancing Machine Readout[a]
$^1/_2U_{max}$	*5.1*
$1U_{max}$	*10.1*
$2U_{max}$	*20.1*
$4U_{max}$	*40.1*



[a]Use balancing machine readouts to scale graph.

Figure C-2—Sample Sensitivity Check Worksheet

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Equipment (Rotor) No.: _____

Purchase Order No.: _____

Correction Plane (inlet, drive-end, etc.—use sketch): _____

Balancing Speed: _____ rpm

N—Maximum Allowable Rotor Speed: _____ rpm

W—Weight of Journal (closest to this correction plane): _____ kg (lbs)

U_{max} = Maximum Allowable Residual Unbalance =
6350W/N (4W/N)
6350 x _____ kg/ _____ rpm; 4 x _____ lbs/_____rpm _____ gm-mm (oz.-in.)

Trial unbalance (2 x U_{max}) _____ gm-mm (oz.-in.)

R—Radius (at which weight will be placed): _____ mm (inches)

Trial Unbalance Weight = Trial Unbalance/R
_____ gm-mm/ _____ mm; _____ oz.-in./ _____ inches _____ gm (oz.)

Conversion information: 1 ounce = 28.350 grams

Test Data

Position	Amplitude	Amplitude
1		
2		
3		
4		
5		
6		
7		

Rotor Sketch

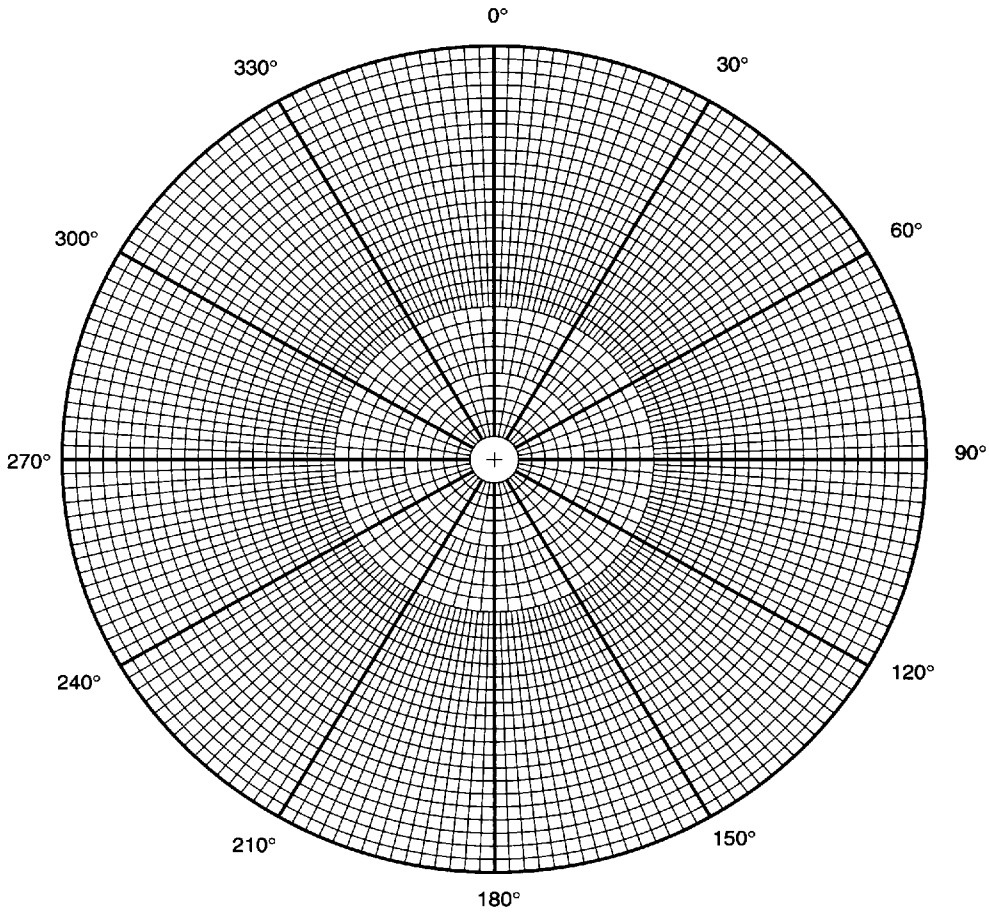
Test Data—Graphic Analysis

Step 1: Plot data on the polar chart (Figure C-3 continued). Scale the chart so the largest and smallest amplitude will fit conveniently.
Step 2: With the compass, draw the best fit circle through the six points and mark the center of this circle.
Step 3: Measure the diameter of the circle in units of scale chosen in Step 1 and record. _____ units
Step 4: Record the trial unbalance from above. _____ gm-mm (oz.-in.)
Step 5: Double the trial unbalance in Step 4 (may use twice the actual residual unbalance). _____ gm-mm (oz.-in.)
Step 6: Divide the answer in Step 5 by the answer in Step 3. _____ Scale Factor

You now have a correlation between the units on the polar chart and the gm-in. of actual balance.

Figure C-3—Residual Unbalance Worksheet

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The circle you have drawn must contain the origin of the polar chart. If it doesn't, the residual unbalance of the rotor exceeds the applied test unbalance. Proceed with the balancing machine sensitivity check before rebalancing is attempted.

If the circle does contain the origin of the polar chart, the distance between origin of the chart and the center of your circle is the actual residual unbalance present on the rotor correction plane. Measure the distance in units of scale you choose in Step 1 and multiply this number by the scale factor determined in Step 6. Distance in units of scale between origin and center of the circle times scale factor equals actual residual unbalance

Record actual residual unbalance _____ (gm-mm)(oz.-in.)

Record allowable residual unbalance (from Figure C-3) _____ (gm-mm)(oz.-in.)

Correction plane _____ for Rotor No. _____ (has/has not) passed.

By _____ Date _____

Figure C-3—Continued

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Equipment (Rotor) No.: C-101

Purchase Order No.: _____

Correction Plane (inlet, drive-end, etc.—use sketch): A

Balancing Speed: 800 rpm

N—Maximum Allowable Rotor Speed: 10,000 rpm

W—Weight of Journal (closest to this correction plane): 908 ~~kg~~ (lbs)

U_{max} = Maximum Allowable Residual Unbalance =
6350W/N (4W/N)
~~6350 x _____ kg/ _____ rpm;~~ 4 x 908 lbs/ 10,000 rpm: 0.36 ~~gm-mm~~ (oz.-in.)

Trial unbalance (2 x U_{max}): 0.72 ~~gm-mm~~ (oz.-in.)

R—Radius (at which weight will be placed): 6.875 ~~mm~~ (inches)

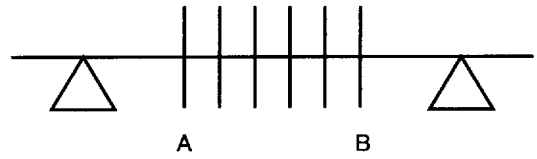
Trial Unbalance Weight = Trial Unbalance/R
~~_____ gm-mm/ _____ mm;~~ 0.72 oz.-in./ 6.875 inches: 0.10 ~~gm~~ (oz.)

Conversion information: 1 ounce = 28.350 grams

Test Data

Position	Amplitude	Amplitude
1	*0*	*21.0*
2	*60*	*24.0*
3	*120*	*22.5*
4	*180*	*13.5*
5	*240*	*10.5*
6	*300*	*19.5*
7	*0*	*21.0*

Rotor Sketch



C-101

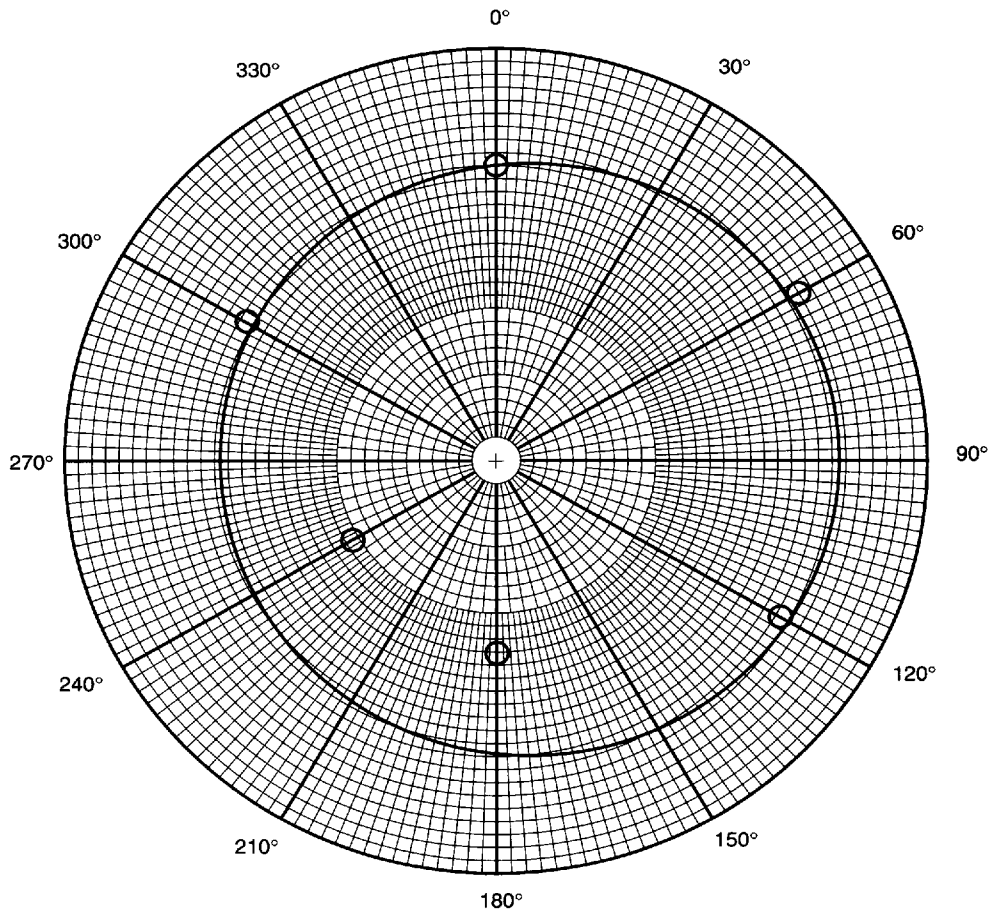
Test Data—Graphic Analysis

Step 1: Plot data on the polar chart (Figure C-4 continued). Scale the chart so the largest and smallest amplitude will fit conveniently.
Step 2: With the compass, draw the best fit circle through the six points and mark the center of this circle.
Step 3: Measure the diameter of the circle in units of scale chosen in Step 1 and record. 35 units
Step 4: Record the trial unbalance from above. 0.72 ~~gm-mm~~ (oz.-in.)
Step 5: Double the trial unbalance in Step 4 (may use twice the actual residual unbalance). 1.44 ~~gm-mm~~ (oz.-in.)
Step 6: Divide the answer in Step 5 by the answer in Step 3. 0.041 Scale Factor

You now have a correlation between the units on the polar chart and the gm-in. of actual balance.

Figure C-4—Sample Calculations for Residual Unbalance

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The circle you have drawn must contain the origin of the polar chart. If it doesn't, the residual unbalance of the rotor exceeds the applied test unbalance. Proceed with the balancing machine sensitivity check before rebalancing is attempted.

If the circle does contain the origin of the polar chart, the distance between origin of the chart and the center of your circle is the actual residual unbalance present on the rotor correction plane. Measure the distance in units of scale you choose in Step 1 and multiply this number by the scale factor determined in Step 6. Distance in units of scale between origin and center of the circle times scale factor equals actual residual unbalance

Record actual residual unbalance 6.5 (0.041) = 0.27 ~~(gm-mm)~~(oz.-in.)

Record allowable residual unbalance (from Figure C-4) 0.36 ~~(gm-mm)~~(oz.-in.)

Correction plane A for Rotor No. C-101 (has/~~has not~~) passed.

By John Inspector Date 11/16/89

Figure C-4—Continued

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APPENDIX D—VENDOR DRAWING AND DATA REQUIREMENTS

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INDUCTION MOTOR VENDOR DRAWING AND DATA REQUIREMENTS

JOB NO. _____ ITEM NO. _____
PURCHASE ORDER NO. _____ DATE _____
REQUISITION NO. _____ DATE _____
INQUIRY NO. _____ DATE _____
PAGE 1 OF 3 BY _____
FOR _____ REVISION _____
SITE _____ UNIT _____
SERVICE _____ NO. REQUIRED _____

Proposal[a] Bidder shall furnish _____ copies of data for all items indicated by an X.

Review[b] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.

Final[c] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.
Vendor shall furnish _____ operating and maintenance manuals.

DISTRIBUTION RECORD

Final—Received from vendor
Final—Due from vendor[c]
Review—Returned to vendor
Review—Received from vendor
Review—Due from vendor[c]

			DESCRIPTION					
			A.1 Dimensioned outline drawings with major and minor connections.					
			a. Primary equipment.					
			b. Auxiliary Equipment.					
			c. Maintenance weights.					
			d. Size of shipping sections.					
			e. Sole plates.					
			f. Heat exchanges.					
			g.					
			h.					
			i.					
			j.					
			A.2 Foundation loading diagrams.					
			a.					
			b.					
			c.					
			A.3 Schematic wiring and/or flow diagrams.					
			a. Speed sensor.					
			b. Space heaters.					
			c. Locked rotor protection package.					
			d. Cooling and exchanger.					
			e. Lubrication (if applicable).					
			f. Vibration monitoring.					
			g. Temperature sensors.					
			h. Differential current transformers.					
			i. Phase current transformers.					
			j.					
			k.					
			B.1 Detail drawings and cross-sectional drawings.					
			a. Shaft and details.					
			b.					
			c.					

[a]Proposal drawings and data do not have to be certified or as built. Typical data shall be clearly identified as such.
[b]Purchaser will indicate in this column the time frame for submission of materials using the nomenclature given at the end of this form.
[c]Bidder shall complete these two columns to reflect his actual distribution schedule and include this form with his proposal.

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INDUCTION MOTOR VENDOR DRAWING AND DATA REQUIREMENTS

JOB NO. _____ ITEM NO. _____

PAGE 2 OF 3 BY _____

REVISION NO. _____ DATE _____

Proposal[a] Bidder shall furnish _____ copies of data for all items indicated by an X.

Review[b] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.

Final[c] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.
Vendor shall furnish _____ operating and maintenance manuals.

DISTRIBUTION RECORD

Final—Received from vendor
Final—Due from vendor[c]
Review—Returned to vendor
Review—Received from vendor
Review—Due from vendor[c]

			DESCRIPTION					
			B.2 Erection/assembly drawings.					
			a.					
			b.					
			c.					
			B.3 Calculations-torsional/lateral response.					
			a.					
			b.					
			c.					
			B.4 Predicted performance curves.					
			a. Power factor versus speed.					
			b. Motor and rotor heating (I^2T).					
			c. Torque versus speed at rated voltage.					
			d. Torque versus speed at ___ percent voltage.					
			e. Current versus speed at rated voltage.					
			f. Current versus speed at ___ percent voltage.					
			g. Expected efficiency.					
			h. Acceleration time curves.					
			i.					
			j.					
			k.					
			C.1 Vendor's data reports (as built).					
			a. X_m					
			b. R_s					
			c. R__					
			d. X_s					
			e. X__					
			f. X__					
			g. T__					
			C.2 Performance test reports.					
			C.3 Installation, maintenance, and operating instructions.					

[a]Proposal drawings and data do not have to be certified or as built. Typical data shall be clearly identified as such.
[b]Purchaser will indicate in this column the time frame for submission of materials using the nomenclature given at the end of this form.
[c]Bidder shall complete these two columns to reflect his actual distribution schedule and include this form with his proposal.

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INDUCTION MOTOR VENDOR DRAWING AND DATA REQUIREMENTS

JOB NO. _____ ITEM NO. _____
PAGE 3 OF 3 BY _____
DATE _____ REVISION _____

Proposal[a] Bidder shall furnish _____ copies of data for all items indicated by an X.

Review[b] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.

Final[c] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.
Vendor shall furnish _____ operating and maintenance manuals.

DISTRIBUTION RECORD

Final—Received from vendor
Final—Due from vendor[c]
Review—Returned to vendor
Review—Received from vendor
Review—Due from vendor[c]

			DESCRIPTION					
			D.1 Recommended spare parts list.					
			a. Construction.					
			b. Start-up.					
			c. Maintenance (2-year).					
			E.1 Complete set of Assembly Photographs (see 6.2.4.5)					

[a]Proposal drawings and data do not have to be certified or as built. Typical data shall be clearly identified as such.
[b]Purchaser will indicate in this column the time frame for submission of materials using the nomenclature given at the end of this form.
[c]Bidder shall complete these two columns to reflect his actual distribution schedule and include this form with his proposal.

Notes:
1. Send all drawings and data to _____
2. All drawings and data must show project, appropriation, purchase order, and item numbers in addition to the plant location and unit. In addition to the copies specified above, one set of the drawings/instructions necessary for field installation must be forwarded with the shipment.

Nomenclature:
_____ S—number of weeks prior to shipment.
_____ F—number of weeks after firm order.
_____ D—number of weeks after receipt of approved drawings.

Vendor _____
Date _____ Vendor Reference _____
Signature _____
(Signature acknowledges receipt of all instructions)

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APPENDIX E—DATA SHEET GUIDE

The material in this appendix offers a guide to the information requested by the data sheets in Appendixes A and B.

E.1 Purchaser's Data Sheets (Appendix A)

The data sheets in Appendix A represent the purchaser's induction motor data sheet. They request information on the motor design, on accessory equipment, and on analysis, shop inspection, and tests.

E.1.1 GENERAL INFORMATION

"Applicable to: Proposal, Purchase, or As Built"—Check "proposal" when the data sheet is sent out for quotation, "purchase" when an order is placed, and "as built" to reflect the completed data sheet after all design details and changes during the manufacture of the motor have been completed.

"For"—This indicates the name of the plant (e.g., Plant 18) or the equipment train identification.

"Site"—This lists the plant name or other identification of the location.

"Service"—This describes the function of the equipment to be driven, such as "boiler feedwater," "FCC blower," or "Alky plant feed pump."

"Vendor"—This refers to the manufacturer of the motor.

"Model"—This is the vendor's model identification number.

"Unit"—This is the motor identification number, such as "MP-501."

"Driven Equipment"—This is the driven equipment identification number.

"No. Required"—This is the quantity of motors required.

"Serial No."—This is the manufacturer's serial number of the motor as assigned by the motor manufacturer.

E.1.2 MOTOR DESIGN DATA

E.1.2.1 Data on Page 1, Left Column, Line 2

"Type of Motor"—This data sheet covers all induction motors generally rated 2300 volts and above and rated 250 HP and larger. See 1.1.1, Note 2, of API Standard 541 for a discussion of what "special purpose" means. This data sheet may also be applied to induction generators with appropriate attention to the requirements of such applications.

E.1.2.2 Data on Page 1, Left Column, Line 3

"Volts, Phase, Hertz"—Normally specified voltages are 2300, 4000, or 13200 supplied by 60 Hz, three-phase power systems of 2400, 4160, and 13800 volts. Occasionally, a motor voltage of 460 volts might be specified for a motor of "form-wound" winding construction supplied by a 480 volts power system.

E.1.2.3 Data on Page 1, Left Column, Line 4

"Nameplate HP and Service Factor"—This refers to the horsepower output rating of the motor, where known. Add a note when this is to be determined by the driven-equipment vendor. A service factor of 1.0 is recommended. If output beyond the 1.0 service factor output rating is required, the next higher motor rating should be chosen. This is primarily done to assure adequate torque margin for the motor (accelerating, pull-up, and breakdown torques). If a 1.15 service factor is specified, API Standard 541 limits the winding temperature rise to 10°C above Class B rise. This ensures long insulation life when applied to a Class F winding insulation system.

E.1.2.4 Data on Page 1, Left Column, Line 5

"Synchronous RPM"—This corresponds to the no-load speed of the motor. Available speeds on 60 Hz power systems include 3600, 1800, 1200, 900, and 720 RPM. Lower speeds are available but rarely used for induction motor application. Synchronous speed can be calculated using the following: Speed in RPM = 120f/p, where "f" is the electrical frequency and "p" is the number of poles in the motor (2, 4, 6, 8, ...).

E.1.2.5 Data on Page 1, Left Column, Line 6

"Insulation Class and Type"—This pertains to the winding electrical insulation. This standard specifies "Class F" insulation as a minimum. It is the standard of the industry and is rated for 155°C operation. For "Type," this standard specifies "VPI Epoxy." For decades, the VPI insulation system has been proven as a reliable insulation system. Epoxy is used in preference over polyester due to its better mechanical and chemical-resistance characteristics. Polyester resins are commonly used in the motor repair business due to their longer "tank" life. Another "type" of insulation is the "resin-rich" system. The resin-rich system has the epoxy catalyst and resin present in the insulation tape and the insulation cures upon baking in an oven. This system is available from several manufacturers. It is acceptable, but is the second choice after the VPI system.

E.1.2.6 Data on Page 1, Left Column, Lines 7–8

"Temperature Rise"—This is the increase in temperature of the windings permitted over ambient air temperature. There are two methods to determine the temperature rise: by RTD

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API STD*541 95 ■ 0732290 0545826 286 ■

and by measurement of the resistance change (RESISTANCE), with temperature, of the winding itself. Use Table E-1 to fill in the blank. Temperature rise "by RTD" is recommended when the motor is equipped with RTDs. If not equipped with RTDs, specify "by resistance."

The Class B rise given above is specified by this standard even though the insulation system is rated for the higher Class F temperature. This is to ensure long insulation life. The costs associated with losses (LCC- see Page 2, Left Column, Line 6) also result in lower than Class B rise in most cases, since losses result in heat. The above values are for a maximum ambient air temperature of 40°C. If the maximum ambient temperature is above 40°C up to and including 50°C, reduce the table values by 10°C. If the maximum ambient temperature is above 50°C up to and including 60°C, reduce the table values by 20°C.

E.1.2.7 Data on Page 1, Left Column, Line 9

"Min % Overspeed"—The normal overspeed capability of two-pole motors is 20 percent and 25 percent for motors with more than two poles. Higher capability, up to 100 percent, may be required for tandem-driven motors or those motors driving pumps without a check valve.

E.1.2.8 Data on Page 1, Left Column, Line 10

"Bearing Type"—Most motors of the size covered by this standard should be "hydrodynamic" type. A "tilting-pad-journal" type is common for two-pole motors rated above 3000 HP. Many vertical motors will have "antifriction" bearings. If antifriction bearings are selected, API Standard 541 specifies the requirements.

E.1.2.9 Data on Page 1, Left Column, Lines 11–12

"Max Driven Equipment Vertical Thrust"—For vertical motors that will be subjected to an external thrust load application (e.g., from an in-line pump), enter the maximum up and/or down thrust. The thrust bearing should be sized for an L-10 life of 5000 hours at 200 percent of the maximum thrust.

Table E-1—Stator Temperature Rise for Induction Motors for All Horsepower and Voltage Ratings

Motor Rating	Class B Rise (Above 40°C Ambient)
All HP ratings—by resistance	80°C
1500 HP and less—by RTD	90°C
Over 1500 HP—by RTD	
(1) 7000 volts and less	85°C
(2) Over 7000 volts	80°C

E.1.2.10 Data on Page 1, Left Column, Line 13

"Differential Protection to Be Applied"—Specifying this will require that both ends of each phase be brought out so that the differential current transformers can be applied.

E.1.2.11 Data on Page 1, Left Column, Line 15

"Area Classification"—Use the Class, Division, and Group as defined in Chapter 5 of the National Electrical Code. Areas are normally defined for existing plants, or projects. The most commonly specified area classification is Class I, Division 2, Group D (Cl. I, Div. 2, Gr. D) for process areas. "Class I" means a flammable gas or liquid, "Division 2" is where the gas or vapor is present only during abnormal conditions, and "Group D" is a category of materials including gasoline. Almost all areas are "Division 2." A "Division 1" area means the gas or vapor is always present and special enclosures or provisions for ventilation must be used.

E.1.2.12 Data on Page 1, Left Column, Line 17

"Ignition Temperature, Temp. ID No."—See Page 3, left column, lines 15-17.

E.1.2.13 Data on Page 1, Left Column, Line 18

"Elevation"—This is significant if over 3300 feet above sea level. Machines must be derated for elevations above this due to the decreased air density.

E.1.2.14 Data on Page 1, Left Column, Line 19

"Ambient Temperature"—This is the minimum and maximum ambient air temperature. It is significant if below -15°C (5°F) or above 40°C (104°F). Minimum temperature may determine the need for bearing-housing oil heaters. Maximum temperature may determine a derating factor for the motor design or may dictate a special oil cooling system.

E.1.2.15 Data on Page 1, Left Column, Line 20

"Relative Humidity"—This indicates the maximum and minimum humidity.

E.1.2.16 Data on Page 1, Left Column, Lines 21–22

This refers to indoors, outdoors, or sheltered with or without heat, and affects the selection of weatherproofing.

E.1.2.17 Data on Page 1, Left Column, Lines 23–24

"Nonmassive Foundation Description"—The foundation should be designed to meet the criteria of a massive foundation. There may be certain installations where this is not pos-

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sible, such as on offshore platforms. Detail these structures on the data sheet. A nonmassive foundation may affect the mechanical dynamic performance of the motor. Refer to the description of a massive foundation in API Standard 541.

E.1.2.18 Data on Page 1, Left Column, Line 25

"Max Sound Pressure Level"—The normal value is 85 dBA at a distance of 3 feet. This is consistent with OSHA rules so that hearing protection is not required while the motor is in operation. Remote, unattended equipment may not require 85 dBA. Certain rare installations near noise-sensitive areas, such as residential dwellings, may require reduced noise levels of 80 dBA or 75 dBA. Consider the alternative of accepting manufacturer standard noise levels and enclosing the entire drive train in a sound enclosure. Consult a local safety engineer for more guidance.

E.1.2.19 Data on Page 1, Left Column, Line 26

"Unusual Conditions"—Other special site data that may affect motor operation or material selection are recorded here. Examples include abrasive dust, adhering dust, salt spray, corrosive agents, and hose down.

E.1.2.20 Data on Page 1, Left Column, Line 28

"External Forces and Moments"—This lists the significant forces that will be imposed on the motor frame, such as heavy ducting, coolers, silencers, filters, or piping. A request may also be made here for the manufacturer to list the maximum allowable forces or moments.

E.1.2.21 Data on Page 1, Left Column, Line 33

"Max. Short Circuit (SC) KVA at Motor Bus"—This is to record the maximum momentary or ½ cycle, short circuit current (RMS amperes) that can be delivered by the power system. Typically the maximum supply switchgear or circuit breaker rating should be used. This is for determining terminal box pressure release or withstand requirements.

E.1.2.22 Data on Page 1, Left Column, Line 34

"I^2t Let-Through Energy"—If the motor is fed from a fused contactor motor starter, you should enter the maximum let-through energy (ampere squared seconds) for the fuse selected in the motor starters that results from the maximum system short-circuit capacity.

E.1.2.23 Data on Page 1, Left Column, Line 35

"Min. Short Circuit (SC) KVA at Motor Bus"—This is to record the minimum short circuit contribution from the power system with no short circuit current contribution from other motor sources. This reflects the capacity of the utility company or the generation system and is used to calculate the voltage drop during a motor start.

E.1.2.24 Data on Page 1, Left Column, Line 36

"X/R Ratio"—This is the ratio of reactance to resistance for the minimum short circuit KVA condition from the power system.

E.1.2.25 Data on Page 1, Left Column, Lines 37–39

"Type of System Grounding"—This is to specify how the power system source to the motor is grounded. Most applications will be either low-resistance grounded (50 to 400 amperes) or high-resistance grounded (10 amperes maximum). Some systems have "delta" connected transformers at their source with no intentional connection to ground and are "ungrounded."

E.1.2.26 Data on Page 1, Left Column, Lines 40–44 ("Starting")

"Torques in Excess of NEMA MG 1-20.41 Required"—This is not frequently required unless a very low starting voltage will be used (less than 80 percent rated), or if the torque imposed by the load is unusual.

"Full Voltage"—This applies in most cases. This is where the motor starter or circuit breaker is closed to start the motor with nothing intentionally inserted in the circuit to reduce the voltage to the motor, i.e., full voltage is applied to start the motor.

"% Reduced Voltage and Type"—This is where the starting voltage is intentionally reduced by use of an autotransformer, reactor, or resistor. You should indicate the value of reduced voltage that is provided by the starting method. For example, the voltage would typically be reduced to either 80 percent or 65 percent with an autotransformer starter.

"Other"—Other starting methods may include captive transformer (a single transformer feeding only the motor), adjustable frequency, or adjustable voltage starters and shunt capacitor (switched during starting), or series reactor/shunt capacitor starting methods.

E.1.2.27 Data on Page 1, Right Column, Line 2

"Load Curve"—This is to fill in the load curve number of the driven equipment. You should list or check off what type of equipment it will be driving. For driven equipment, indicate whether the start is loaded or unloaded (i.e, with valves, dampers, or guide vanes closed).

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E.1.2.28 Data on Page 1, Right Column, Line 3

"Voltage at Locked Rotor"—The motor is normally designed to be capable of accelerating its shaft load with 80 percent of its rated voltage applied. This is usually a conservative approach, and 80 percent should be specified unless the "stiffness" of the power system ensures the higher voltage of 85 percent or 90 percent. This should first be verified by a motor starting voltage drop calculation with the minimum capability from the supply transformers, lines, utility company, or generation including the effects of the starting method. For captive transformer applications, the voltage available may be less than 80 percent, typically 70 percent to 75 percent for normal transformer ratings in relation to the motor rating. For autotransformer starting, an 80 percent autotransformer start may actually result in 75 percent voltage available to the motor, depending on the power system and autotransformer impedance.

E.1.2.29 Data on Page 1, Right Column, Line 4

"Number of Full-Voltage Starts, if not 5000"—Unless the motor is applied in a repetitive start application, such as a coker jet pump or pipeline pump/compressor motor, specify 5000.

E.1.2.30 Data on Page 1, Right Column, Lines 5–6

"Capability in Excess of Table 2"—If the motor will have very severe starting requirements, specify the exact requirements here. Table 2 already imposes requirements usually in excess to NEMA, so this should not be specified often.

E.1.2.31 Data on Page 1, Right Column, Lines 7–9

"Mounting"—This refers to either horizontal or vertical shaft orientation. For vertical motors, indicate whether the drive end of the shaft is up or down and if the motor is a solid-shaft type or a hollow-shaft type. Most motors are horizontal type, but vertical motors are used for such things as in-line process pumps, turbine pumps, and down-hole well pumps. For vertical type, indicate if it is a solid-shaft motor where the coupling between the motor and driven load is below the motor, or if it is a hollow-shaft type where the driven-load shaft is brought through the hollow center of the motor shaft and coupled to the top of the motor shaft. Hollow-shaft motors are most commonly applied to down-hole well pumps and are recommended for vertical turbine pumps due to greater ease of alignment and coupling to the driven equipment.

E.1.2.32 Data on Page 1, Right Column, Lines 10–11

"Baseplate Furnished by, Soleplate Furnished by"—When an adaptor, mounting plate, or sole plate for embedding in concrete is to be supplied, indicate it here and give a description.

E.1.2.33 Data on Page 1, Right Column, Lines 12–19 ("Enclosure")

"Open Dripproof (ODP)"—This is the minimum protection normally applied for an indoor environment with clean air. It is not recommended for outdoors.

"Weather Protected (WP)"—Air from outside the motor is passed through its interior for cooling active parts. Use the Weather Protected Type II (WP II) for most outdoor applications. The WP II machine is constructed so that high-velocity air and dirt blown into the motor by wind can be discharged without entering the internal air passages to the electric parts of the motor. Use the Weather Protected Type I (WP I) for sheltered locations that may be subject to some weather intrusion or water spray. The WP II and WP I enclosures may not be an appropriate choice where adhering dust is present or if the area does not have free air exchange. The hot air discharged from the motor can cause a closed-in area to become unbearably hot.

"Totally Enclosed Fan Cooled (TEFC)"—This is a construction where free exchange of air is prevented between the inside and outside of the motor. The motor is cooled by a shaft-mounted fan external to the main frame or enclosure, which forces air past the outside of the frame. It is only available in the smaller ratings covered by this specification, which are less than about 800 HP. It is recommended for severe environments. An alternative for larger sizes is the TEAAC.

"Totally Enclosed Water-Air Cooled (TEWAC)"—This is usually the most expensive construction. Use it in environments with adhering dust or dirt, where it is desired to remove the motor-loss heat from a building, or if the motor is critical and none of the other totally enclosed constructions are applicable. A source of cooling water is needed, usually 1 GPM for each kilowatt of motor loss [(0.746 x HP x (100—efficiency %)]. The material chosen for the cooler is usually 90/10, copper/nickel.

Single-tube cooler construction is usually specified with drip trays and leak detectors within the motor. "Single tube" means that the motor cooling air is in direct contact with the finned tube through which cooling water flows. When a water leak occurs, the motor must be shut down.

Double-tube cooler construction is warranted for nonspared service. "Double tube" means that every tube through which water flows is enclosed within a second tube. The clearance

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between the tubes is small and empties into a separate header. If a water leak should develop in an inner tube, the leak is enclosed in the second outer tube and collects in the header. This header is usually equipped with a water detector and will trigger an alarm circuit. Both sets of tubes are rated for the operating water pressure, so no leakage occurs in the air path used for the motor cooling. The motor can continue in operation until a shutdown can be scheduled to repair the cooler.

A water flow switch and/or air outlet temperature RTD is recommended to alarm a loss of cooling water.

"Totally Enclosed Pipe Ventilated (TEPV)"—There is no free exchange of air between the inside of the motor and the air immediately outside the motor enclosure. This is used where the motor is located in very dirty locations or if the motor is installed in a Division 1 hazardous location. It requires air inlet and outlet ducts to duct air to and from the motor, inlet air filters, and usually inlet air blowers.

"Explosion Proof"—This is not often used for this size range. Some manufacturers have listed motors available to 1000 HP. It is only applicable to Class I, Division 1, Groups C and D. For Class II, Division 1 explosive dust environments, designate in "other".

"Other Type"—This may include other NEMA or international enclosure designations. You also could designate use of a "dust-ignition-proof" (DIP) motor for Class II (explosive dust) environments. The DIP motor is totally enclosed and is constructed so that ignitable amounts of dust will not enter the enclosure. It also prevents heat or sparks inside the enclosure from causing ignitions outside the motor.

"Totally Enclosed Air-to-Air Cooled (TEAAC)"—This is similar in function to the TEFC type. It has an air-to-air heat exchanger, usually mounted on the top of the motor, to remove heat from the internal air of the motor by blowing outside air through the exchanger tubes. Use this for locations with severe environments not involving adhering dust. Choose the heat exchanger material based on what is most compatible with air contaminants. Copper-free aluminum is less expensive than stainless steel.

"AISI 300 Series Hardware"—This is recommended for corrosive environments. Common zinc-plated steel fasteners will corrode and make machine disassembly and maintenance difficult. Stainless steel fasteners may cost more, depending on the machine size and the motor manufacturer's pricing policy.

E.1.2.34 Data on Page 1, Right Column, Lines 20–30

"TEWAC Heat Exchanger"—See discussion under Enclosure (TEWAC), page 1, right column, lines 11-17.

E.1.2.35 Data on Page 1, Right Column, Lines 31–32 ("Drive System")

"Direct Connected"—Check if the motor is directly coupled (at the same speed) to the driven load. (It is not coupled through gears, belts, etc.)

"Gear"—List the ratio of speed-increasing or speed-decreasing gearboxes. If this is to be completed by the driven-equipment vendor, add a note to that effect.

E.1.2.36 Data on Page 1, Right Column, Lines 33–34

"Type of Coupling, Furnished by"—Indicate the type or model of coupling; for example, a flexible diaphragm type or a gear type. If it is to be supplied by the driven-equipment supplier, indicate that here.

E.1.2.37 Data on Page 1, Right Column, Lines 35–36

"Rotation"—Indicate the direction of rotation as viewed from the opposite-drive end (the "outboard" end) of the motor. Indicate clockwise (CW) or counterclockwise (CCW).

E.1.2.38 Data on Page 1, Right Column, Lines 37

"Driven-Equipment Wk2"—List the load inertia, including all parts such as couplings, gears, and driven-equipment rotors. These are usually referenced to the motor speed. If this is to be completed by the driven-equipment vendor, add a note to that effect.

E.1.2.39 Data on Page 1, Right Column, Lines 38–42

"Load Reacceleration Required"—List all data for reaccelerating a motor and load following a voltage dip, if acceleration is required.

E.1.2.40 Data on Page 1, Right Column, Lines 43–44

"Other"—This may include belt or clutch drives, for example.

E.1.2.41 Data on Page 2, Left Column, Lines 1–2

"Reciprocating Compressor Load, Crank Effort Diagram Curve No."—Used by the motor suppliers to design the motor with adequate flywheel effect and to minimize line current variation (keep within the NEMA 66 percent variation, unless specified lower).

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E.1.2.42 Data on Page 2, Left Column, Lines 3–5 ("Miscellaneous")

"Special Low Temperature Material Requirements"—If application will be at or below –20°F (–29°C), the purchaser should indicate any special requirements here.

E.1.2.43 Data on Page 2, Left Column, Line 6

"Evaluation Factor"—See below to determine the "$/kilowatt" to be used for the evaluation of bids based on quoted efficiency. This should be completed for all motors. You should also list the load operating point. If you are unsure, list the motor nameplate horsepower.

Motors should be purchased on the basis of LCC. LCC is the purchase price of the motor plus the value of losses over the life of the motor using the evaluation factor (EF) shown.

$$LCC = P + EF\,[0.746 \times BHP \times [(100/E) - 1]]$$

Where:

EF	=	Evaluation Factor ($/KW) = C x N x PWF.
BHP	=	Brake HP of Driven Equipment at Operating Point.
C	=	Energy Cost in $/KWH (Dollars per Kilowatt Hour).
E	=	Motor Efficiency in Percent, at the Driven-Equipment Operating Point.
N	=	Operating Time in Hours Per Year.
P	=	Purchase Price of the Motor in Dollars.
PWF	=	Cumulative Present Worth Factor. (This factor typically ranges from 2 to 4 for the purpose of bid valuations.)

E.1.2.44 Data on Page 2, Left Column, Lines 7–9

"Federal, State, or Local Codes or Special Rules"—List and supply copies of any local, special codes. Southern California installations frequently call for special electrical codes, for example.

E.1.2.45 Data on Page 2, Right Column, Lines 2–5 ("Miscellaneous [Continued]")

"List of Materials to Be Identified With Applicable ANSI, ASTM, or ASME Numbers"—If you want certain parts identified by standard designations, list them here. This may include flanges, minor hardware, shaft material, and bearing babbitt alloy.

E.1.2.46 Data on Page 2, Right Column, Lines 6–9

"Special Identification"—Any special identification for drawings, correspondence, etc. should be entered here.

E.1.3 ACCESSORY EQUIPMENT

E.1.3.1 Data on Page 2, Left Column, Lines 11–15

"Winding Temperature Detectors"—These detectors are installed in intimate contact with the winding insulation and give an accurate measurement of the operating temperature of the winding. They provide better protection for the motor than current sensitive overload relays. They also improve protection against clogged air filters, which can cause high-winding temperatures in weather-protected (WPI and WPII) enclosures. It is generally recommended on all motors covered by this standard. Two detectors should be specified for each phase of the motor winding. A temperature detector type should be specified, depending on the monitoring system design.

E.1.3.2 Data on Page 2, Left Column, Lines 16–21

"Bearing Temperature Detectors"—These detectors should be applied consistent with the entire equipment train. They are usually applicable to large (1000 HP and greater) and special purpose equipment trains. They can provide early warning of lube oil loss or impending bearing failure. API Standard 670 outlines requirements that will ensure accurate bearing-metal temperature measurement. When radial temperature detectors are used for shutdown systems, either resistance temperature detectors (type 100-ohm platinum at 0°C or 120-ohm nickel at 0°C) or thermocouples (type iron constantan, for example) can be specified. Separate terminal heads are sometimes specified with an external conduit run to each head. An alternative is a terminal box, either separate or in combination with the winding RTDs.

"Bearing Housing Temperature Indicators Required"— This introduces temperature indicators into the oil sump of each bearing to give a general indication of the oil discharge temperatures. It would generally be used by operators on regular rounds. Bearing temperature detectors (above) give much better indication of the bearing condition.

E.1.3.3 Data on Page 2, Left Column, Lines 22–39

"Lubrication System"—The various features of the lubrication system are selected in this section. Only the significant items are described.

a. "Self Lube, Pressure Lube"—Specify the type of lubrication system. Where "self lube" is acceptable, select that option. Many motors can be run with self- or oil-ring lubrication up to about 1500 HP, depending on the manufacturer's recommendation. Pressure-fed lubrication is recommended wherever possible for more reliable motor operation. A general purpose lubrication system is recom-

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mended where the motor has "back-up" oil-ring lubrication. The special purpose system of API Standard 614 is recommended for bearing designs that do not have oil rings. Where the lube oil system is common with driven equipment, indicate such (also in this section).

b. "Oil Mist"—This should be considered where antifriction (ball or roller) bearings are applied.

c. "Special Seals for Gas Purge"—Specify this for motors that will operate in severe dust or dirt environments. A clean, dry air or other gas source must be available.

d. "Oil Heater Required" -This applies to the sump of the lube system where heaters may be required to bring the oil to minimum operating temperature.

E.1.3.4 Data on Page 2, Left Column, Lines 40–45 and Right Column, Lines 11–18 ("Vibration Detectors")

a. "Noncontacting (Shaft) Probes"—Specify this consistent with the equipment train. It is recommended for critical motors with sleeve or tilting-pad bearings with less than six poles. Normally, a phase-indicating probe is specified whenever probes are specified. This provides a phase reference for filtered vibration and speed measurements. Two probes per bearing are normal (90 degrees apart), but four probes per bearing may be desired if the probes are not accessible during the motor operation and vibration monitoring is considered important (usually only two oscillator/demodulators are supplied for each bearing since the probe wires can be exchanged during operation).

b. "Bearing Housing Seismic Sensors"—Specify this consistent with the equipment train. Usually applied on critical sleeve bearing motors rated with six or more poles and on motors with antifriction bearings of all speed ratings. Accelerometers should be specified using API Standard 678, or velocity pickups may be specified. Specify acceleration, velocity, or displacement transducer output for accelerometers (velocity output is recommended) and velocity or displacementtransducer output for velocity pickups. Spot-faced surfaces can be specified for future mounting of transducers, usually two on each bearing housing configured 45° off vertical.

E.1.3.5 Data on Page 2, Right Column, Lines 19–46; Page 3, Left Column, Lines 1–9 ("Main Conduit Box")

"Box Location"—Indicate top, bottom, or side (left or right, facing a designated end of the motor).

"Oversized Box"—Specify if there are special needs. The user and supplier must agree on how large a box to provide.

The box should include the following:

1. "Main Supply Leads: Conductor Size/Type"—Indicate conductor size; number per phase; and whether the cable is shielded or unshielded, type MC (metal-clad) cable, type TC (tray-cable) cable, or type MV (medium voltage) cable. Coordinate the choice with the project or responsible electrical engineer. "Enter From"—Indicate the position of the main power conductor entry. This depends on the physical configuration of the cable/conduit system. "Conduit Type, Size, Number"—Select from available types and materials for conduit or the use of type MC cable, for example.

2. "Current Transformers for Differential Protection"— This is recommended for critical or special purpose motors or all motors rated 1500 HP and larger. Appropriate protective relays must be provided in the motor starter or switchgear. Core-balance or window-type current transformers are the most common. Either specify the specific type (e.g. type BYZ) or the current transformer accuracy class. A "C 10" accuracy class is usually adequate, but "C 20" may be required depending on the protective relay type. A "C 10 accuracy class" for "type" and a ratio of 50 to 5 (50:5) are recommended. Bar-type current transformers are only applied for very high continuous current ratings and are specified with appropriately high ratios and accuracy classes to match a set of three current transformers in the supply switchgear. For either the core-balance or bar-type current transformers specified, indicate whether they will be supplied by you (or the switchgear vendor) or the motor manufacturer.

3. "Surge Capacitors"—This is recommended for critical motors, those connected through one transformer or directly to a bare overhead line, or those that have switched capacitors on the same voltage level. Under these conditions, it is recommended for each motor individually. Specify 0.5 microfarad for motor voltage ratings through 4160 volts and 0.25 microfarad for ratings 6600 volts and above.

4. "Surge Arresters"—This is recommended for the same-conditions given for surge capacitors. For those motors connected to a bare overhead line through at least one transformer (protected on its primary with arresters), one set of surge arresters applied on the main switchgear to protect a group of motors is adequate. Specify 2.7 kV rated arresters for 2.3 kV rated motors, 4.5 kV for 4.0 kV motors, 7.5 kV for 6.6 kV motors, and 15.0 kV for 13.2 kV motors.

5. "Phase Current Transformers for Local Ammeter"—If a single current transformer is needed for load current indication local to the motor, specify the type (usually a window type) and ratio with a maximum current rating of 150 percent of the rated load current of the motor. An alternative is to use a current transducer mounted in the switchgear wired to an ammeter mounted at the motor.

6. "Voltage Transformers for Local Voltmeter"—This is rarely specified, except for captive transformer applications where switchgear usually is not available at the motor uti-

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lization voltage to extract a voltage. It adds complication, space requirements, and a potential failure point within the terminal box.

7. "Bushing Studs or Receptacles"—This is rarely specified. It is used where a fast change of motors is necessary in the event of failure. Specify the type of stud or receptacle.

8. "Space for Stress Cones"—If shielded conductors are used, specify the length of the cable termination.

9. "(Interior Top Side) Thermal Insulation"—This is recommended for motors with large terminal boxes (enclosing surge protection and instrument transformers) in locations where moisture condensation on cool metal surfaces is common. Use this along with terminal box space heaters.

10. "Space Heaters"—See temperature limits for "motor space heaters." This is usually specified as 120 or 240 volt single phase due to typical low power requirement. These space heaters should always be energized, even while the motor is in operation. The heater element surface temperature limits are the same as for the machine space heaters.

11. "Breathers"—This should be specified as below for "drains," usually combined with drains.

12. "Drains"—This should be included for all smaller, non-space-heated terminal boxes to allow drainage of condensation.

13. "Provision for Purging"—Purging is required by the National Electrical Code for terminal boxes having surge arresters mounted within the terminal box when the motor is installed in a Class I, Division 1 area. Specify for this condition.

14. "Removable Links"—This is recommended for motors with larger terminal boxes to permit isolation of each phase of the motor from the incoming cable and surge protection. This permits maintenance high-potential or insulation tests to be done without tedious untaping and disconnection of devices in order to isolate the motor windings.

15. "Silver-Plated Bus Joints"—This is recommended for bus links and cable terminations to reduce contact resistance from corrosion.

16. "Ground Bus"—This should be specifed for all motors.

17. "Fault Withstand"—The extreme temperature of an arcing fault within a sheetmetal terminal box can result in internal air pressure that causes latched or bolted covers to blow off during a fault. Indicate if this "fault withstand" protection is required in this space. You can indicate if this is to be done with a pressure rupture disc or through reduction of fault levels (to ground-fault levels) through phase segregation within the terminal box.

18. "Other Terminal Box Requirements"—Specify any other terminal box features or requirements here.

E.1.3.6 Data on Page 3, Left Column, Lines 12–13

"Air Filters"—Filter provisions (mounting hardware) are provided as standard on WP II enclosures, but should be specified for WP I and TEPV unless the TEPV motor has a separate filtered air supply. The cost of filter provisions is low. Specify filters for dirty or dusty environments, but keep in mind that either an air filter differential pressure switch or winding temperature detectors must be wired to alarm operators when the filters become dirty. When filters are specified, order a set of spares so they can be exchanged from the motor and cleaned.

E.1.3.7 Data on Page 3, Left Column, Lines 14–15

"Differential Pressure Service Required"—This should be specified for weather-protected (WP I or WP II) motors ordered with air filters. It gives advance warning when the filters are clogged (when wired to an alarm) before winding temperature detectors will register a significant change.

E.1.3.8 Data on Page 3, Left Column, Lines 17–18

"Space Heaters"—All motors should include space heaters. Normally, the "manufacturer's standard" sheath material is acceptable, but stainless steel or other material could be specified. Specify 120 or 240 volt, single-phase or 208 or 480 volts, three-phase power, depending on what power source would be available with the motor shut down. Three-phase power for the space heaters is usually only needed for motors over approximately 4000 HP.

"Maximum Sheath Temperature" is the maximum temperaturepermitted at the surface of the heater element, which could be a source of ignition in an explosive atmosphere if the element is too hot. For Class I (gas or vapor), Group A, B, C, or D locations, specify the lesser of 200°C or 80 percent of the autoignition temperature from NFPA 497M for the specific gas or vapor involved.

Table E-2 provides a list of selected flammable gases and vapors of liquids having an auto ignition temperature (AIT) of less than 250°C, requiring space heaters to have sheath temperatures less than 200°C.

(Extracted from NFPA 497M-1991.)

E.1.3.9 Data on Page 3, Left Column, Lines 19–21

"Bearing Heaters"—This is not specified very often because most motors in extremely cold climates are installed in heated shelters. Where needed to bring lube oil in the bearing housing sump up to the minimum temperature recommended by the manufacturer, specify a type that can conform with API Standard 541 specifications.

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E.1.3.10 Data on Page 3, Left Column, Lines 22–30

"Miscellaneous"—Select desired items from this section.

E.1.3.11 Data on Page 3, Left Column, Lines 32–35

"Miscellaneous—Other"—One example of what could be entered here follows:

"Other"—Oil sump temperature detectors may be specified for self-lubricated bearings, or oil inlet temperature detectors may be specified for forced lubricated bearings. These would provide early warning of oil temperature problems, but no indication of bearing problems. Specify only for large, critical motors. Air temperature sensors are recommended for large motors with TEWAC enclosures to detect cooling water loss of flow. This could be used instead of a water flow switch. Specify the type of detector (RTD or thermocouple).

E.1.3.12 Data on Page 3, Right Column, Lines 1–8

"Coupling"—If a coupling will be supplied to the motor manufacturer for mounting, specify it here. This is recommended for two-pole motors over 2000 HP and may be specified for critical drivers of lower speeds. Be sure to specify an idling adapter if the motor is to be run uncoupled with the coupling mounted or if the motor will have any rotor-dynamic tests at the factory. Indicate all coupling information in this section.

E.1.3.13 Data on Page 3, Right Column, Lines 9–11

"Painting"—The standard painting is normally accepted unless the motor will be installed in a very severe environment,

Table E-2—List of Liquids With an Auto Ignition Temperature (AIT) of Less Than 250°C, Requiring Space Heaters With Heater Element Surface Temperature Less Than 200°C (Extracted From NFPA 497M-1991)

Material	Group	AIT°C	80% AIT°C
Acetaldehyde	C	175	140
Acrolein (inhibited)	B(C)	235	188
n-Butyraldehyde	C	218	174
Carbon Disulfide		90	72
Crotonaldehyde	C	232	186
Cyclohexane	D	245	196
Cyclohexene	D	244	195
Cyclohexanone [a]	D	245	196
Decene [a]	D	235	188
Diethyl Ether	C	160	128
1,4-Dioxane	C	180	144
Dipentene [a]	D	237	190
Ethylene Glycol Monoethyl Ether [a]	C	235	188
2-Ethyl Hexanol [b]	D	231	185
2-Ethylhexaldehyde [a]	C	191	153
Fuel Oils [a]	D	210-407	168
Heptane	D	204	163
Hexane	D	225	180
Hexenes	D	245	196
Hydrazine [a]	C	23-270	—
Isobutyraldehyde	C	196	157
Isoprene	D	220	176
Iso-octyl Aldehyde [a]	C		
Kerosene [a]	D	210	168
Methyl Formal	C	238	190
Monomethyl Hydrazine	C	194	155
Nonane	D	205	164
Octane	D	206	165
Octene	D	230	184
Pentane	D	243	194
Propionaldehyde	C	207	166
n-Propyl Ether	C	215	172
Propyl Nitrate	B	175	140
Unsymmetrical Dimethyl Hydrazine	C	249	199
Valeraldehyde	C	222	178

[a]Flash point of these materials is between 100°F (37.8°C) and 140°F (60°C). Special electrical equipment is required only if these materials are stored or handled above their flash points.

[b]Flash point of these materials is between 140°F (60°C) and 200°F (93.3°C). Special electrical equipment is required only if these materials are stored or handled above their flash points.

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such as offshore. If special paint is specified, surface preparation and paint specifications should be included with the quotation request to the manufacturer and consideration should be given to the painting system used on the inside surfaces of a WP enclosure.

E.1.3.14 Data on Page 3, Right Column, Lines 12–17 ("Shipment")

a. "Domestic"—Depending on the protection inherently offered by the enclosure, specifying this would include the protection described in API Standard 541 and coverage of the motor by a tarpaulin or plastic.

b. "Export"—This would normally include complete, sealed coverage with desiccant. The manufacturer should describe the preparation included when the proposal is presented.

c. "Export Boxing Required"—A substantial box is built around the motor in addition to the sealing and desiccant described above.

d. "Special Shipping Bearings"—Most motor suppliers will ship the motor with the bearings installed with some temporary protection. If the motor is likely to experience abuse during shipment, or if extra insurance against bearing damage during shipment is desired, select this feature that requires the supplier to install special bearings for shipment only (these may be made of wood).

e. "Mounted on Skid"—If the motor will be handled by forklift at the site prior to installation, specify this option.

E.1.3.15 Data on Page 3, Right Column, Lines 18–34 ("Other Requirements")

E.1.3.16 Data on Page 3, Right Column, Line 25

"Special Weather Protection Requirements"—If there are special weather conditions that will exist during storage, start-up, or operation, list the conditions. Specifying this requires the manufacturer to quote protective items.

E.1.3.17 Data on Page 3, Right Column, Lines 26–27

"Vendor's Review and Comments on the Purchaser's Foundation Drawings"—This should be specified for applications where rotor dynamics or vibration may be a problem, such as large, two-pole motors or motors to be mounted on elevated structures.

E.1.3.18 Data on Page 3, Right Column, Line 28

"Guaranteed Efficiency and Power Factor Required"— This should be requested of all proposals. To compare bids, it is suggested that all vendors quote to one method, such as IEEE 112, Method F.

E.1.3.19 Data on Page 3, Right Column, Lines 31–35

"Other"—Examples of what may be specified here follow:

"Fit-up/Assembly of Machine-Mounted Components"—You should specify when the motor will be supplied with oil piping or water coolers.

"Certified Test Reports Required Prior to Shipment"— This is not normally specified, but it would give the manufacturer incentive to get the reports out quickly.

E.1.3.20 Data on Page 3, Right Column, Lines 37–46

"Comments"—Fill in any additional or explanatory information.

E.1.4 ANALYSIS, SHOP INSPECTION, AND TESTS

E.1.4.1 Data on Page 4, Line 1

"Lateral Critical Speed Analysis"—This analysis is usually done by the motor manufacturer and gives the system "critical speeds" of the rotor, bearing, bearing support, and foundation system. A critical speed is usually excited by rotor unbalances or misalignment. It is important to predict where the critical speeds are and how the rotor will react to excitations that can result in lateral shaft vibration. This analysis is recommended for two-pole motors 800 HP and larger or for four-pole motors 5000 HP or larger unless the motor is specifically designed to operate below its first system resonance. Most other ratings operate well below the first critical speed.

E.1.4.2 Data on Page 4, Line 2

"Design Review Meeting"—This is a comprehensive review meeting where the detailed electrical and mechanical designs are discussed and any electrical/mechanical analyses are presented. This is recommended for critical motor driven equipment trains with ratings above 3000 HP, or for new manufacturer designs (prototypes). This is not the "order-coordination meeting."

E.1.4.3 Data on Page 4, Line 3

"Torsional Analysis"—This is normally done by the driven-equipment manufacturer but could also be verified by the motor manufacturer or a third party. This analysis is usually performed only on drive trains with speed-increasing gearboxes to the driven equipment or those supplied by adjustable frequency drives. Usually it is not specified for direct-driven loads.

E.1.4.4 Data on Page 4, Line 4

"Shop Inspection"—Suppliers usually charge for the extra inspections included in this specification.

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E.1.4.5 Data on Page 4, Line 6

"Material Certifications"—This is recommended for all forgings, such as shafts for two-pole motors. When this is specified, the wording of API Standard 541 includes major forgings and castings.

E.1.4.6 Data on Page 4, Line 7

"Assembly, Maintenance, and Running Clearances"—It is recommended that this be specified for special purpose motors or if your maintenance organization needs this information for its data base. These data should be available at minor cost.

E.1.4.7 Data on Page 4, Line 8

"Painting Deferred (on Parts)"—The purchaser must list the specific part, such as a casting, welded shaft, or rotor assembly. It is not usually specified.

E.1.4.8 Data on Page 4, Lines 9–13

"Surface and Subsurface Inspection of Parts"—Specify for motors over 3000 HP in very critical applications with welding on the shaft or those motors with forged shafts. Unless acceptance standards are established or agreed upon at the time of purchase, the motor manufacturer is free to determine what is acceptable or not acceptable. Normally, inspections might include liquid penetrant or magnetic particle inspection for welds or ultrasonic inspection of forged shafts. The part to be inspected and the nondestructive examination to be made must be filled in.

E.1.4.9 Data on Page 4, Line 14

"Hardness Test"—This is rarely specified except for arctic applications. Standards and acceptance criteria must be established or agreed upon.

E.1.4.10 Data on Page 4, Line 15

"Optional ASTM Tests to Be Proposed by Vendor"—This requests that the motor supplier propose tests to ensure that the materials are suitable for service. Specify for severe environments or critical service.

E.1.4.11 Data on Page 4, Line 16

"Balance in Minimum of 3 Planes"—Specify this for motors that will operate above their first system resonance, generally two-pole motors above approximately 800 HP. The additional third plane at the center of the rotor generally reduces vibration as the motor passes through the first system resonance.

E.1.4.12 Data on Page 4, Line 17

"Check Balance With Half Coupling"—This is usually warranted only for two-pole motors of large size (2000 HP and above) to determine the effects of mounting the coupling on the balance of the rotor. This is done in the same time frame and using the same procedures as the residual unbalance test. Check "required" when this is to be done.

E.1.4.13 Data on Page 4, Line 18

"Inspection for (Lube Oil System) Cleanliness per API Standard 614"—This requires the lube oil system cleanliness of API Standard 614. Specify when the special purpose API Standard 614 lube oil system is specified for the drive train.

E.1.4.14 Data on Page 4, Line 19

"Review of Quality Control Program"—Normally this is not specified unless the motor manufacturing plant has undergone change or if recent problems have been experienced.

E.1.4.15 Data on Page 4, Line 20

"Stator Inspection Prior to Vacuum Pressure Impregnation"—This is a physical inspection of the iron core of the motor and the winding insulation taping before the winding is put through VPI, a procedure to seal and solidify the winding insulation system. This inspection should be specified for very critical, unspared motors or for motors with voltage ratings 6600 volts and above. It is frequently omitted when the purchaser has confidence in the manufacturers. When specified, it should be "witnessed." The cost impact is usually minimal.

E.1.4.16 Data on Page 4, Line 21

"Residual Unbalance Test"—This balance verification procedure is recommended on motors 600 HP and larger with two or four poles and is required on all special purpose motors. This test should not be performed on motors having eight poles or more.

E.1.4.17 Data on Page 4, Line 22

"Routine Test"—These tests are required on all motors. For motors approximately 1000 HP and greater, both the "routine" and "complete" tests are frequently specified, which includes a more extensive series of tests to prove the efficiency and winding temperature rise of the motor.

E.1.4.18 Data on Page 4, Line 23

"Polarization Index Voltage"—Normally you would specify 2500 or 5000 volts. If not filled in by the user, the supplier may perform the PI test at a lower voltage.

E.1.4.19 Data on Page 4, Line 24

"Vibration Recording"—Evaluate the vendor's proposed data acquisition system for vibration tests. This is very useful and should be used for motors that will undergo unbalance response tests or the "dual-frequency" type of rated temper-

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ature test. This requirement primarily applies to two-pole motors 800 HP and above but may also be applicable to slower speed motors of ratings above 3000 HP. Use of a recording system actually saves the manufacturer test-floor time since an unbalanced response test, which takes up to 30 minutes for each "coast down," need only be done once and the data accessed rather than being done four times (once for each shaft vibration probe). During the dual-frequency test, rapidly changing vibration quantities can be captured as the motor cools down after the removal of the power source for the second frequency.

E.1.4.20 Data on Page 4, Line 25

"Stator Core Test"—This is a quality test for core-plate insulation integrity or damage. Some suppliers do this test during manufacturing, but it should only be specified by the user for unspared or applications where the motor will be inaccessible for easy repair or replacement.

E.1.4.21 Data on Page 4, Line 26

"Special Surge Test on Sample Coils"—This is recommended for windings rated 6600 volts and above, usually as an "observe."

E.1.4.22 Data on Page 4, Line 27

"Power Factor Tip-up Test"—This is recommended for stator voltages 6600 volts and above for the completed stator. It provides a "baseline" for later maintenance tests to detect corona discharge within the insulation system. It may also be specified on sacrificial coils for development testing but acceptance criteria needs to be agreed upon between the user and supplier. The power factor tip-up test is usually conducted at 10 percent and 110 percent or 20 percent and 120 percent of rated line-to-ground power voltage. The test results are plotted for each motor, and percent deviation is calculated.

E.1.4.23 Data on Page 4, Line 28

"Sealed Winding Conformance Test"—This test should be specified when verification of a sealed winding is desired. This test should be as a "witness" and may be applied to critical, special purpose motors. Some manufacturers do this test as a standard so the "required" option should be considered. The cost of this test is based on the vendor's perceived risk of failing the test.

E.1.4.24 Data on Page 4, Line 29

"DC High-Potential Test"—The motor will already have had a final AC high-potential test for a 1-minute duration to prove the insulation of the motor. Subsequent high-potential tests in the field will most likely be DC tests. If a baseline dc test is desired to compare with tests in the field, specify that here. This test is performed at approximately 75 percent of the equivalent final ac high-potential test, so it does relatively little damage to the insulation at this stress level.

E.1.4.25 Data on Page 4, Line 30

"Complete Test"—This imposes the API Standard 541 "Complete Test" requirements. Specify this when the efficiency and temperature rise are to be determined. Normally this is not specified for smaller motors. This test should be specified for at least one of each motor rating ordered at the same time. It should also be specified where the evaluation factor justifies the test cost to prove the efficiency.

E.1.4.26 Data on Page 4, Line 31

"Rated Rotor Temperature Vibration Test When Complete Test Not Specified"—The test is recommended for all two-pole motors, rated 600 HP or larger, and critical, special purpose motors of slower speed ratings because of the susceptibility of many rotors to changing balance condition with temperature. During this test, the motor is heated to its rated operating temperature and the vibration performance is monitored and compared with acceptable limits. Some exceptions can be made for "general purpose" motors in noncritical applications. This test is usually "witnessed."

E.1.4.27 Data on Page 4, Line 32

"Bearing Inspection at Completion of Test"—This item supplements the normal bearing inspection of 4.3.2.1.j and is recommended for two-pole motors operating above their first mechanical system resonance ("critical speed") to ensure the bearings have had no distress as the rotor speed passed through the resonance. It is also recommended for motor designs where the mechanical performance is sensitive to small changes in bearing clearances.

E.1.4.28 Data on Page 4, Line 33

"Unbalance Response Test"—This test is recommended for all two-pole motors 800 HP and larger and all four-pole motors 5000 HP and larger to verify the motor's performance operating through its first resonant speed or to verify the location of its resonant speed above operating speed. Often it is dependent on the manufacturing process so it should be done on every motor unless the user's experience would dictate relaxation of this test, perhaps to one of each size purchased at the same time.

E.1.4.29 Data on Page 4, Line 34

"Bearing Housing Natural Frequency Test"—Normally this is specified for the first motor manufactured of a certain

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frame size or a uniquely designed motor. The risk of not requiring the test is low due to the low bearing housing vibration limits required by API Standard 541, and if the motor passes the vibration tests, the motor probably does not have a significant resonance.

E.1.4.30 Data on Page 4, Line 35

"Certified Data Prior to Shipment"—This may provide an incentive to provide these data promptly.

E.1.4.31 Data on Page 4, Line 36

"Test Equipment Supplied by Purchaser"—If you desire to provide your own vibration or other test equipment for use during testing, itemize the equipment here. This may be desired for data compatibility reasons for later documentation or analysis.

E.1.4.32 Data on Page 4, Line 37

"Test Equipment/Data Acquisition Accuracy to Be Demonstrated Prior to Test"—The motor supplier should have this documented every 6 months to 1 year and should be able to supply this documentation on request. If the test setup is unique or if the motor supplier does not have the accuracy documented, you may wish to specify this.

E.1.4.33 Data on Page 4, Lines 38–43

"Other"—Examples of other analysis, inspection, or tests that might be specified are as follows:

Analysis—Indicate other analyses to be done by the motor or driven-equipment manufacturer, such as current-pulsation analysis for reciprocating loads.

Tests—These tests are normally "witnessed" if done at the same time as other tests. Specify as applicable:

a. "Cooler Hydrotest"—This applies 150 percent design pressure to the cooler to make sure it does not leak. Only specify this for motors with water coolers (TEWAC). Usually this is specified as "required" or "observed" with the test done at the cooler manufacturing site.

b. "Leak Detector"—This verifies the detector works when water is present. It is a simple test to be specified for TEWAC motors only.

c. "Air Filter DP Switch"—This test for proper operation of the air filter differential pressure (DP) switch should be made when the DP switch is specified on weather-protected motors (see Page 3, Left Column, Line 13).

d. "Reed Critical Frequency"—This is a test to verify the mechanical resonance of the motor structure, which acts as a vibrating "reed." It is only applicable to vertical motors. A test in the motor manufacturer's factory only confirms the calculated value for the motor by itself. Considering the expense of this test, it is not specified often. It is a more meaningful test if the motor is mounted to the actual pump head and the test is performed on the assembled motor and pump in the pump manufacturer's shop. Reed resonances should be at least 15 percent separated from running speed frequency excitation for the assembled motor and pump.

e. "Noise"—Only specify this if the motor is to be installed in a noise-sensitive area. In many cases, certified data can be accepted from a motor of duplicate design.

Inspection of Equipment Lube Oil Piping—This is applied when a motor with a forced lubrication system has a thorough inspection program. It is recommended for forced lubricated bearings where a thorough flush of the equipment lube oil piping will not be made at the installation site.

E.2 Vendor's Data Sheets (Appendix B)

The data sheets in Appendix B are to be used by the vendor. They include the Vendor's Induction Motor Proposal Data Sheet (one sheet) and the Vendor's Induction Motor Contract Data Sheet (two sheets).

Note: Many of the items in the Appendix B data sheets are analogous to ones explained in E.1. Also, the heads of the following subsections do not necessarily match heads in Appendix B's data sheets.

E.2.1 VENDOR'S INDUCTION MOTOR PROPOSAL DATA SHEET

The Vendor's Induction Motor Proposal Data Sheet (the first data sheet of Appendix B) must be completely filled out by each motor manufacturer so a fair evaluation of all bids can be made. The following is included to help you evaluate the information supplied by the vendor. The sequence is as shown on the data sheet.

"Vendor"—This is required for identification.

"Horsepower" is the rated shaft output power at the motor's rated load speed. This is the nameplate horsepower.

"Full Load Speed (RPM)" is the speed of the motor with rated horsepower load applied to the shaft.

"Frame No." is the manufacturer's designation.

"Enclosure" is the NEMA or IEC type designation.

"Locked Rotor current at rated voltage and 80% voltage" is the current demanded by the motor at standstill. It is a value required to make voltage drop calculations during the start.

"Power Factor" is also useful for motor-starting calculations.

"Motor Rotor Wk^2"—Inertia data for the motor is requested. Acceleration time calculations can be made by the purchaser for evaluating the motor starting capability and setting the motor's protective relays.

"Torques: Full Load, Locked, Pull-up, and Breakdown" are key torque values defined for induction motors.

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"Curves of Torque, Current, and Power Factor Versus Speed" are used for motor starting time and voltage drop calculations.

"Estimated Starting Time (with load)" represents the time that must be allowed by power system protective devices for the motor to accelerate. This is often used by the electrical engineer to determine the settings for protective relays.

"Locked-rotor Withstand Time With Motor Cold and at Rated Operating Temperature" gives a measure of how much thermal capability the rotor is designed for and is critical to locked-rotor protective device settings.

"No Load Current" is the current drawn by the motor with rated voltage applied and zero shaft load. It is helpful in determining the maximum power factor correction that can be applied at the motor terminals.

"Amperes at ¼, ½, ¾, Full, and Service Factor Load" is the current demanded by the motor at various shaft loads at rated voltage.

"Efficiency at ¼, ½, ¾, Full, and Service Factor Load" gives other points of efficiency for evaluation.

"Power Factor at ¼, ½, ¾, Full, and Service Factor Load" can be used to estimate the overall plant power factor and estimate power factor correction capacitors.

"Guaranteed Efficiency; IEEE Method" is the efficiency used for LCC evaluations. Insist on one method—other methods are not comparable. For Method F, the manufacturer has the choice of either assuming the stray-load loss or testing for it. If the assumed stray-load loss is used, it is usually greater than the tested stray-load loss. The test for stray-load loss is time consuming.

"Bearing Type" is most commonly a split-sleeve type but could be a tilting-pad-journal type of sleeve bearing. An antifriction bearing could also be indicated if it was specified by the purchaser.

"Special Weather Protection" is outlined by the vendor in response to conditions specified by the user.

"Utility Requirements for oil, cooling water, or ventilation air" are listed by the vendor for evaluation purposes or preliminary design.

"Reciprocating Compressor Load: crank-effort, current pulsations, and calculated efficiency" are requested for applications with nonconstant shaft torque.

"Weights and Dimensions" include catalog or estimated information on weights and dimensions.

"Air Gap" is the radial distance between the rotor and stator of the motor.

"Max Sound Pressure Level at 1 Meter" is the sound level proposed, as measured by IEEE 85 methods.

"Vibration Test Using Responsive Amplification Method" indicates whether or not the vendor is proposing to use this method for vibration acceptance.

"When Rotor Dynamic Analysis Is Specified, List of Foundation Data Required From Purchaser" is the vendor's opportunity to outline data requirements that the user will have to respond to soon after order placement. The data are needed to ensure timely completion of the analysis in case major problems affect the motor or foundation design.

"List of Optional ASTM Tests Proposed by Vendor" is the manufacturer's opportunity to propose tests to make sure the materials in the motor are satisfactory for the specified service.

"Rotor and Shaft Details"—As defined in API Standard 541, details on shaft sealing, bearings, internal construction, rotor construction, and bar-to-end-ring attachment are required with the proposal.

"Comments"—Any additional remarks or notes by the manufacturer during the proposal should be included here.

E.2.2 VENDOR'S INDUCTION MOTOR CONTRACT DATA SHEET

The Vendor's Induction Motor Contract Data Sheet (the second and third data sheets in Appendix B) requests information determined during the motor design. They should be submitted with the requested data not more than 6 weeks after the purchase order has been placed.

E.2.2.1 Motor Limits and Constants

The following information related to motor limits and constants is requested by Appendix B's data sheets:

a. "Locked-rotor Withstand Times" are used to determine the final settings of motor-protective devices.
b. "Allowable Acceleration Times" with the motor at its maximum operating temperature takes into account rotor cooling and other effects during acceleration.
c. "Squirrel-cage Model Constants" are used for motor simulation in power system studies and other calculations.
d. "Phase-to-phase Resistance" is helpful information for at-site efficiency determination.
e. "Subtransient Reactance" is used for power system short circuit studies.
f. "Symmetrical Current Contribution to Three-phase Terminal Fault" is information used to check short circuit study approximations.
g. "Total Winding Capacitance to Ground" is used for power system charging current calculations for the application of high-resistance grounding.
h. "Residual Voltage Open-circuit Time Constant" is needed for application of fast supply bus transfer schemes.

E.2.2.2 Bearing Details

The following information related to bearing details is requested by Appendix B's data sheets:

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a. "Bearing Type; Capable of Self-lubrication" confirms the design of the bearing and whether provisions have to be made for backup of forced lubrication.
b. "Bore Length and Diameter" lists critical bearing dimensions.
c. "Design Clearance With Shaft" lists the minimum and maximum clearances for use during the inspection and testing phase of manufacturing or for future maintenance.
d. "Loading" gives a measure of whether the bearing is in normal design limits.

E.2.2.3 Alarm and Shutdown Values

This lists the manufacturer's recommended settings for alarm and shutdown. It includes bearing temperature detectors, winding temperature detectors, air filter differential pressure switch, or any other detectors.

E.2.2.4 Winding and Rotor Repair Data

This includes data that would be helpful in the event of winding or rotor failure.

E.2.2.5 Submittals

A listing of the more important submittals is given. The lateral and torsional critical speed reports are required where the purchaser has specified them. Where testing is required, the manufacturer must submit a test agenda and sample data sheets.

E.2.2.6 Remarks

Any appropriate remarks are given.

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